Roma Tre

INGEGNERIA
CIVILE, INFORMATICA E DELLE TECNOLOGIE AERONAUTICHE

Information Visualization

Straight-line Grid Drawings

M. Patrignani

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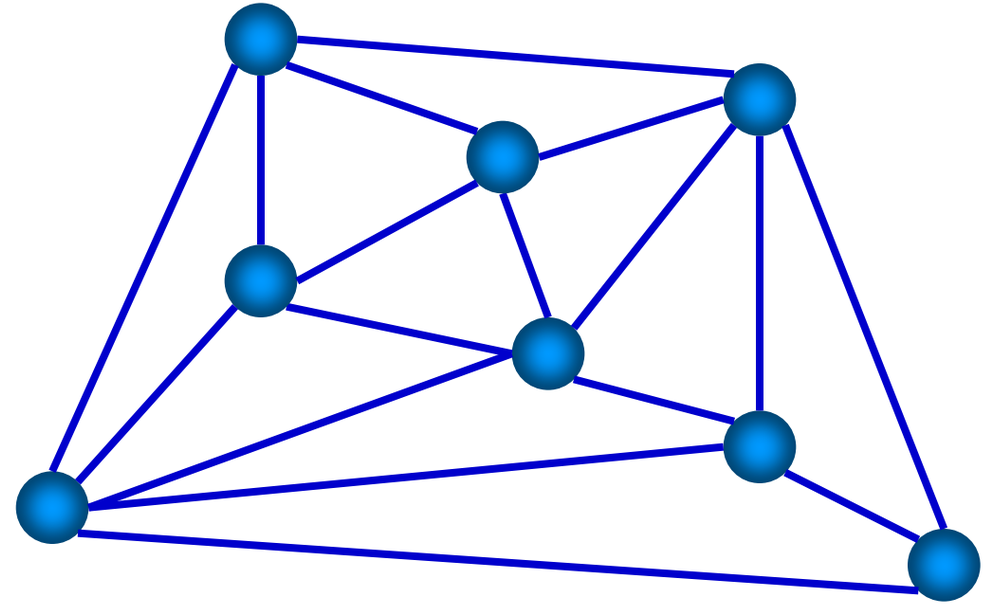
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Definition of straight-line drawing

- An *straight-line drawing* is a planar drawing of a graph where edges are straight-line segments



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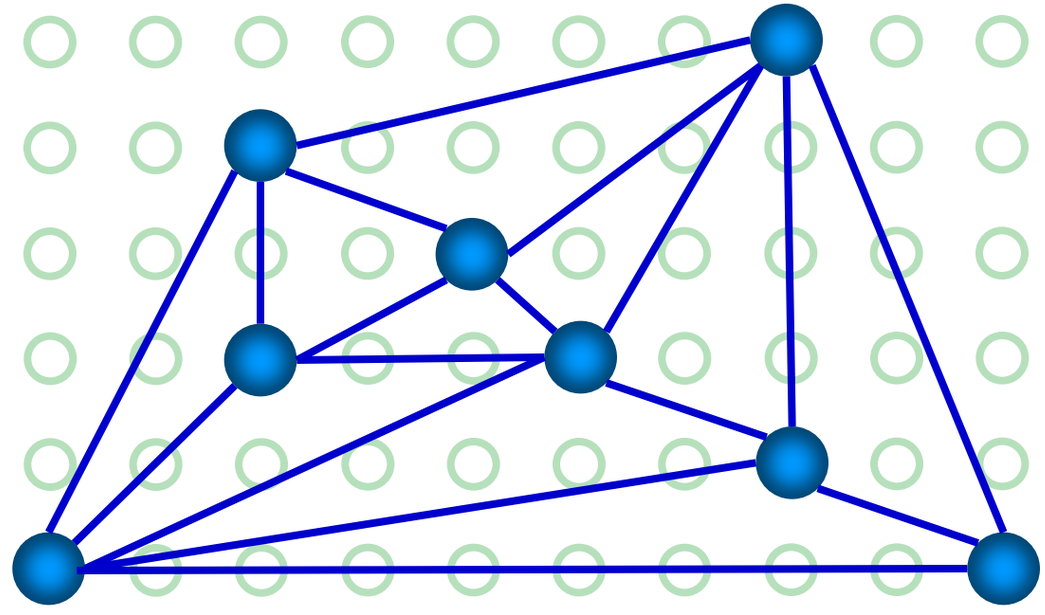
Limitations of Tutte's algorithm

- The baricenter algorithm by Tutte computes a straight-line drawing of a (triconnected, planar) graph
- Drawbacks
  - coordinates are real numbers
  - exponential ratio between the longest and the smallest edge of the graph
  - exponential many bits needed by a computer to store the coordinates of the vertices

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Polynomial size grid drawings?

- Rational coordinates
- Integer coordinates
  - straight-line grid drawings



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General problem

Input	A plane graph G=(V,E)
Output	A straight-line grid drawing of G
Target	Minimize the size of the graph

- Aesthetics that are usually considered
  - minimize the area
    - usually measured in terms of number of horizontal grid lines time number of vertical grid lines
  - minimize the maximum edge length
  - minimize the average edge length

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Assumptions

- The number of nodes is $n \geq 3$
  - the problem would be trivial otherwise
- The input graph is maximal planar
  - all faces have three edges
  - Euler's formula
    - the number of edges is $m = 3n - 6$
    - the number of faces is $f = 2n - 4$
- The input graph has a topological embedding
  - both the rotation scheme and the external face have been chosen

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Obtaining straight-line grid drawings

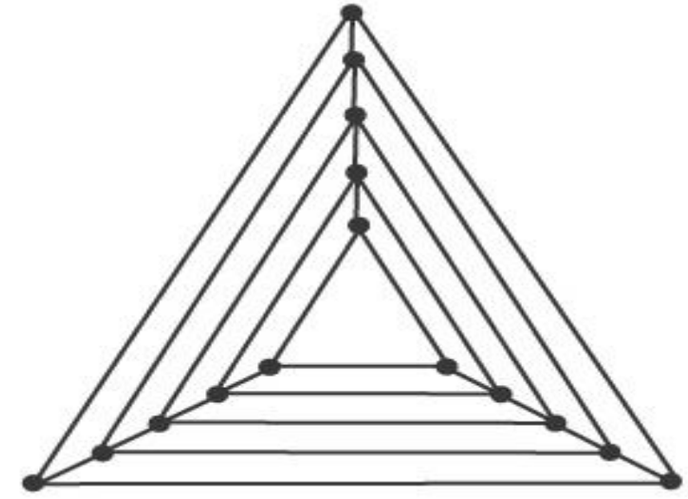
- Was a long-standing open problem
- Two different methods
  - the shift method [de Fraysseix et al. 88]
    - linear time
    - $2n-4 \times n-2$ grid
  - the realizer method [Schnyder 90]
    - linear time
    - $n-2 \times n-2$ grid
- They seem very different but are actually based on the same concepts

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Quadratic area is necessary

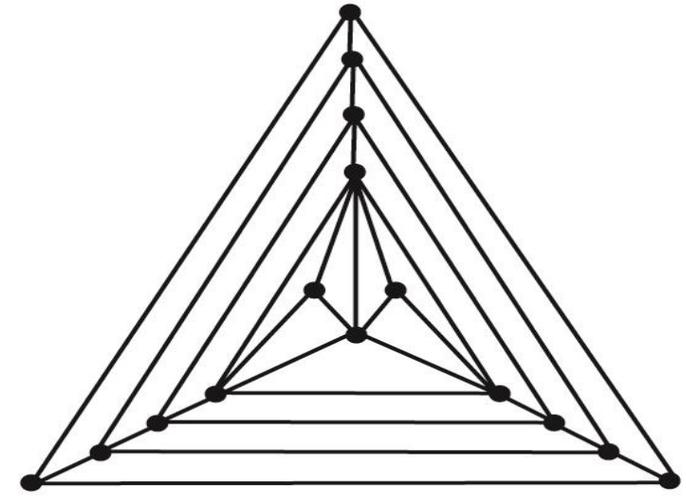
- $\left(\frac{2}{3}n-1\right)\times\left(\frac{2}{3}n-1\right)=\frac{4}{9}n^2-\frac{4}{3}n+1$

lower bound by [de Fraysseix et al. 88]



- $\left(\frac{2}{3}n-1\right)\times\left(\frac{2}{3}n\right)=\frac{4}{9}n^2-\frac{2}{3}n$

lower bound by [Frati & Patrignani 07]



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A problem still open

- Close the gap between the upper bound of [Schnyder 90]

$$(n-2)\times(n-2) = n^2 - 4n + 4$$

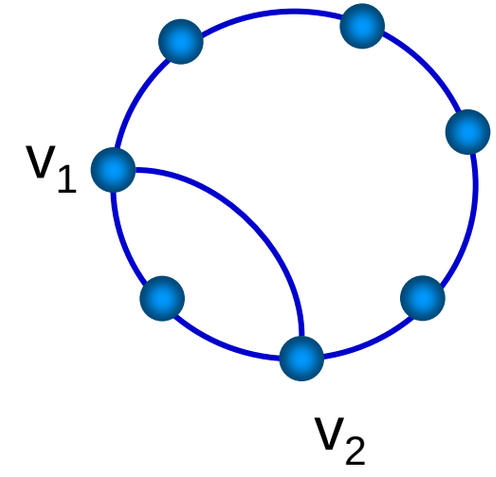
and the best known lower bound

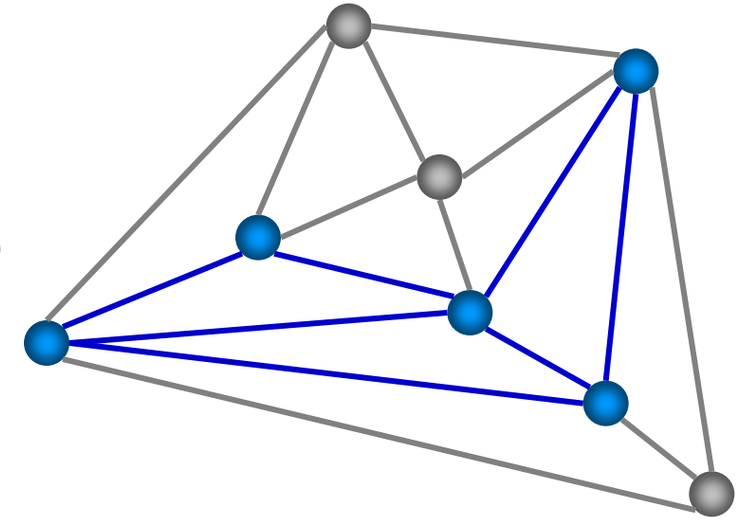
$$\frac{4}{9}n^2 - \frac{2}{3}n$$

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Basic definitions

- Chord
  - given a cycle C and two vertices v_1 and v_2 on C, if edge (v_1,v_2) is not part of the cycle then it is called a *chord*
- Induced subgraph
  - the subgraph of G=(V,E) *induced* by a set of vertices V' $\subseteq$ V is composed by the vertices in V' and all the edges between them

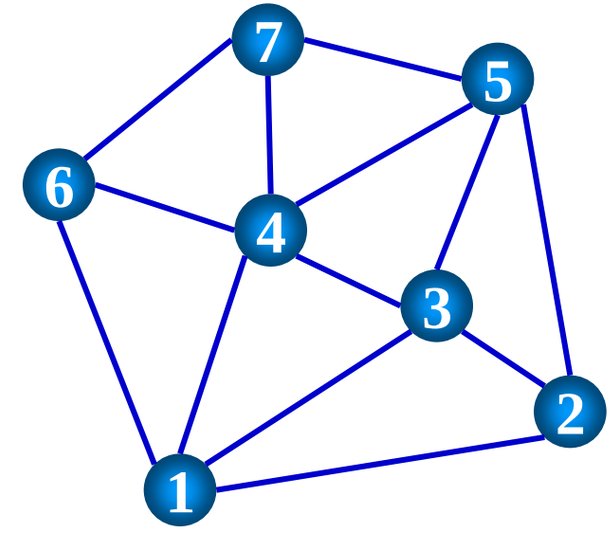


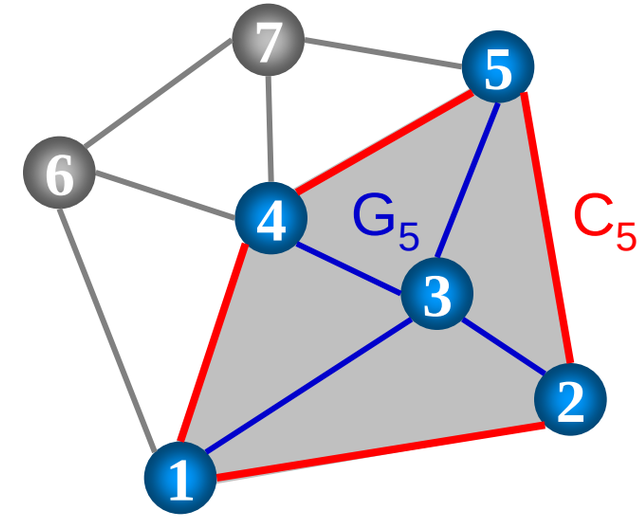


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Orderings and notation

- Given an ordering $v_1, \ldots, v_n$ for the vertices of a plane graph G=(V,E)
  - G_k denotes the graph induced by vertices $v_1, \ldots, v_k$
  - C_k denotes the outer cycle of G_k





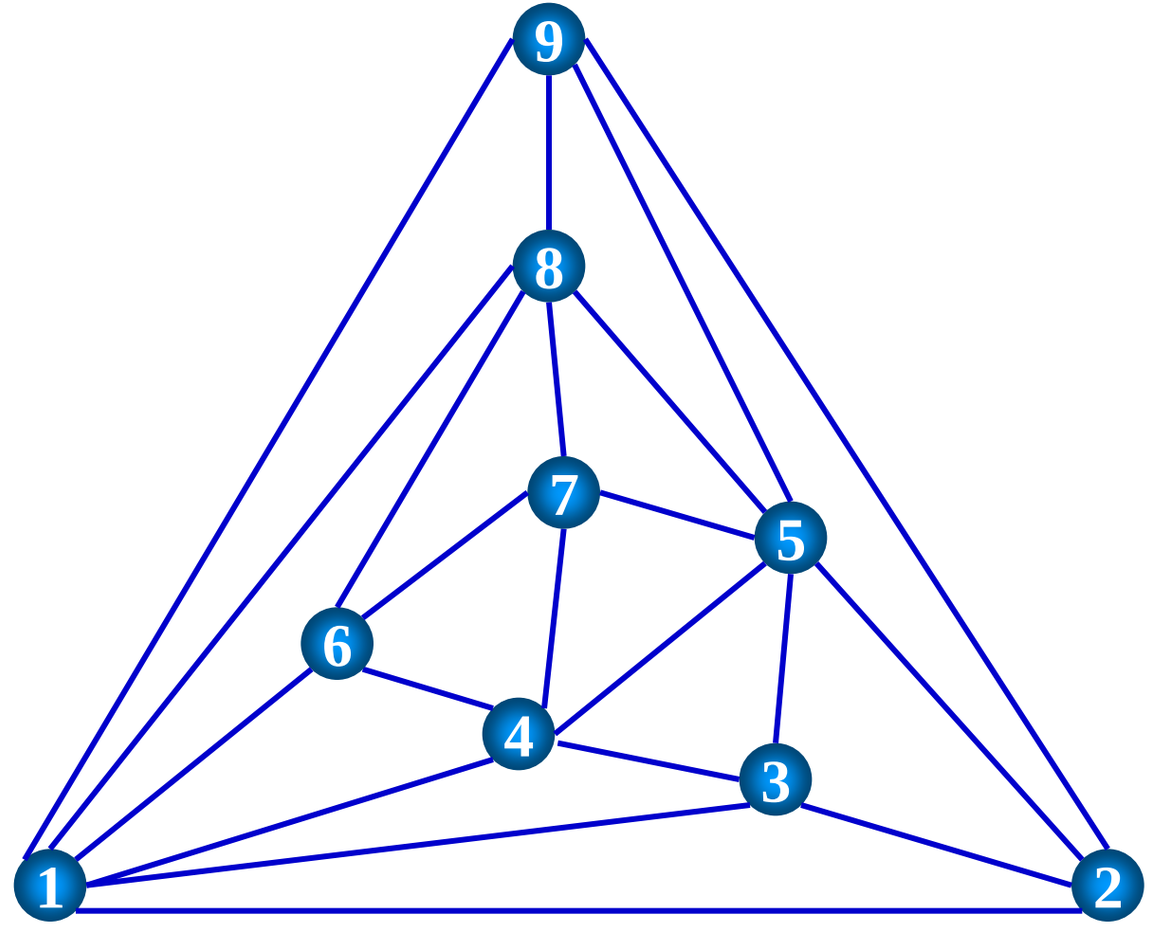
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Canonical ordering

- An ordering of vertices v_1, …, v_n is *canonical* if for each index k, $3 \le k \le n$:
  1. G_k is 2-connected and internally triangulated
  2. edge (v_1, v_2) is on C_k
  3. If $k+1 \le n$, v_{k+1} is in the external face of G_k and its neighbours in G_k appear consecutively on C_k
- A canonical ordering is also called a *shelling ordering*

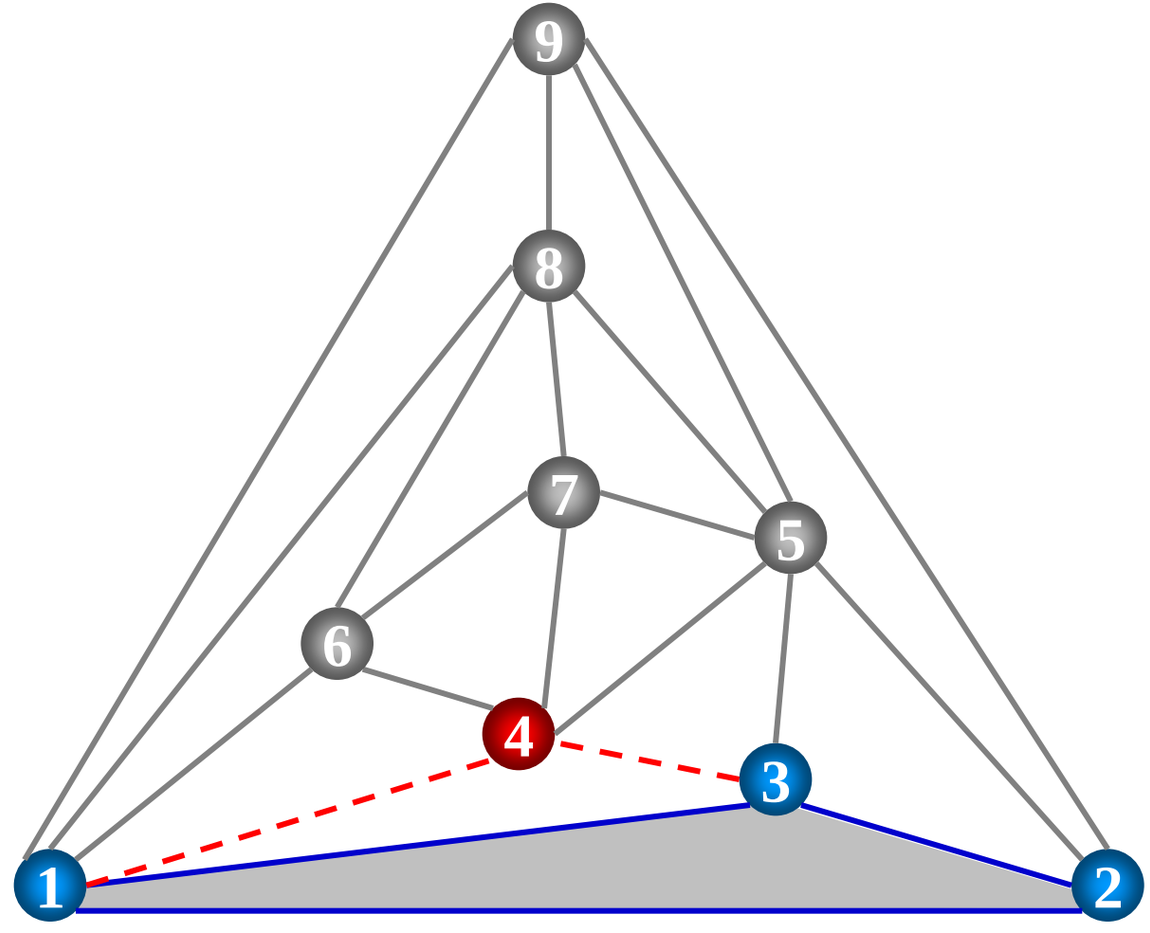
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Example of canonical ordering



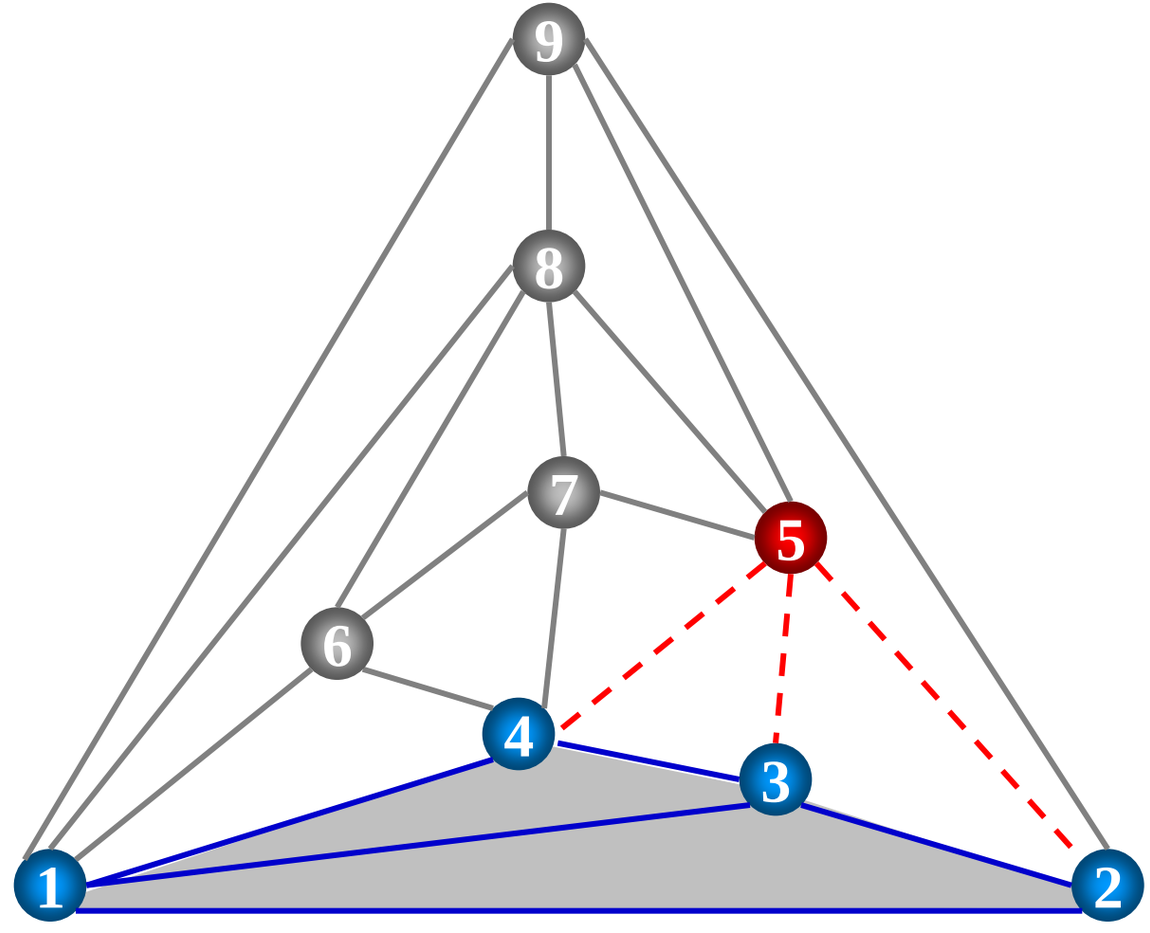
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Example of canonical ordering



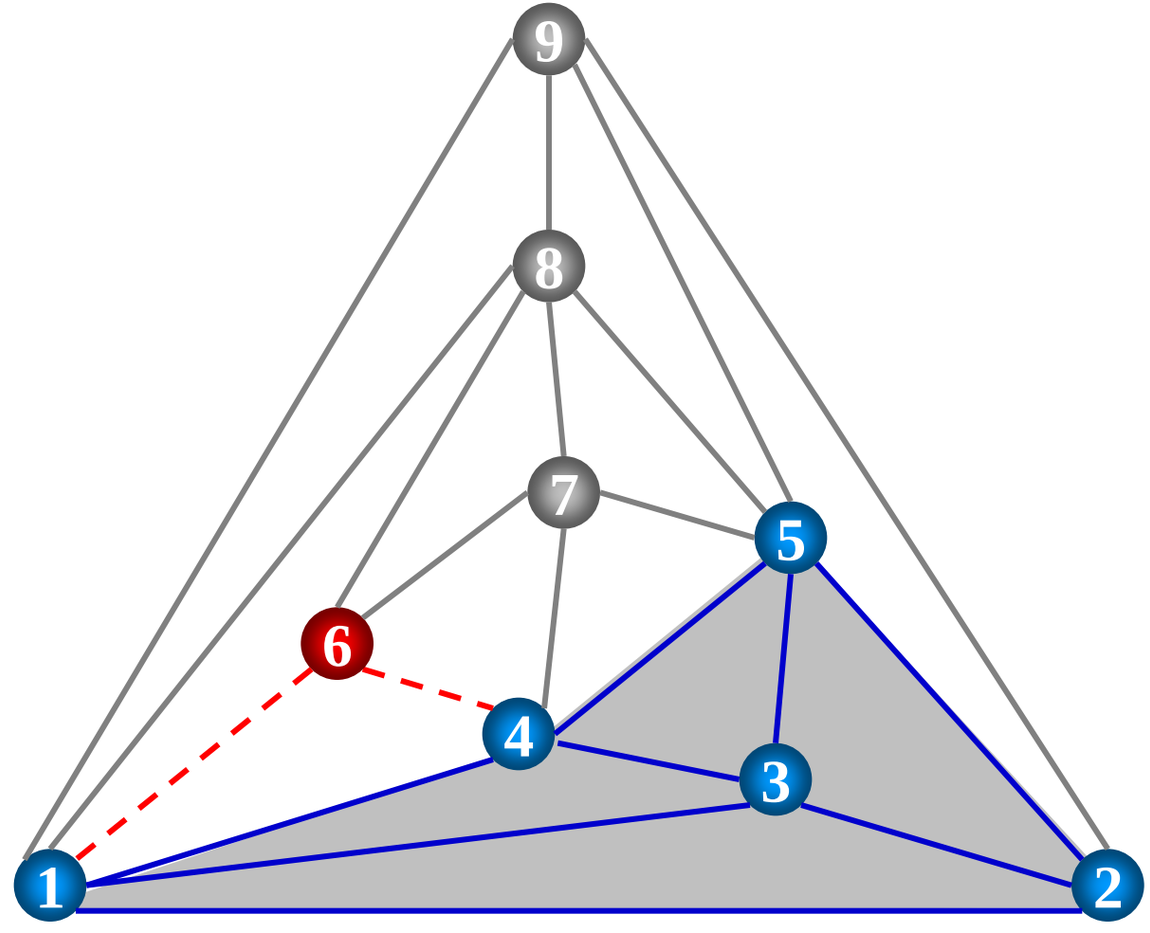
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Example of canonical ordering



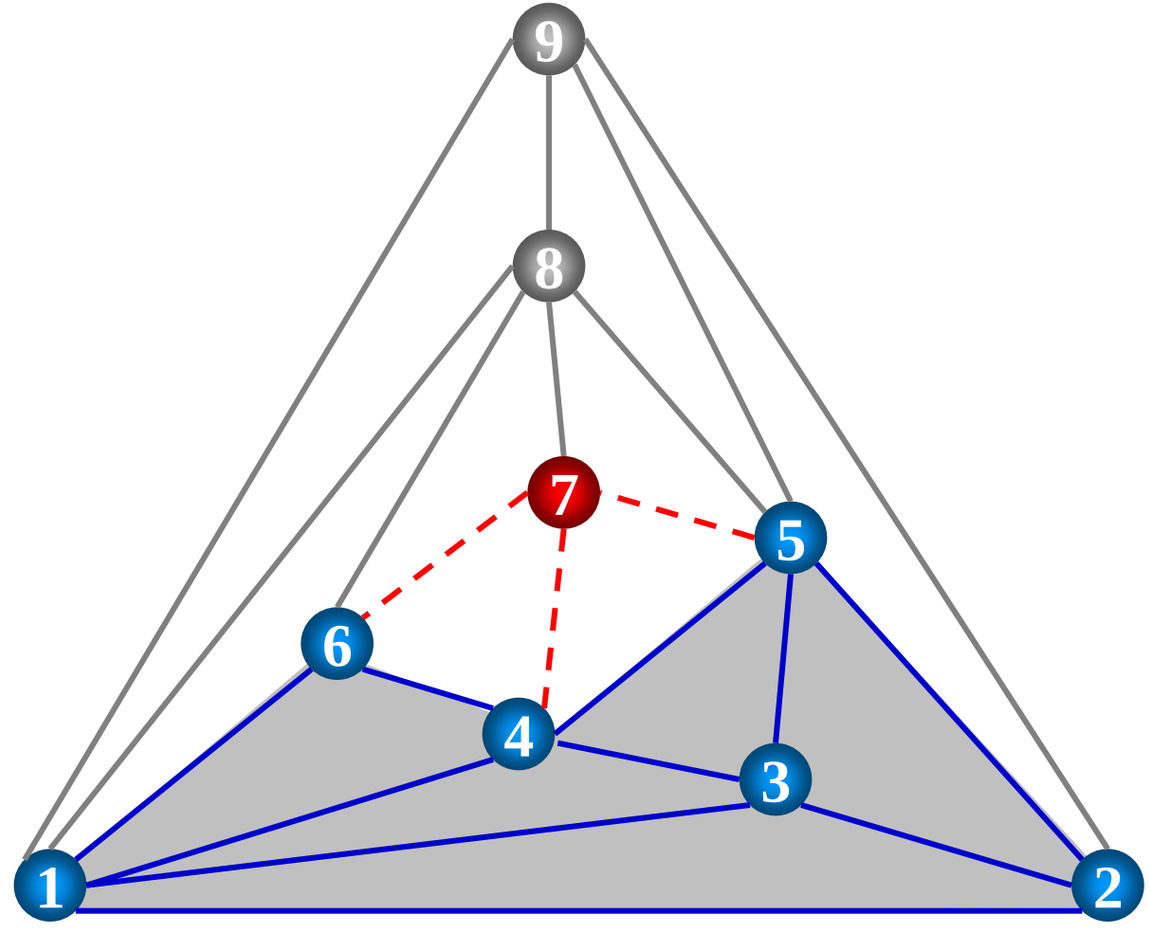
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Example of canonical ordering



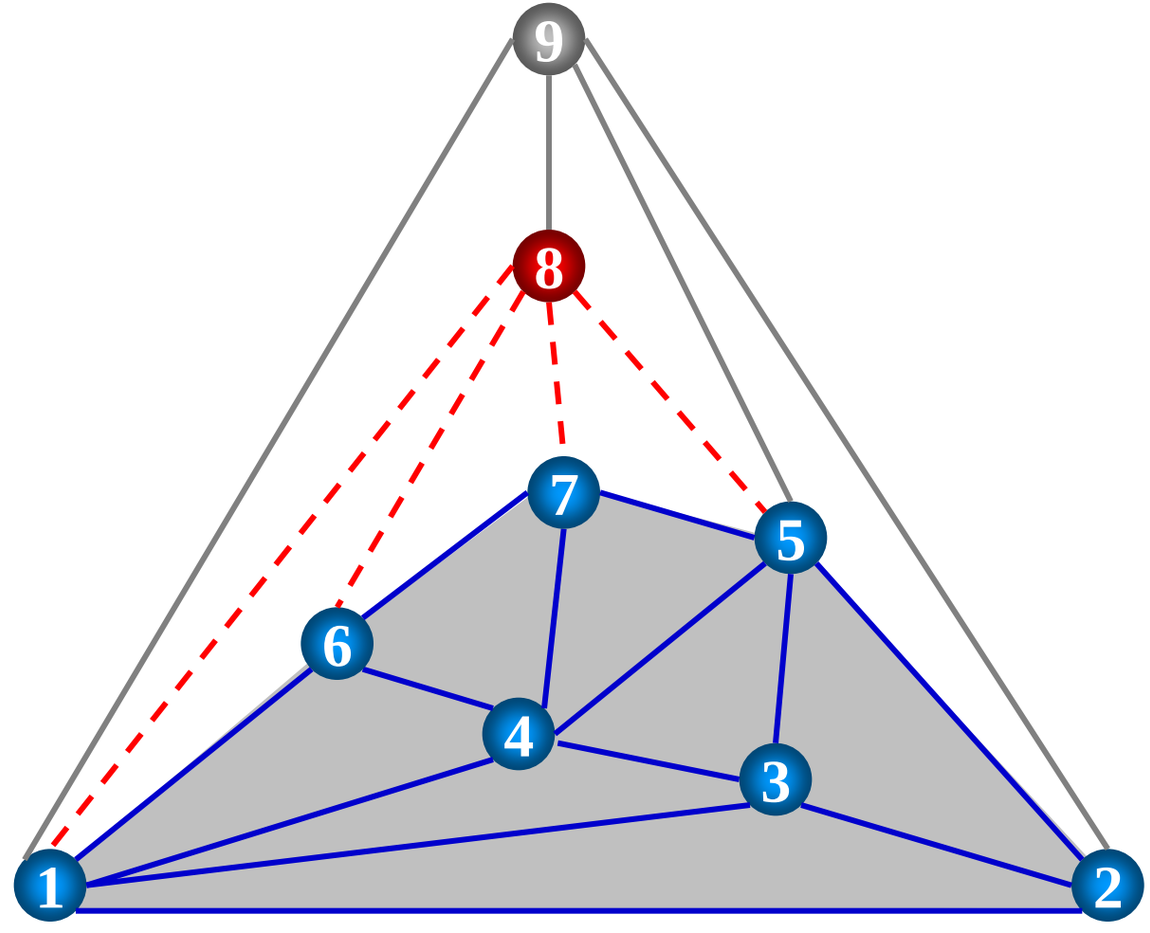
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Example of canonical ordering



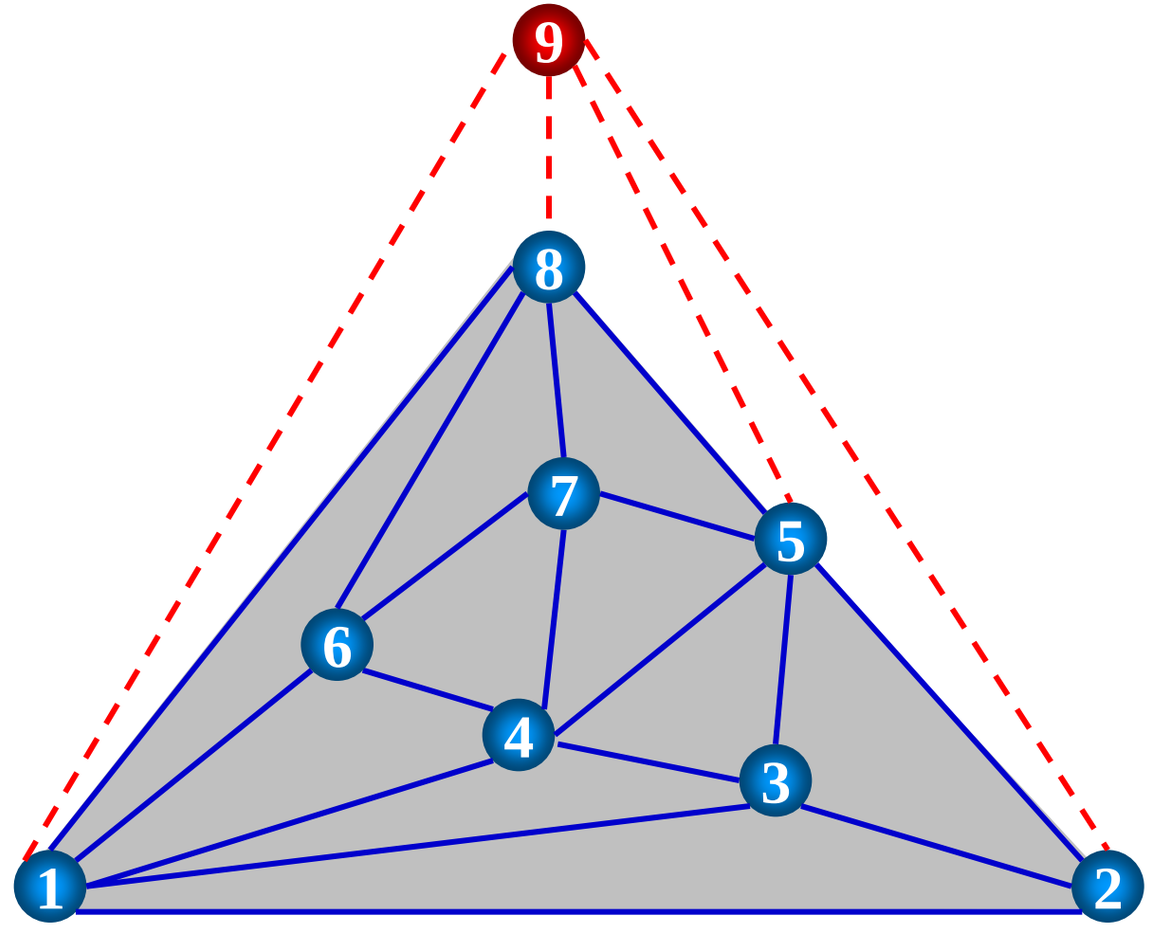
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Example of canonical ordering



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Example of canonical ordering



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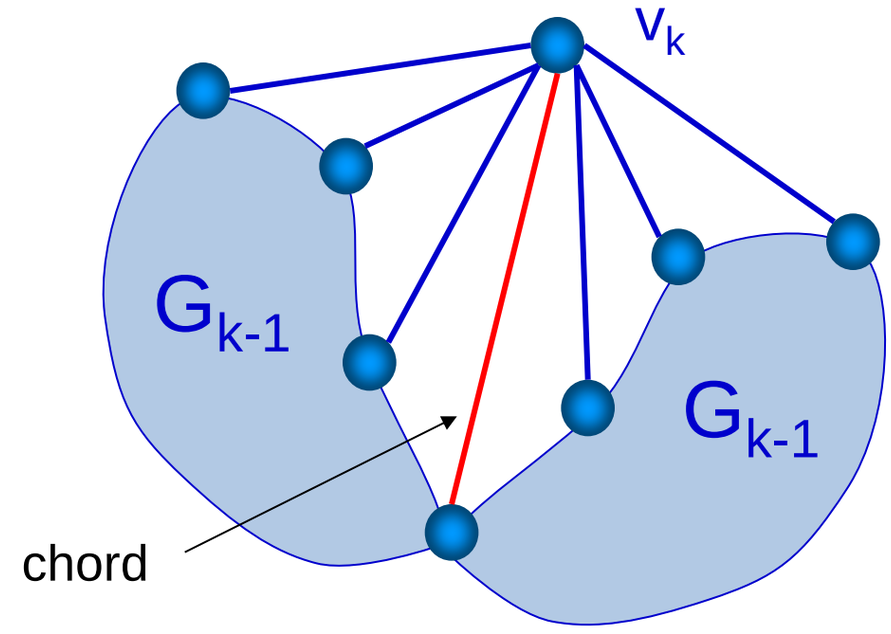
Canonical orderings always exist

- Every maximal plane graph admits a canonical ordering
  - trivial if n=3, so assume n>3
- Proof by reverse induction
  - base case (k=n)
    - conditions hold since $G_n = G$
      - also G_n is triconnected and triangulated
    - C_n consists of v_1, v_2 and v_n
    - there is no v_{n+1}
  - reverse inductive case
    - assume that v_n, v_{n-1}, ..., v_{k+1} have been appropriately chosen so that the conditions hold for n, n-1, … k
  - we have to find a suitable v_k and prove the conditions hold for k-1

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Reverse inductive case

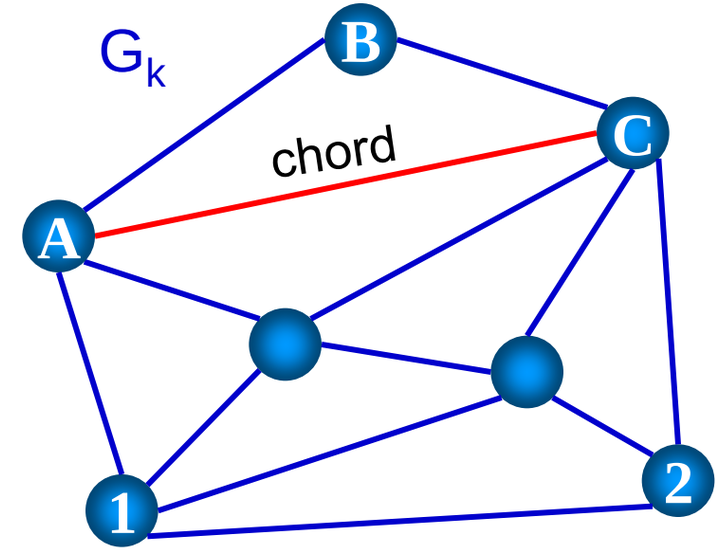
- If we can find a node v_k on C_k which is different from v_1 and v_2 and which is not an end-point of a chord, the conditions hold
  - this follows from $G_{k-1} = G_k - v_k$

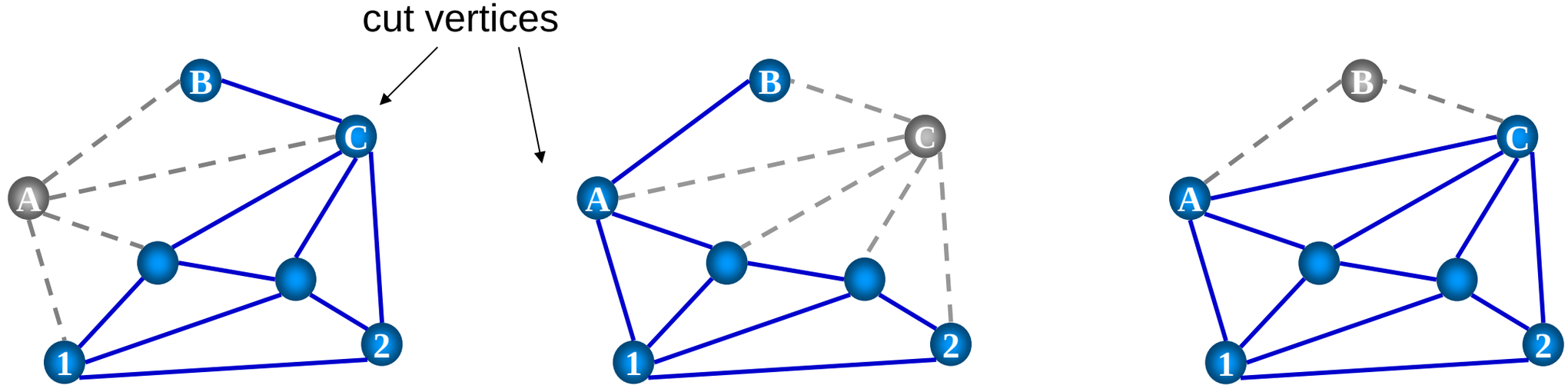


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Example

- v_k = A or v_k = C would leave G_{k-1} simply connected
- v_k = B leaves G_{k-1} biconnected

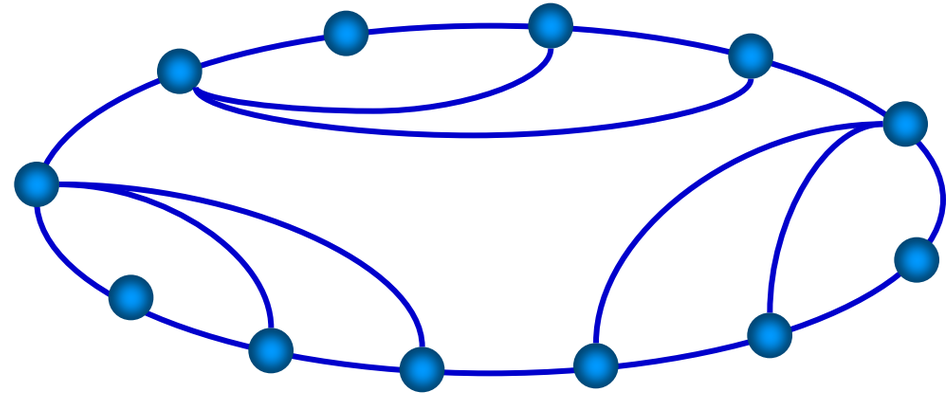




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Reverse inductive case

- There always exist a node v_k on C_k which is different from v_1 and v_2 and is not an end-point of a chord
  - a chord always skips at least one node on the cycle
  - take any chord
    - C_k is divided into two paths
    - consider the two most nested nodes of the two paths
    - the most nested nodes can not be both v_1 and v_2



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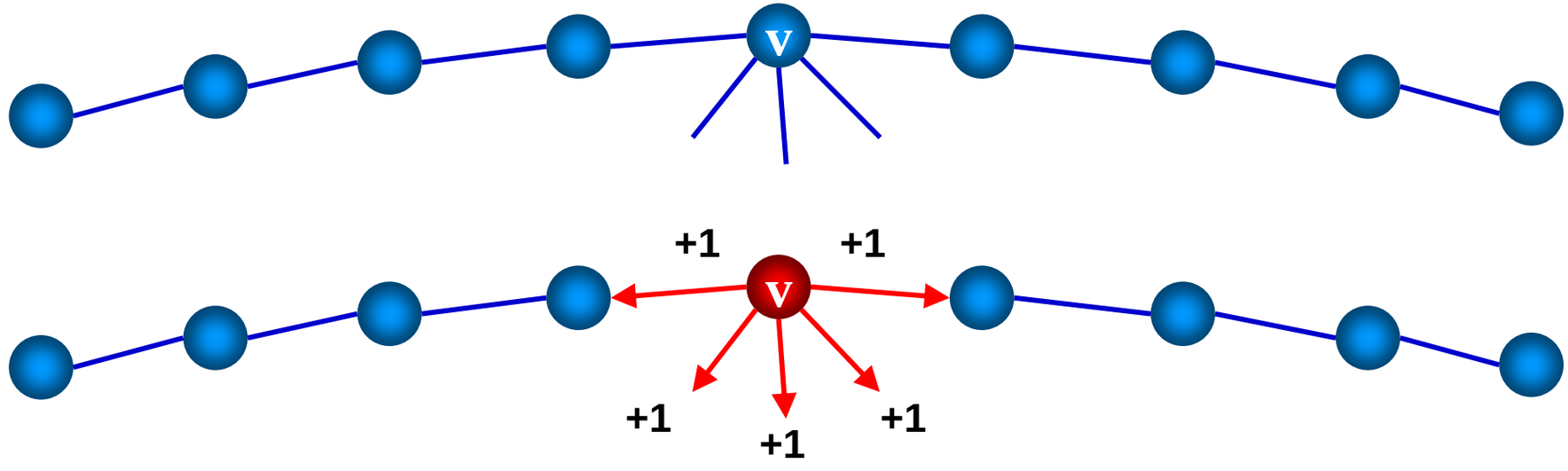
Finding canonical orderings efficiently

- A canonical ordering of a maximal plane graph can be found in linear time with the method by [Kant 1996]
  - maintain a counter for each node *v* of the graph that records the number of nodes on the external face to which *v* is adjacent to
  - maintain a list of the nodes on the external face in clockwise order
  - for k=n-1 to 4 choose a node v_k different from v_1 and v_2 that has exactly two neighbors on the external face
    - since v_k has only two adjacencies on the external face it can not be and end-point of a chord
  - when a node of the external face is chosen, update the counters of all its adjacent nodes

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Initialization

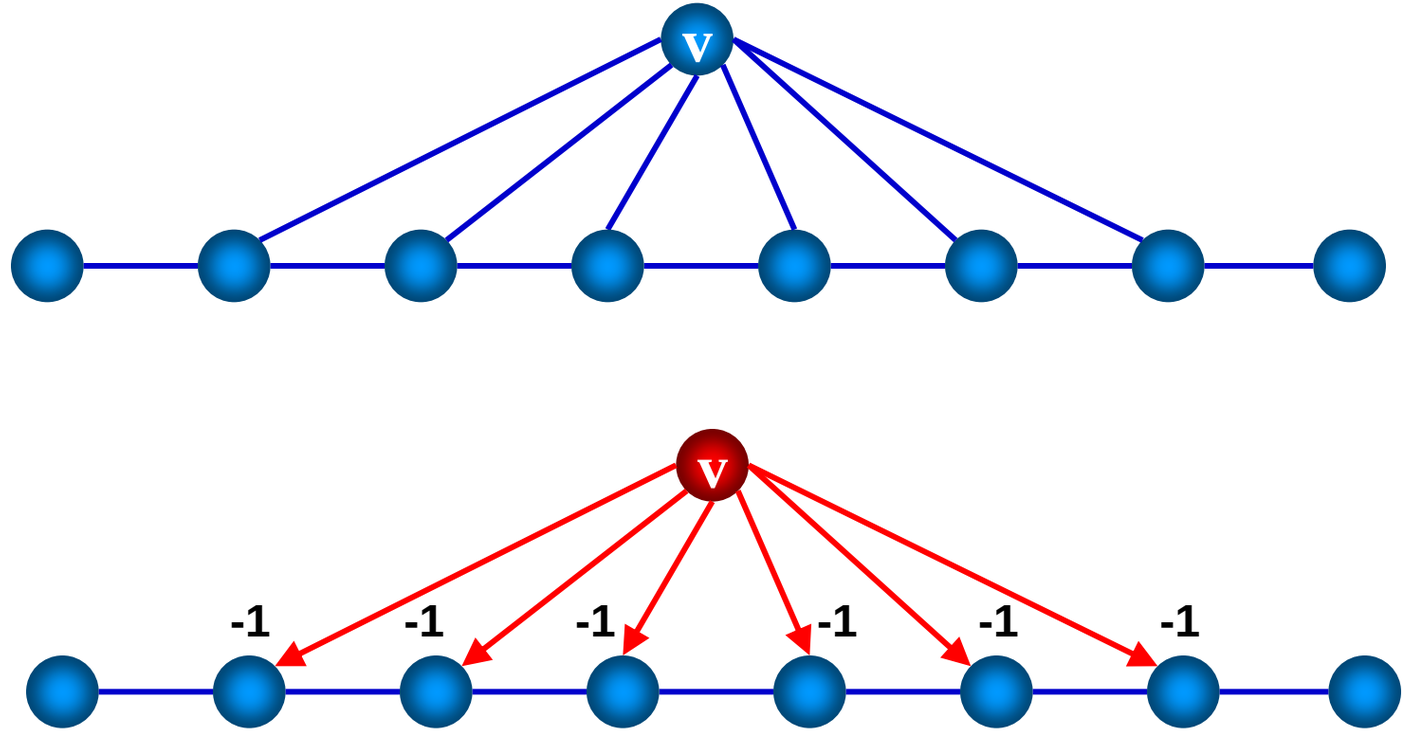
- Initially, set the couter of the number of adjacencies on the external face to zero for each node
- Traverse the external face and increment the counters of the nodes adjacent to the current one
- Now, all counters are correct and the process can start



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Updating rule for the counters

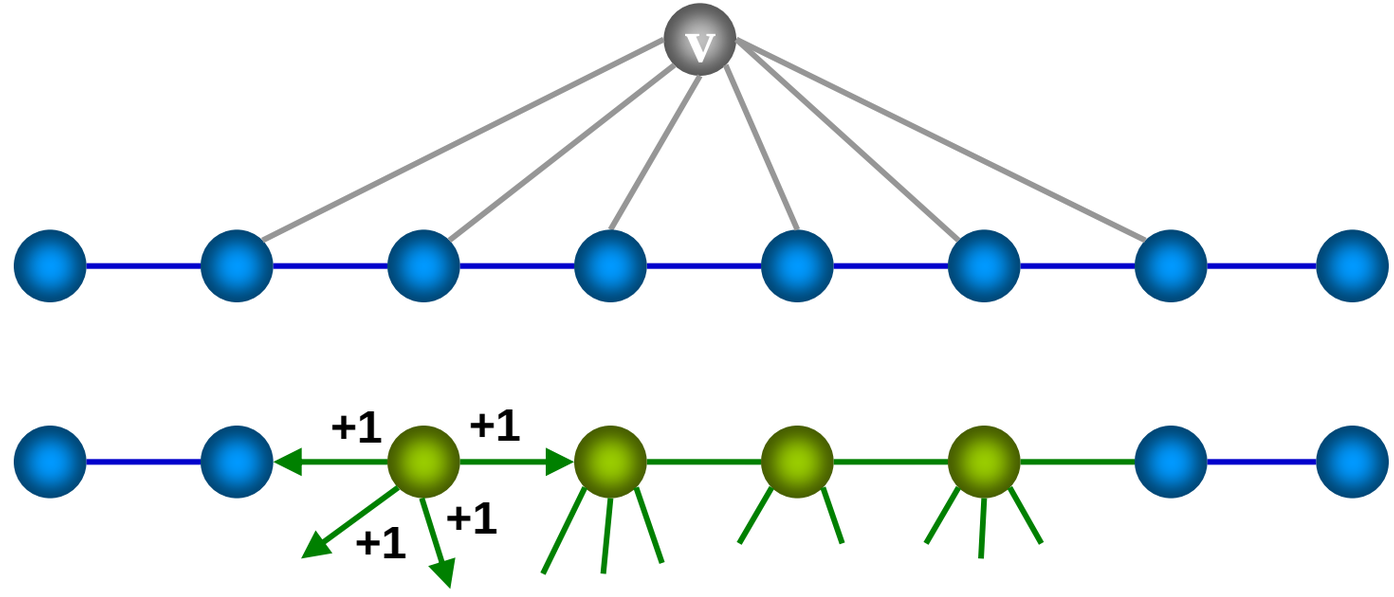
- When node v is removed from the external face all the counters of adjacent nodes are decreased



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Updating rule for the counters

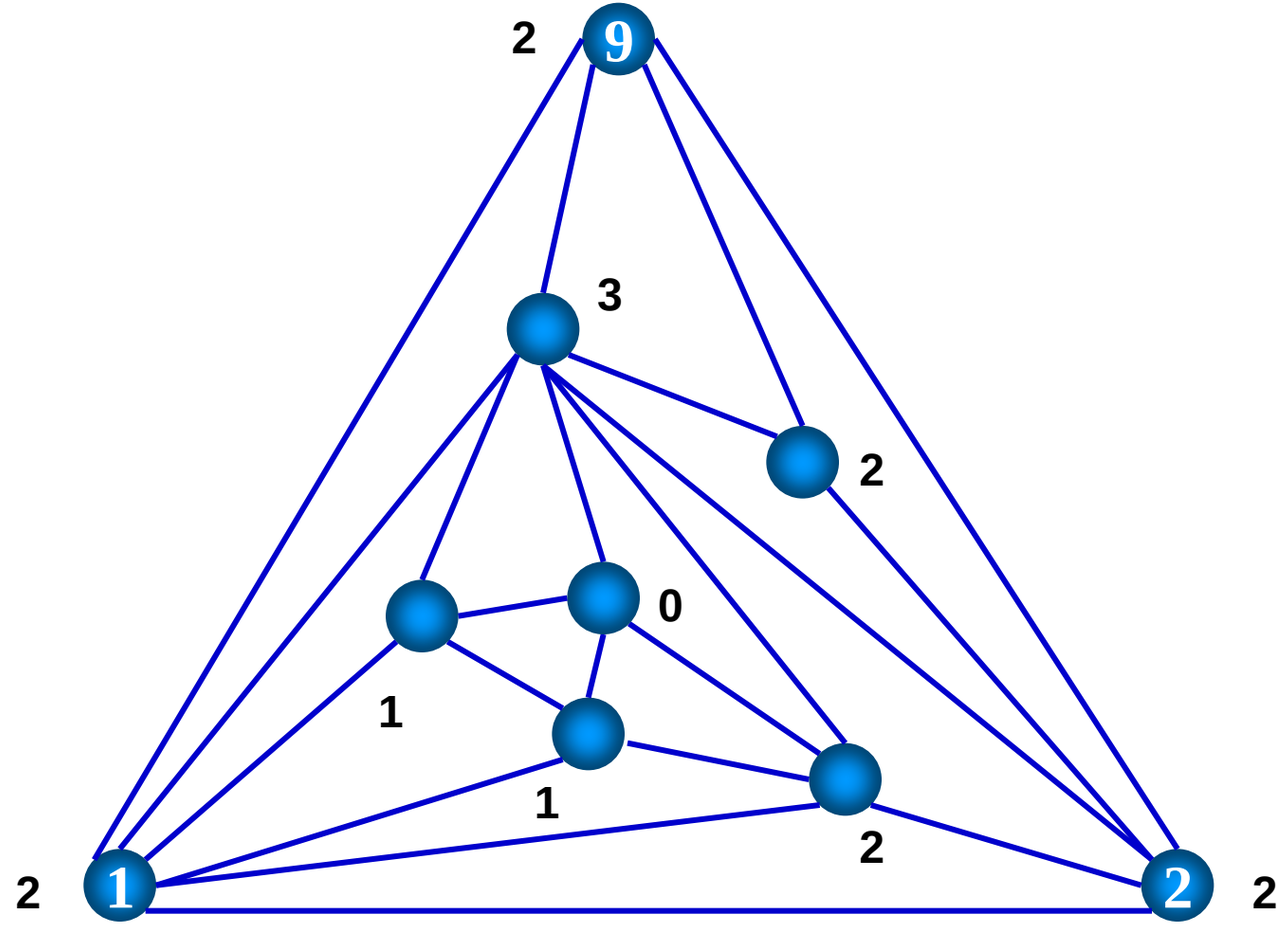
- Nodes that become external after the removal of v increase the counters of their adiacent nodes



- All updates can be performed in linear time

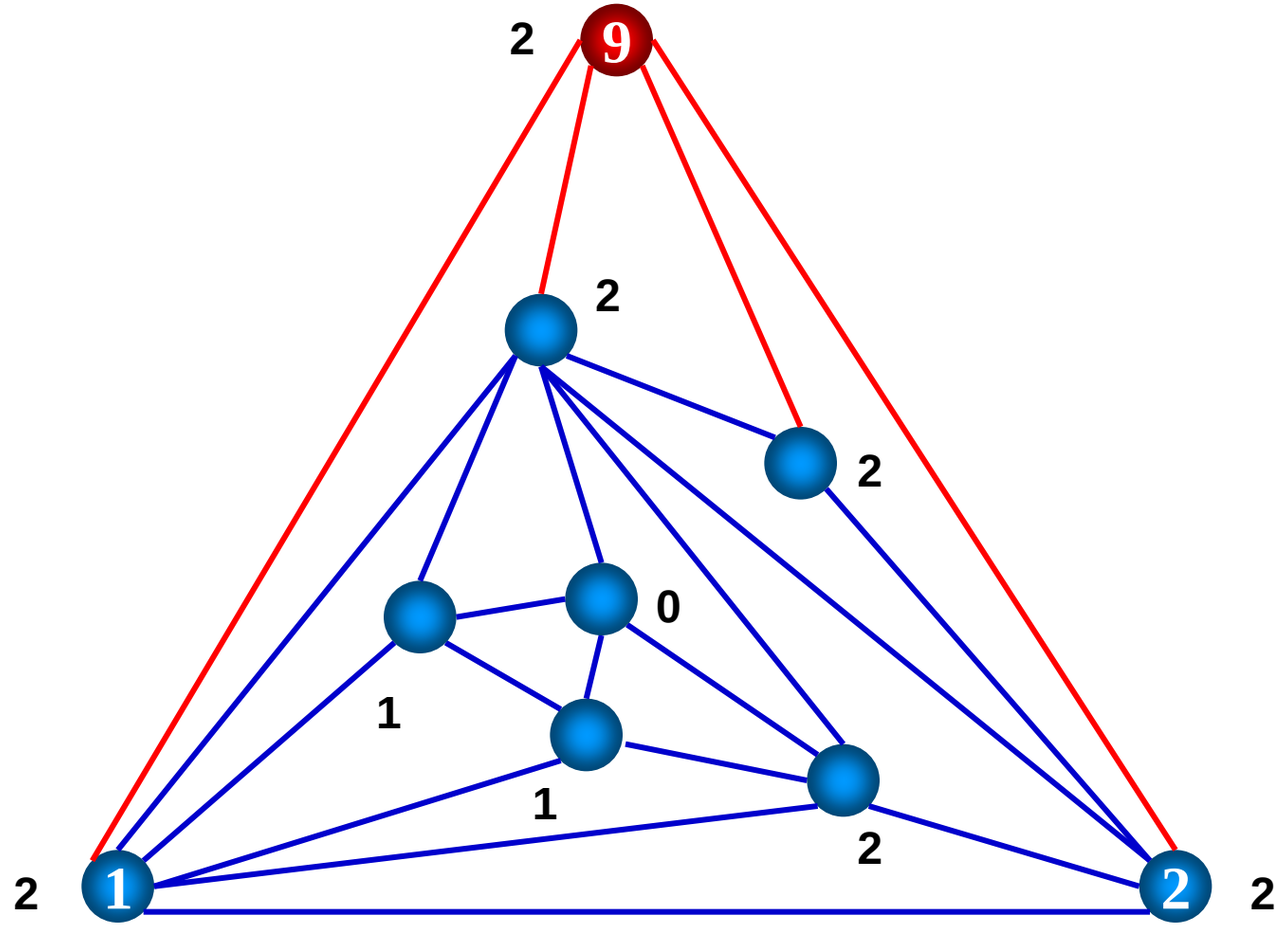
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Computing a canonical ordering



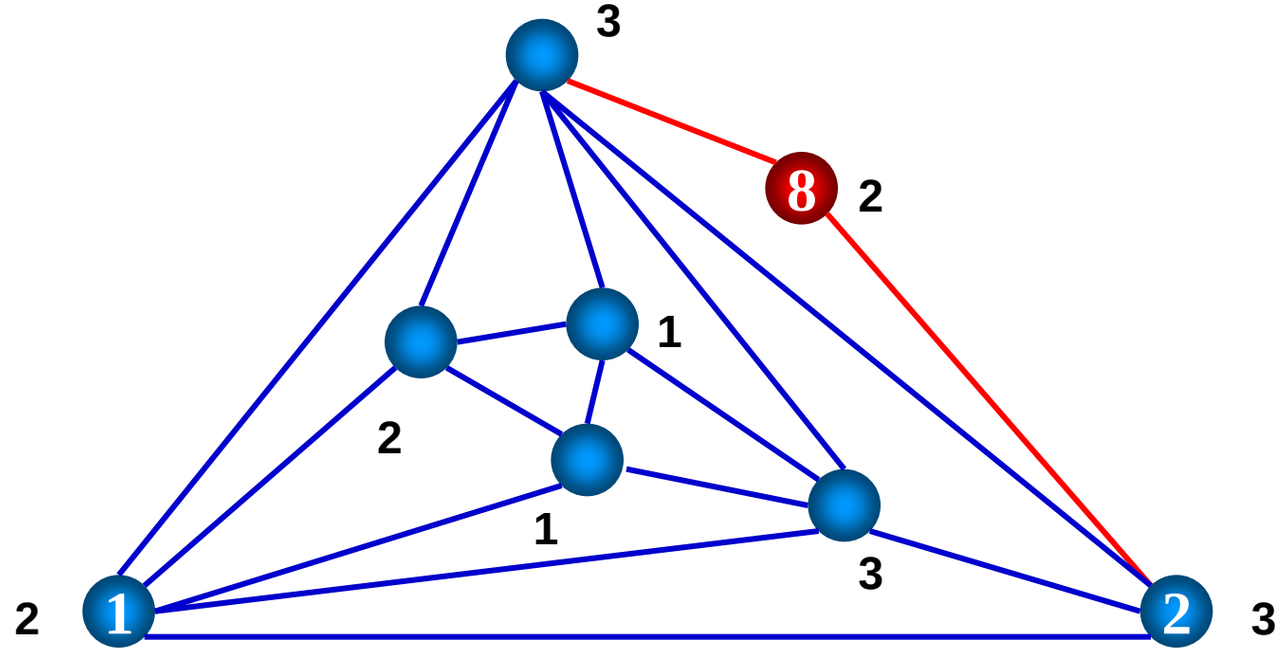
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Computing a canonical ordering



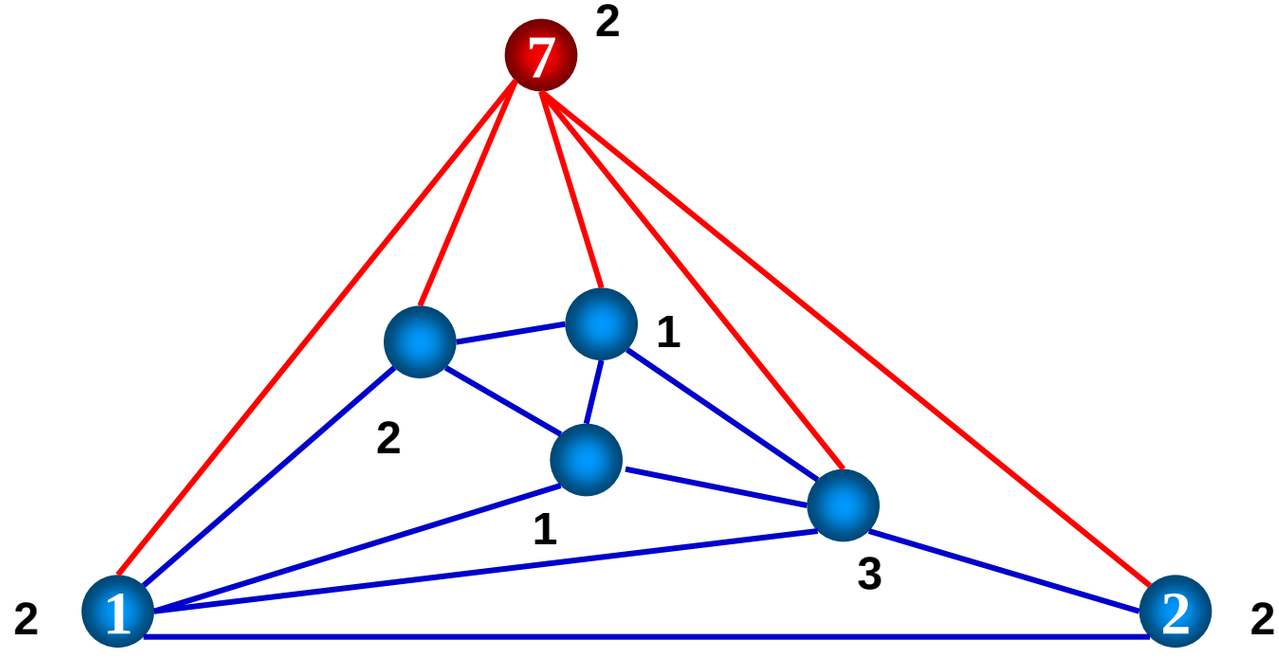
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Computing a canonical ordering



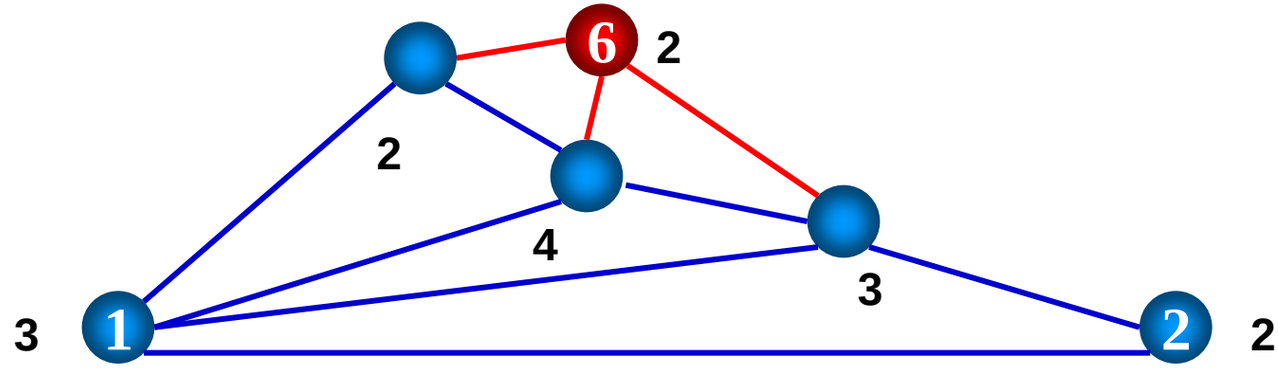
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Computing a canonical ordering



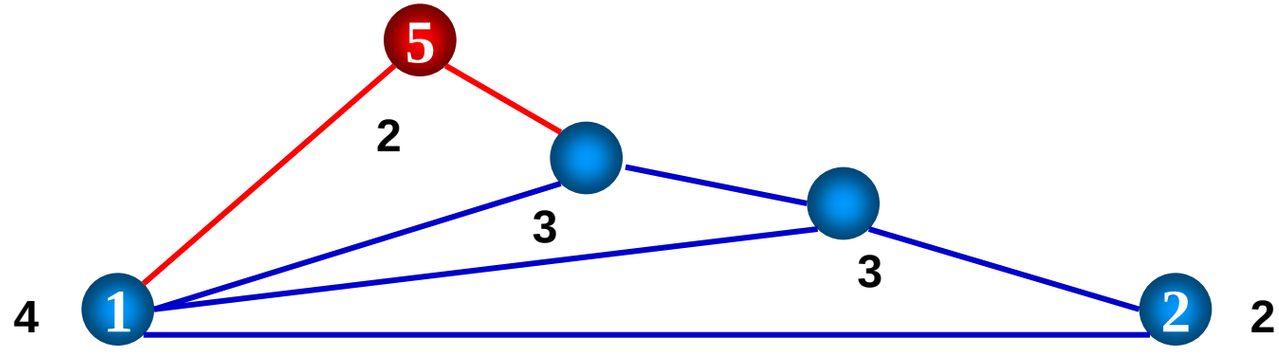
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Computing a canonical ordering



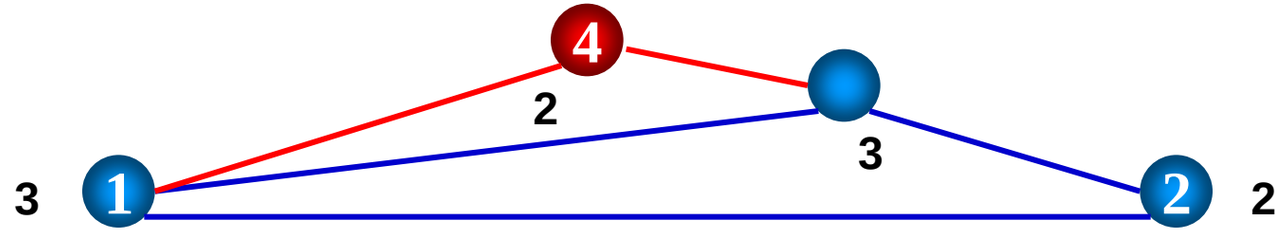
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Computing a canonical ordering



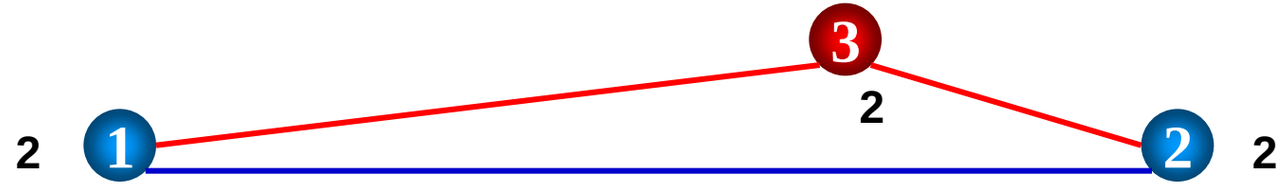
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Computing a canonical ordering



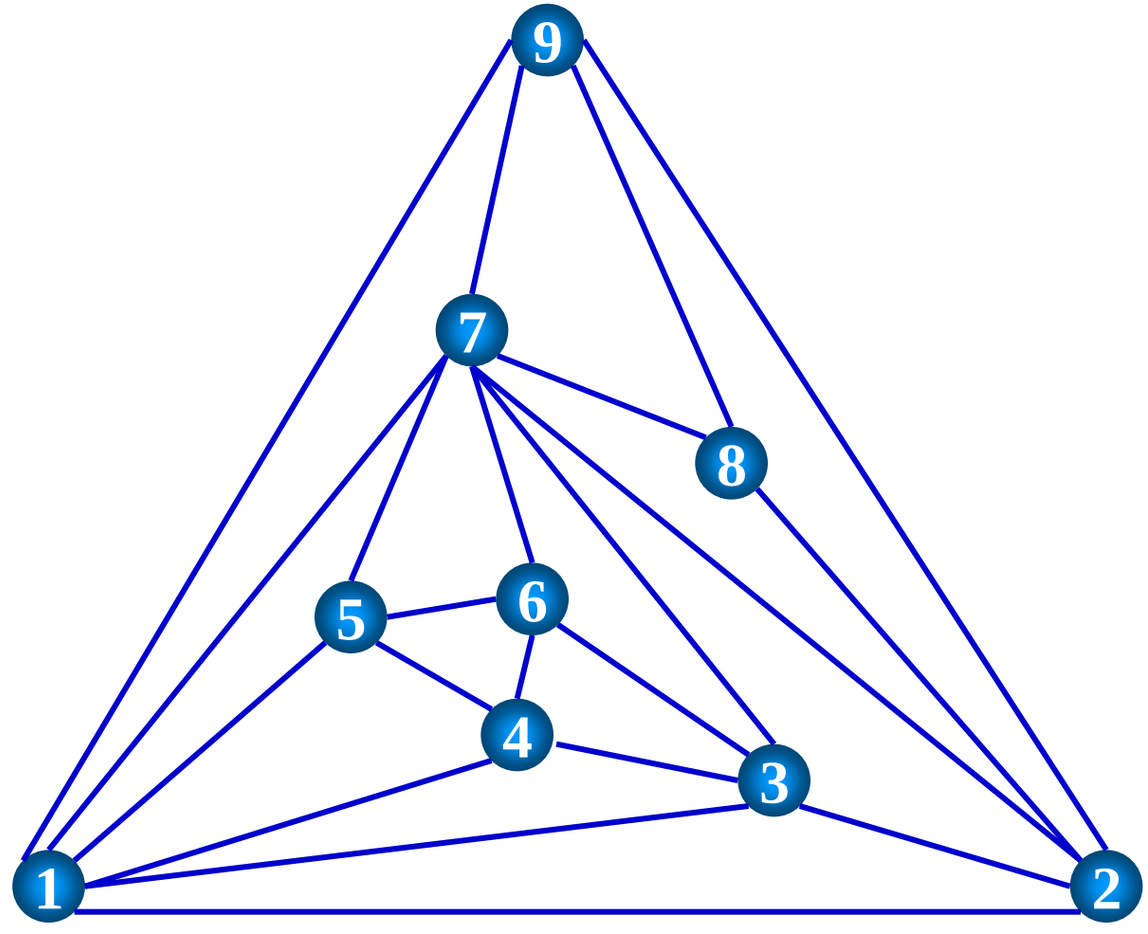
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Computing a canonical ordering



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The computed canonical ordering



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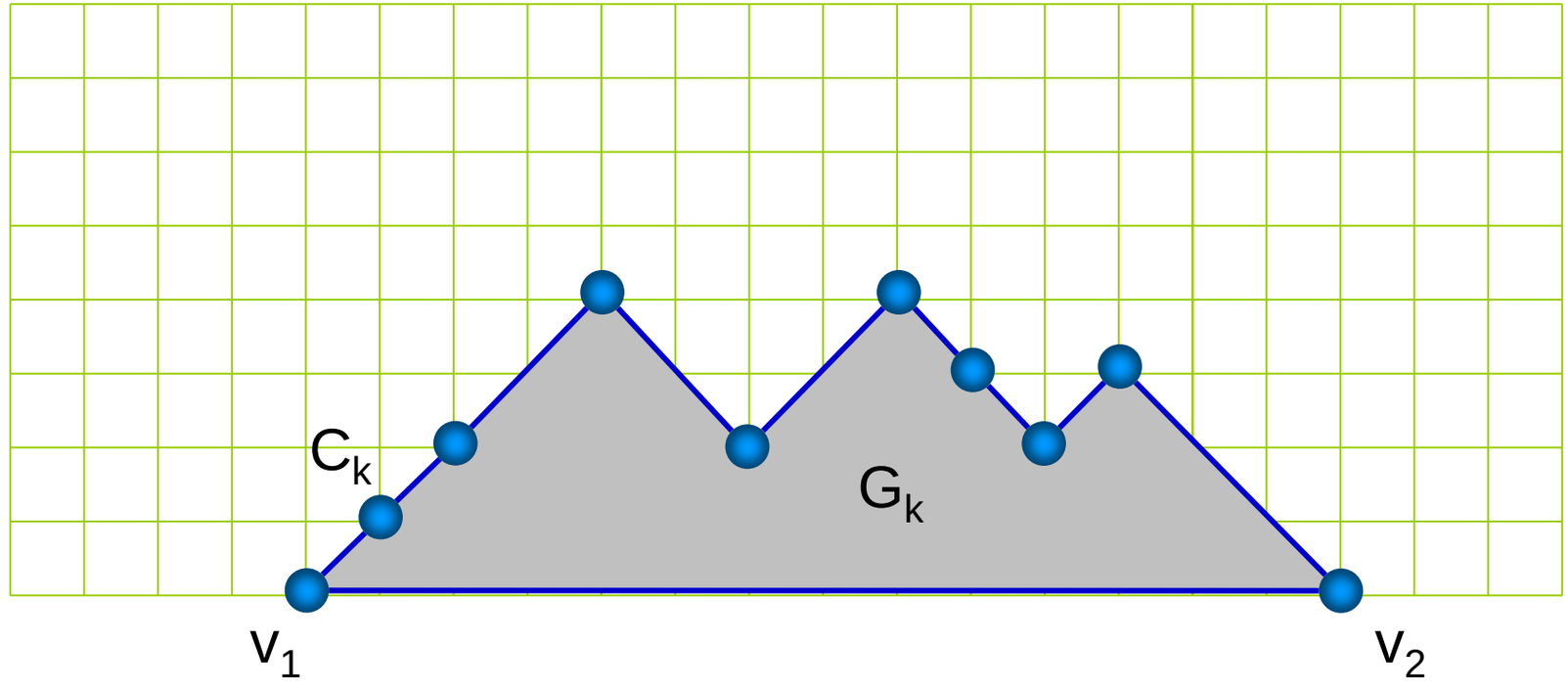
The shift method

de Fraysseix, Pach, and Pollack 1988

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Drawing invariant

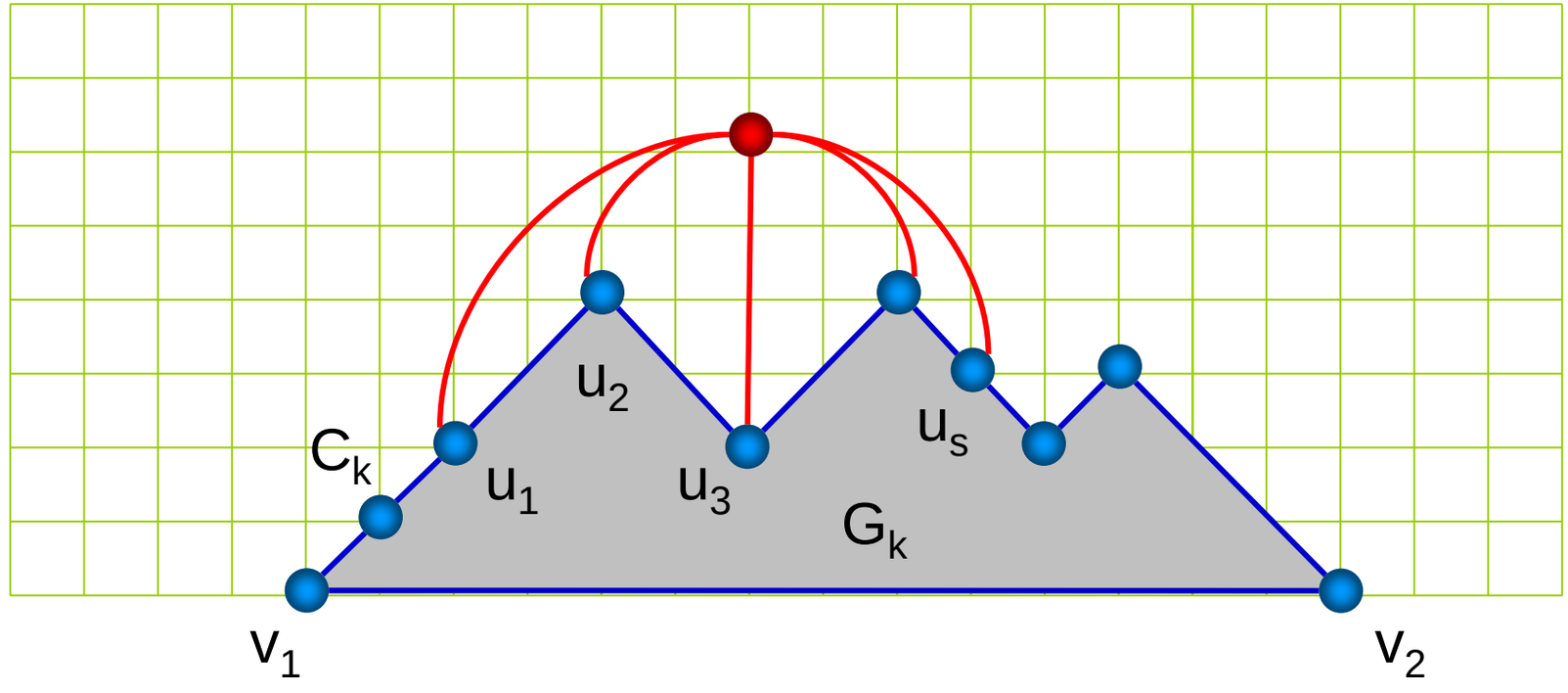
- Each G_k is drawn in such a way that all edges of C_k have slope +1 or -1 with the exception of edge (v_1,v_2)



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Drawing strategy

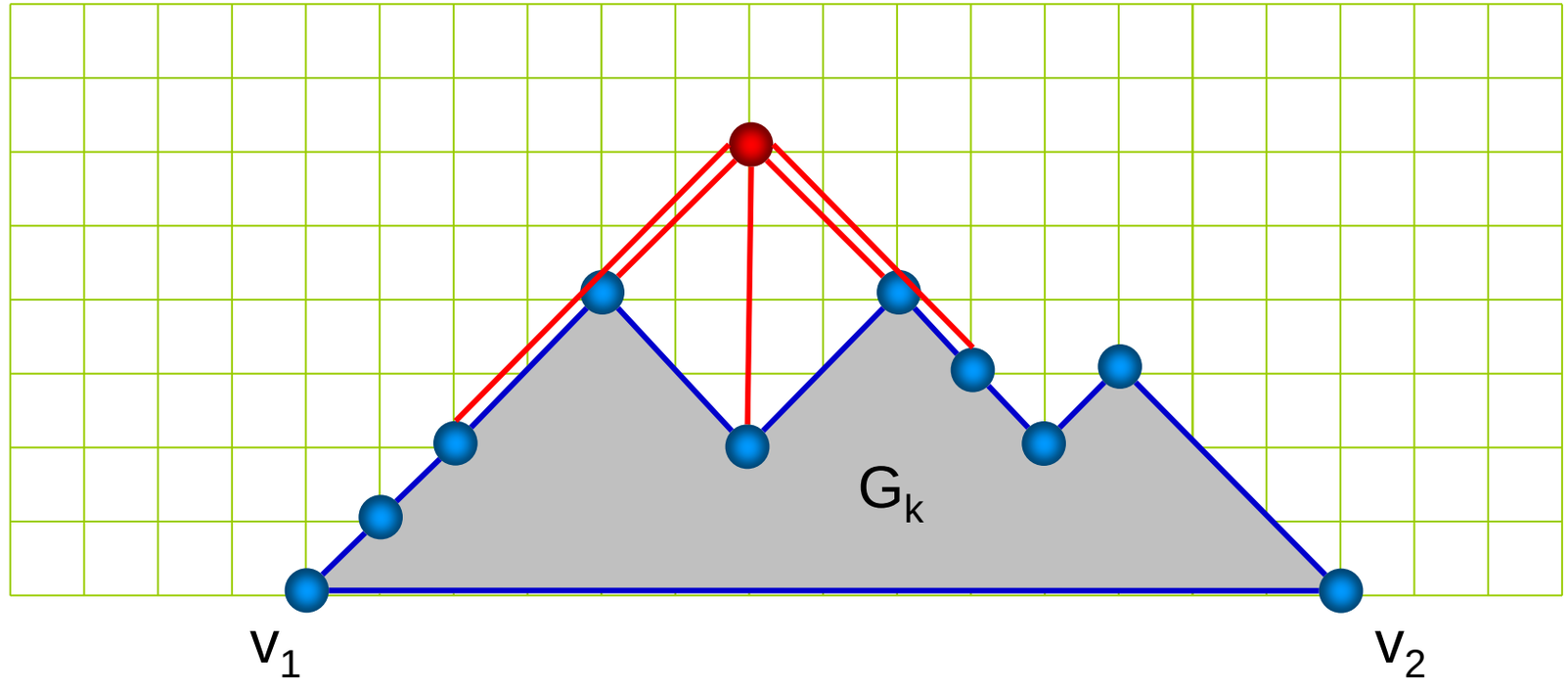
- By the properties of canonical ordering node v_{k+1} attaches to consecutive nodes u_1, u_2, …, u_s along C_k



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Drawing strategy

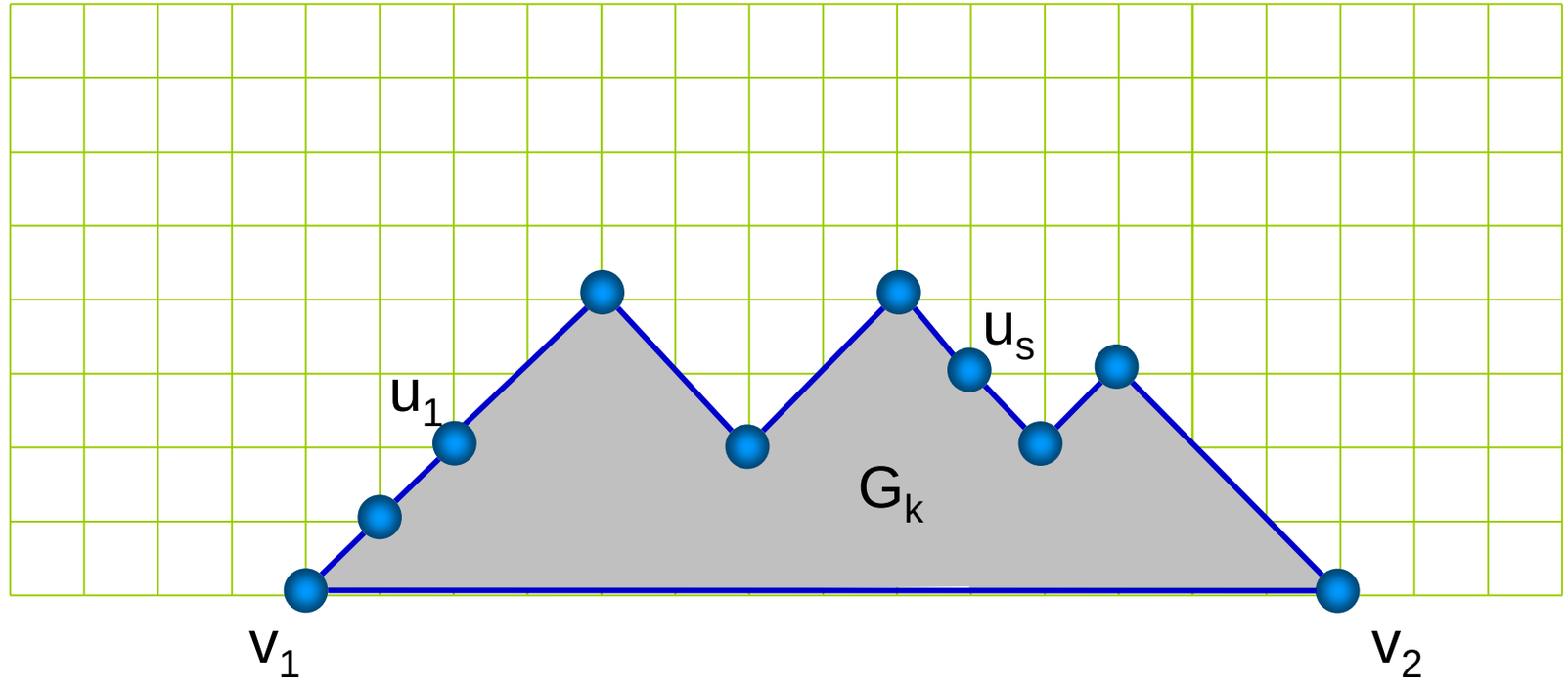
- You could easily find a place for v_{k+1} in such a way that the edges of C_{k+1} have slope +1 or -1
  - however you could create some intersections and empty faces



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Drawing strategy

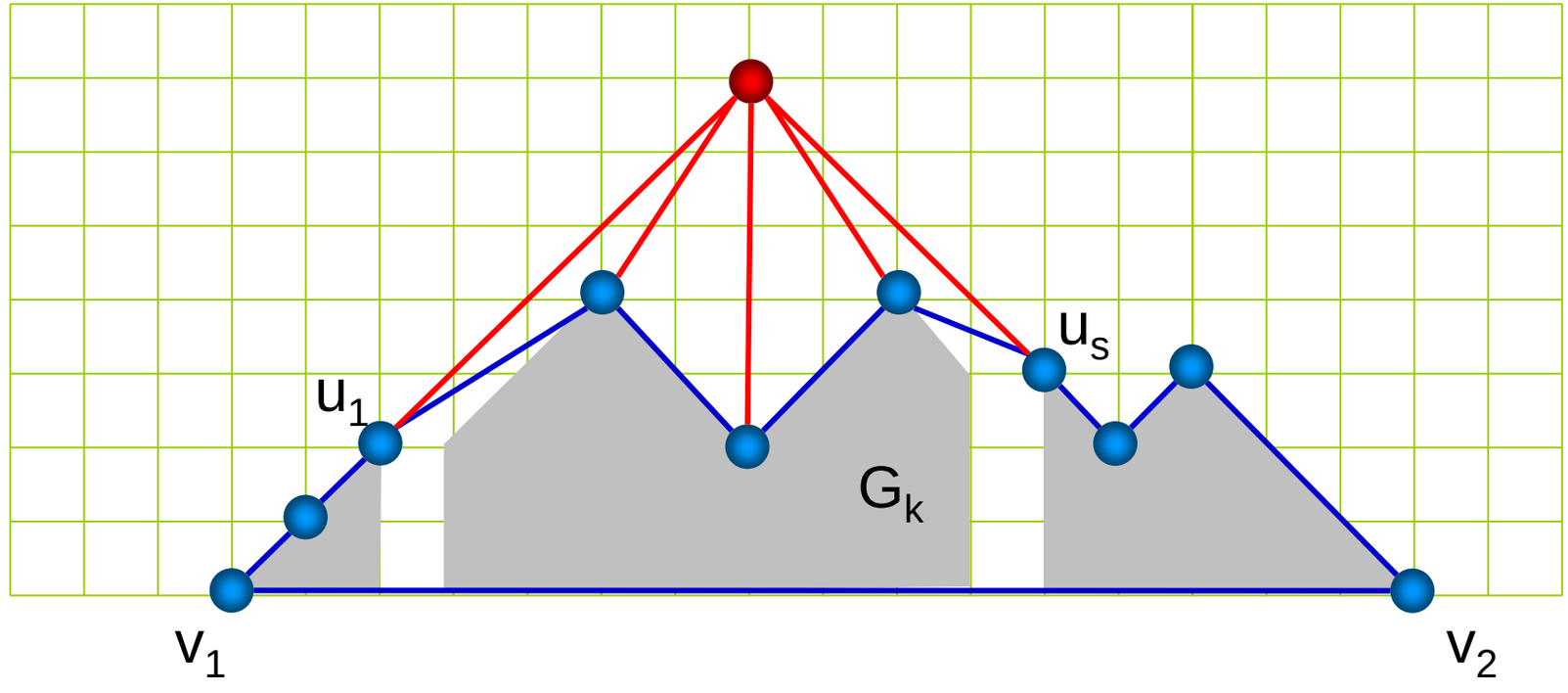
- Before placing v_{k+1}
  - shift left all nodes with x-coordinate $\leq u_1$
  - shift right all nodes with x-coordinate $\geq u_s$



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Drawing strategy

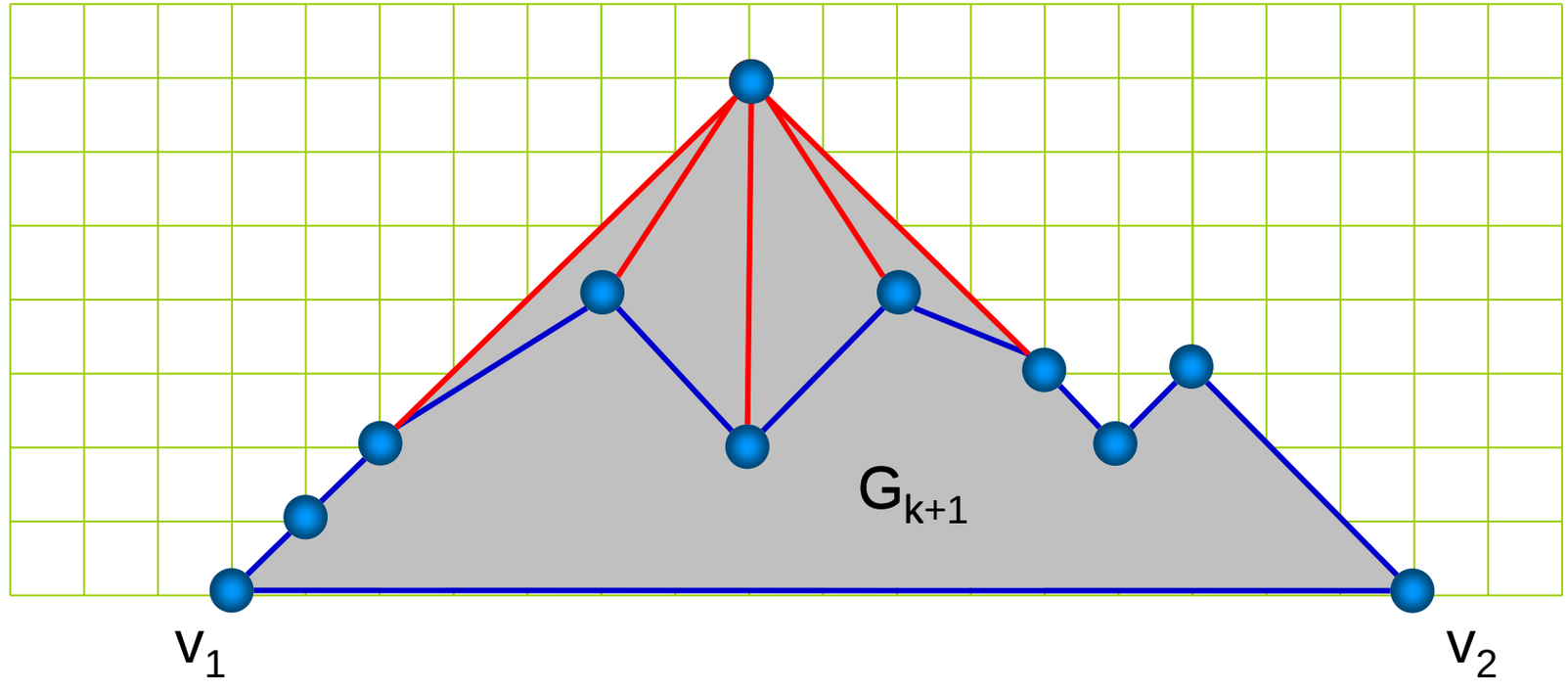
- Before placing v_{k+1}
  - shift left all nodes with x-coordinate $\leq u_1$
  - shift right all nodes with x-coordinate $\geq u_s$



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Drawing invariant is preserved

- All edges of C_{k+1} (with the exception of edge (v_1,v_2)) have slope +1 or -1



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Efficient implementation

- To obtain a linear-time algorithm, x-offsets are used in stead of x-coordinates
- A binary tree is used to represent shift dependencies
  - each node shifts whenever its parent shifts
  - each node stores its x-offset and y-coordinate
- Using the tree, the x-offsets can be accumulated in linear time
  - for an efficient algorithm see [Chrobak & Payne 95]

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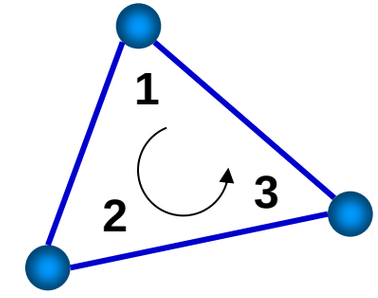
The realizer method

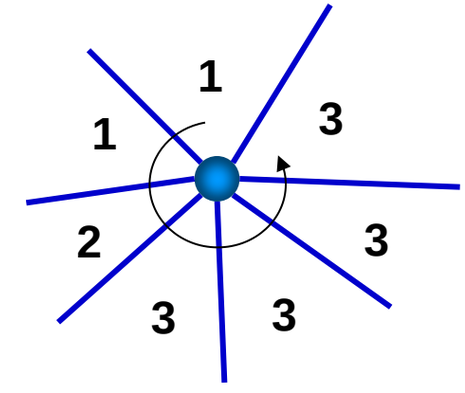
Schnyder 1990

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Schnyder labelings

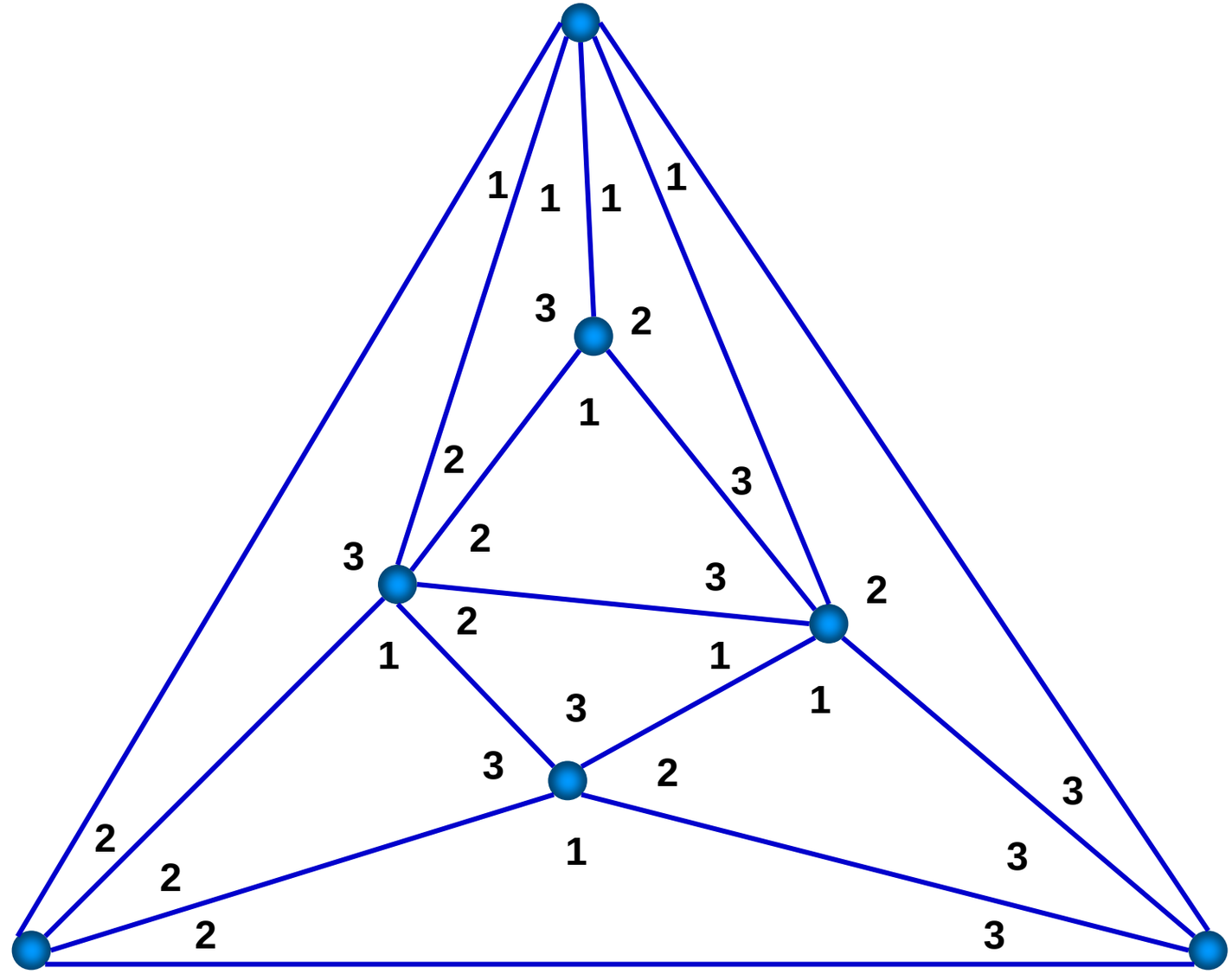
- A Schnyder labeling is a labeling of all angles of all internal faces of a maximal plane graph with labels 1, 2, 3 such that
  - angles of each face are labeled 1, 2 and 3 in counter-clockwise order
  - labels of the angles at each internal node form non-empty intervals of 1's, 2's and 3's in counter-clockwise order





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Example of Schnyder labeling

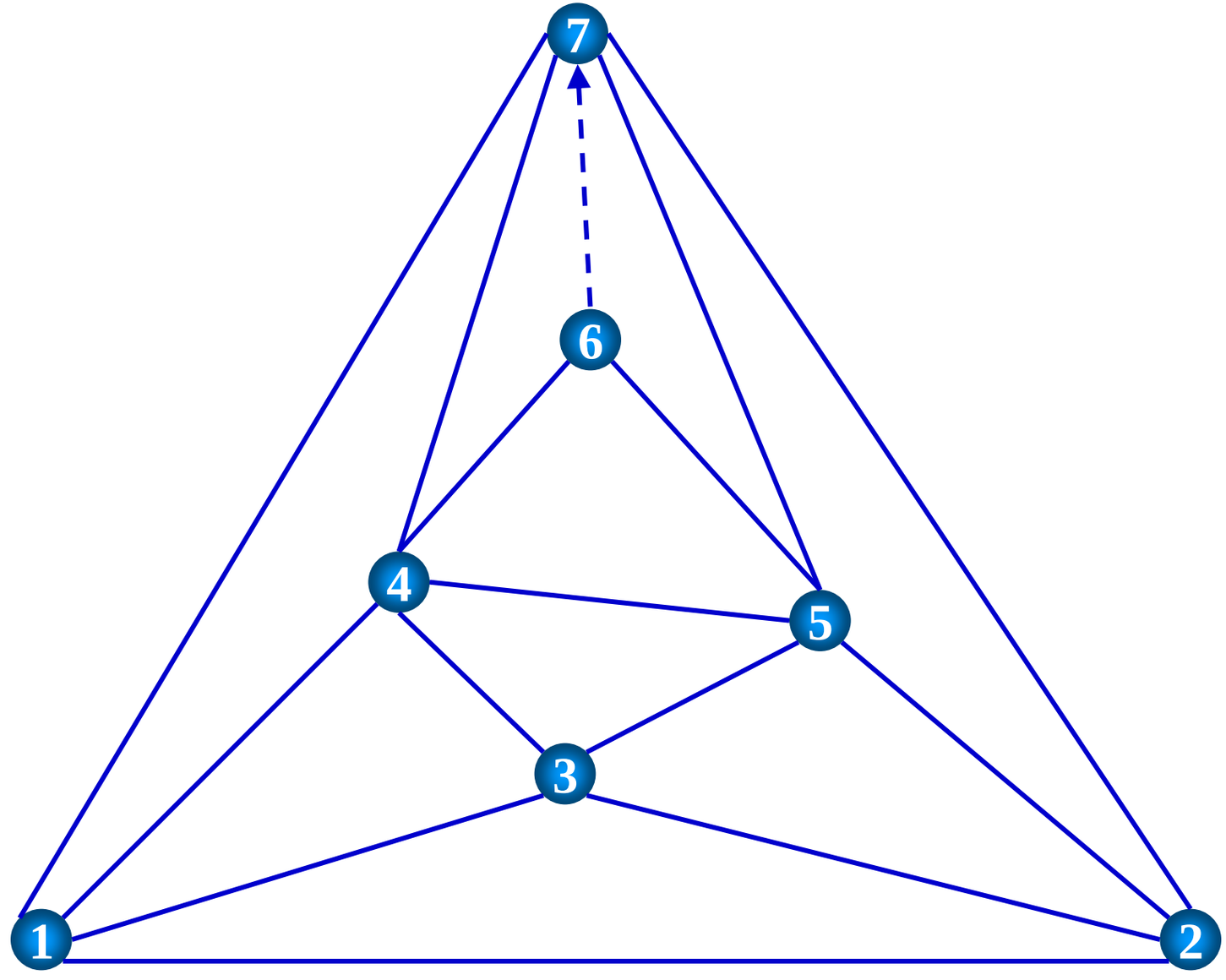


Schnyder labelings always exist

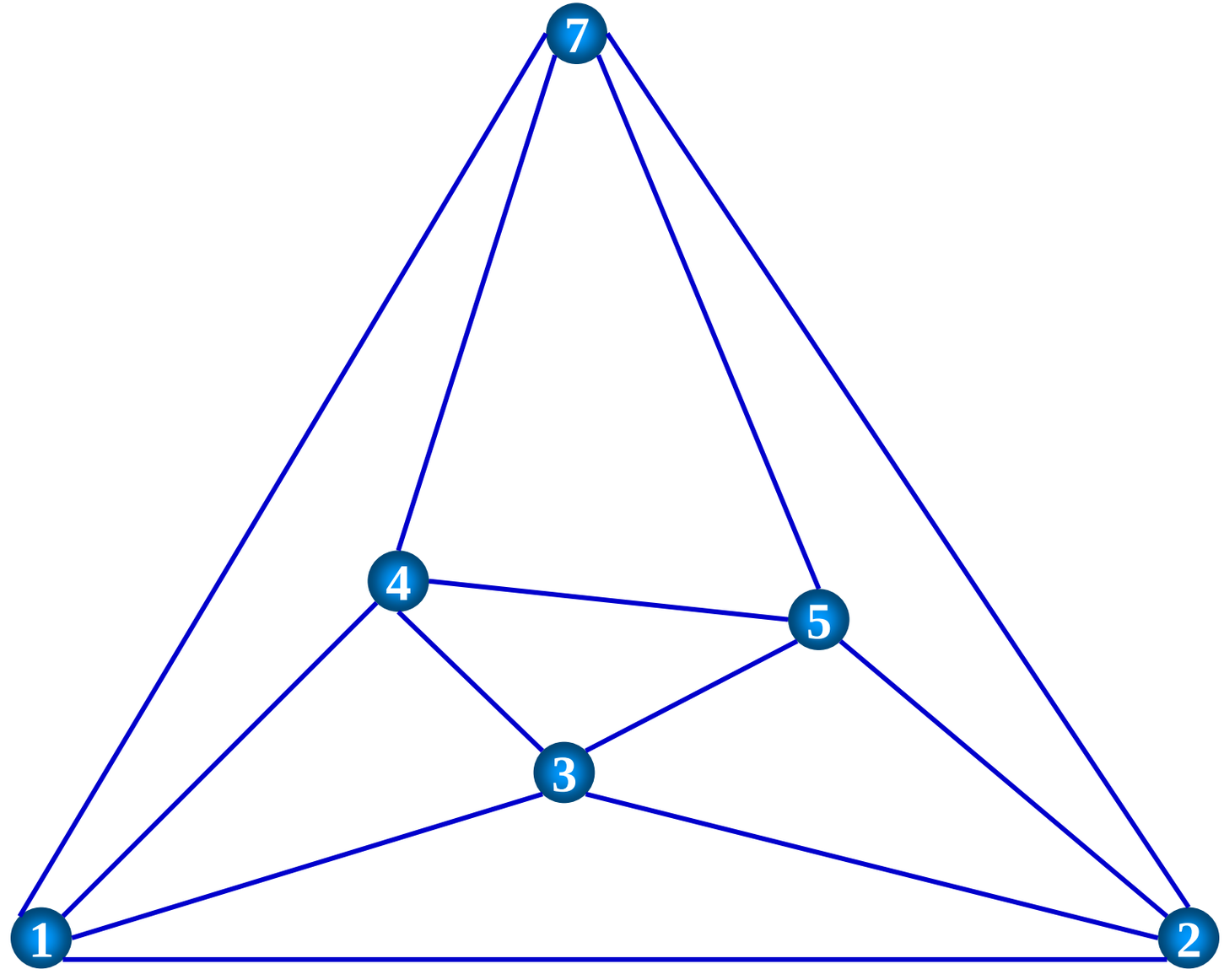
- Every maximal plane graph has a Schnyder labeling
- Exploit a canonical ordering
  - contract inner edges (v_n, v_{n-1}), (v_n, v_{n-2}), …
    - edges are contractible if graph remains triangulated
  - label all angles at v_n with 1
  - expand again in reverse order

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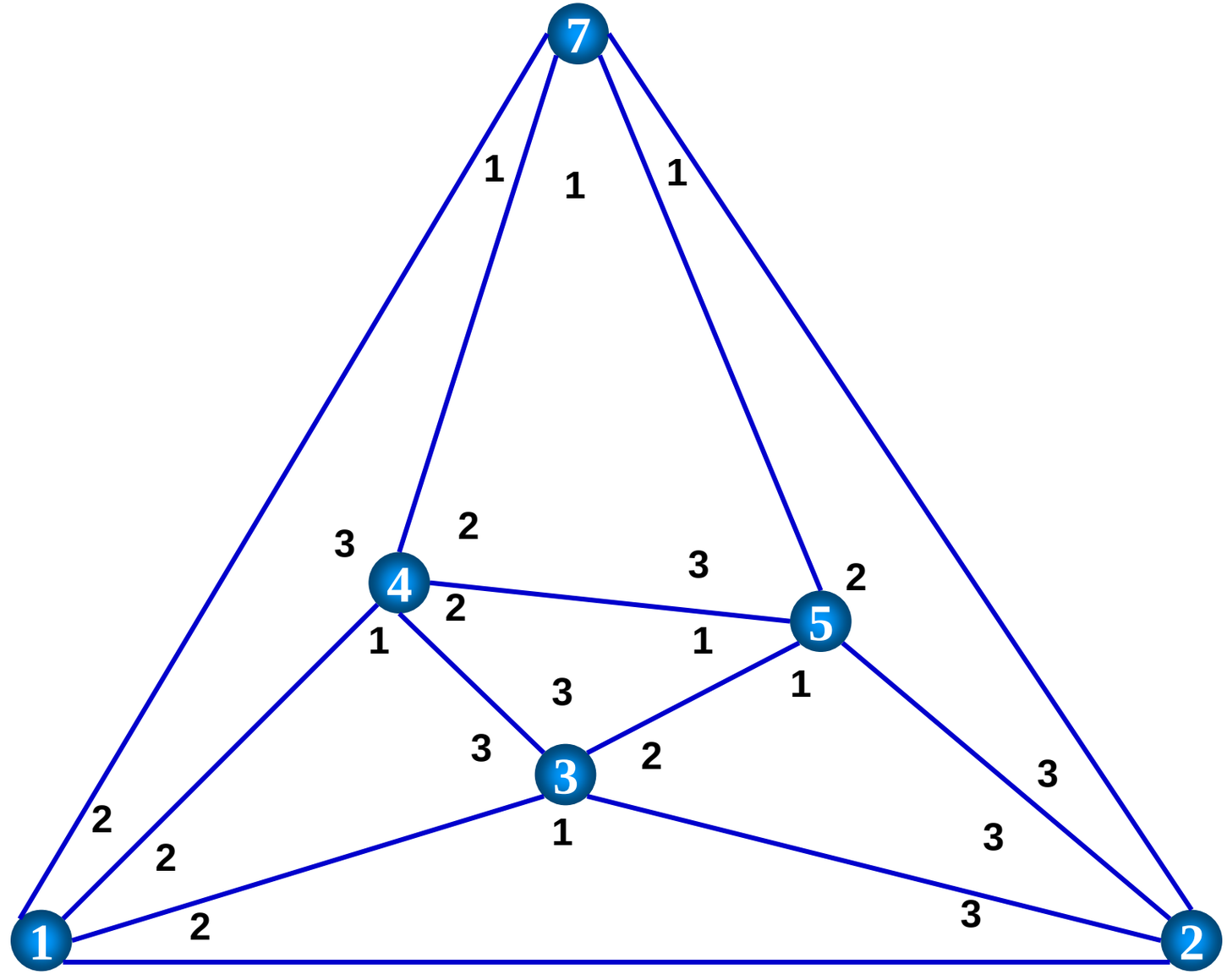
Contraction of edge (v_n,v_{n-1})



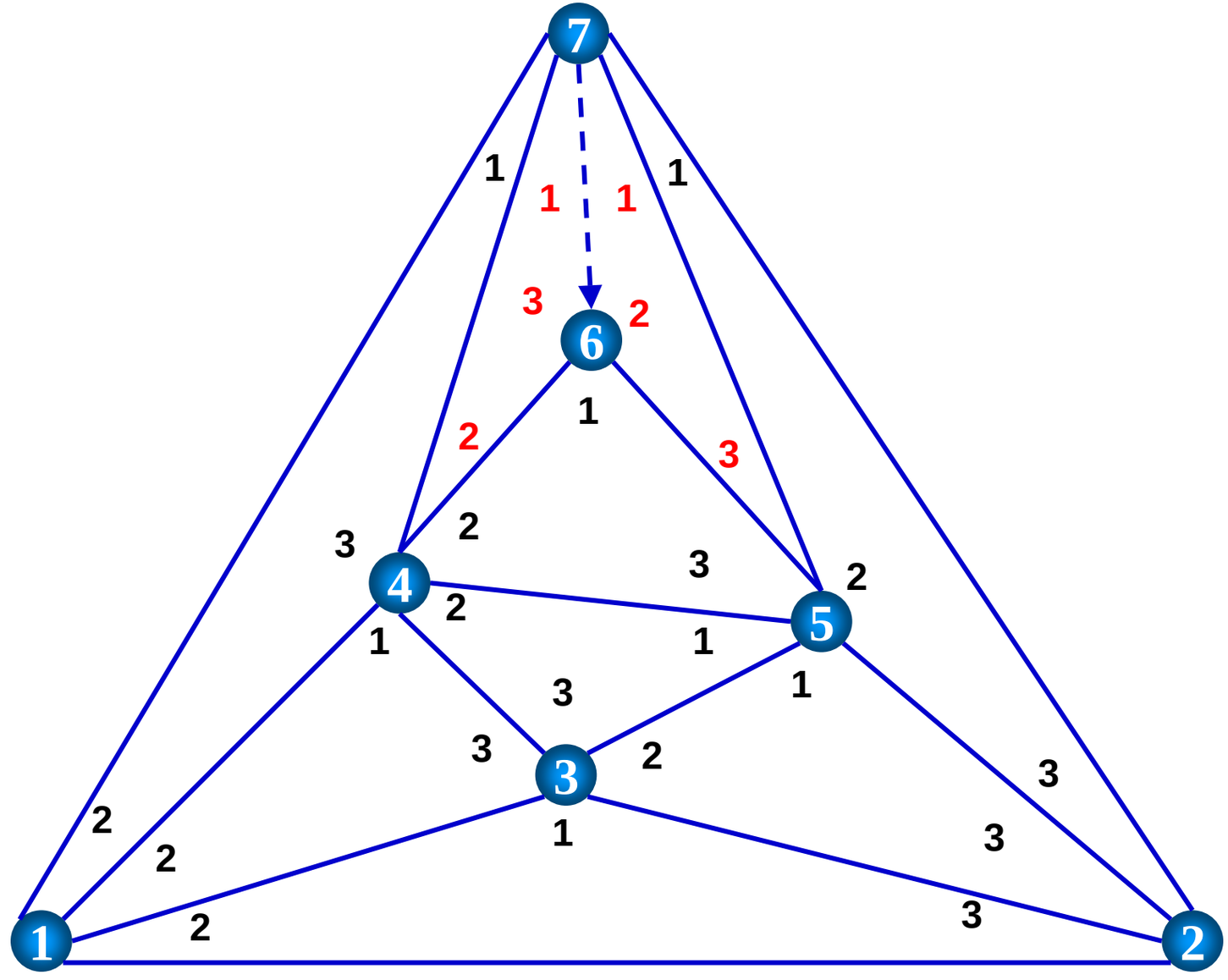
Contraction of edge (v_n,v_{n-1})



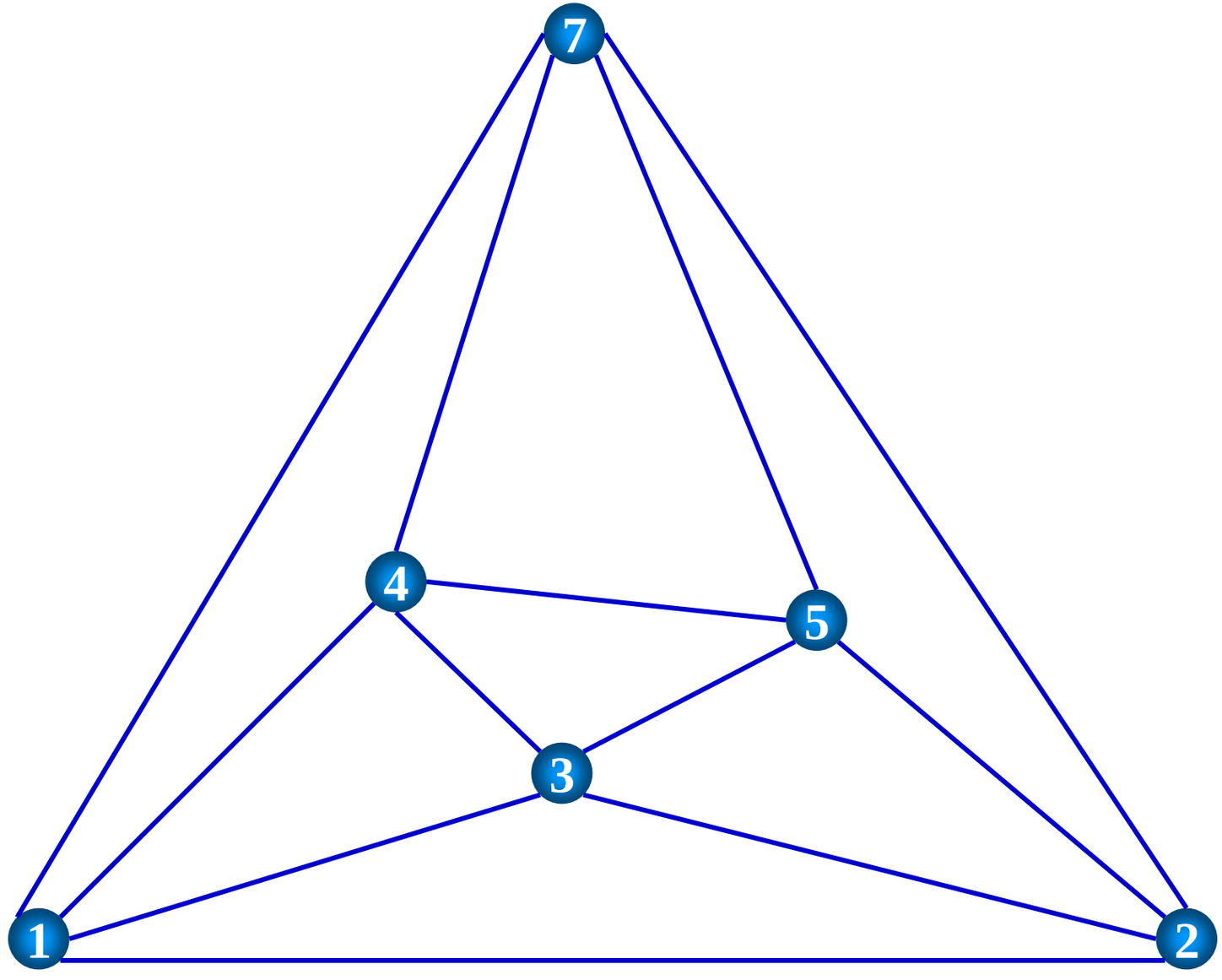
If we had this Schnyder labeling…



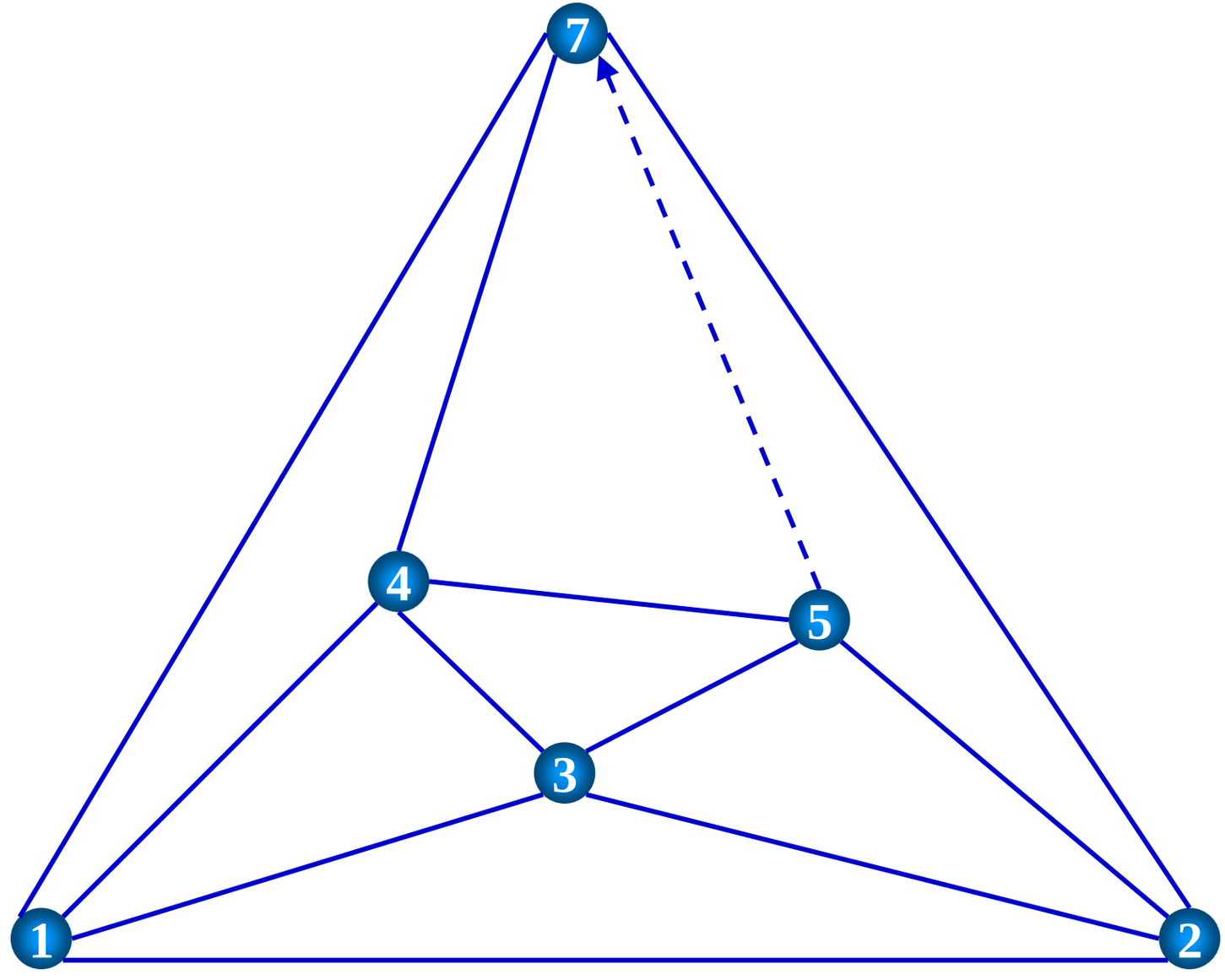
…we could easily expand edge (v_n,v_{n-1})



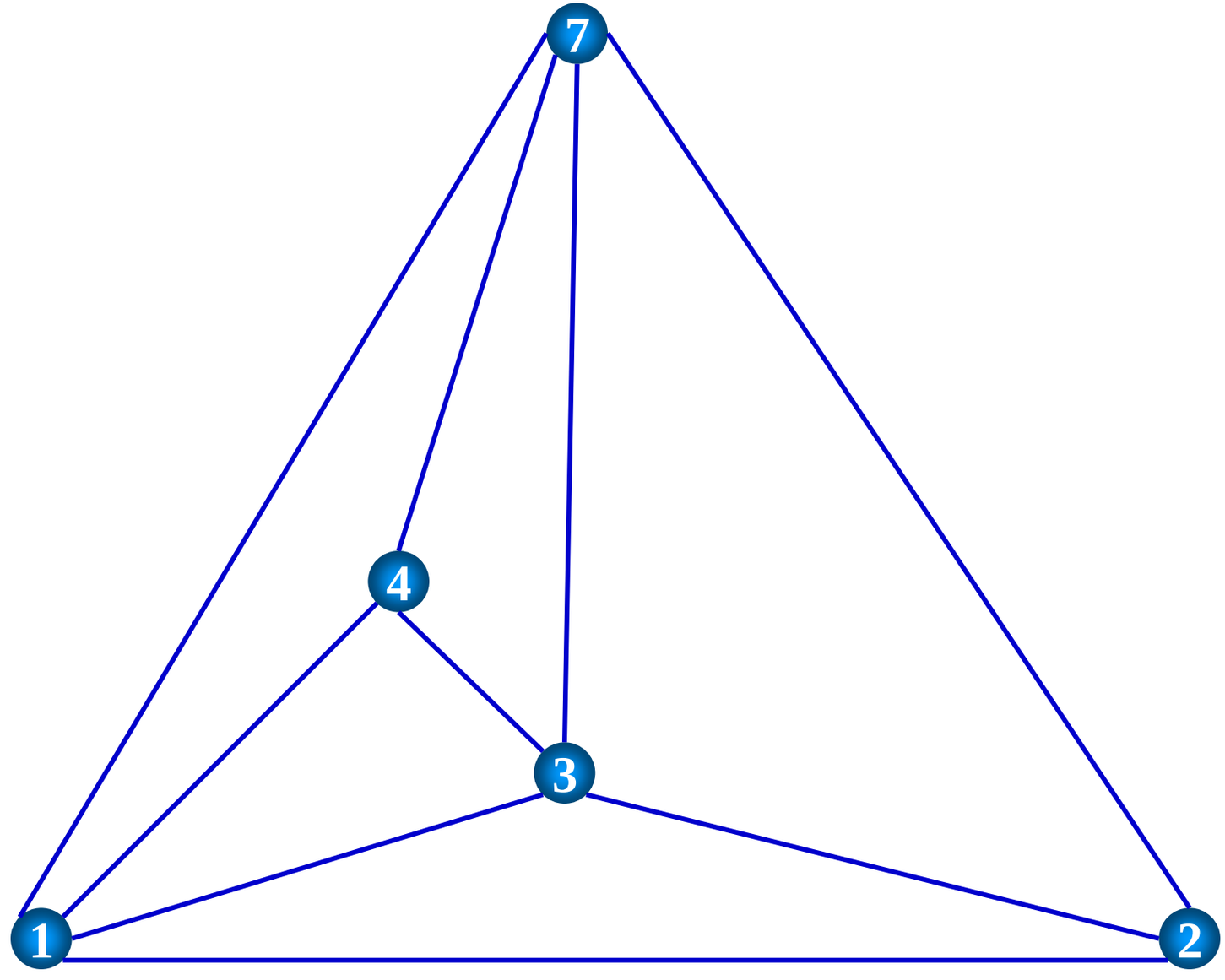
How to get a labeling for this graph?



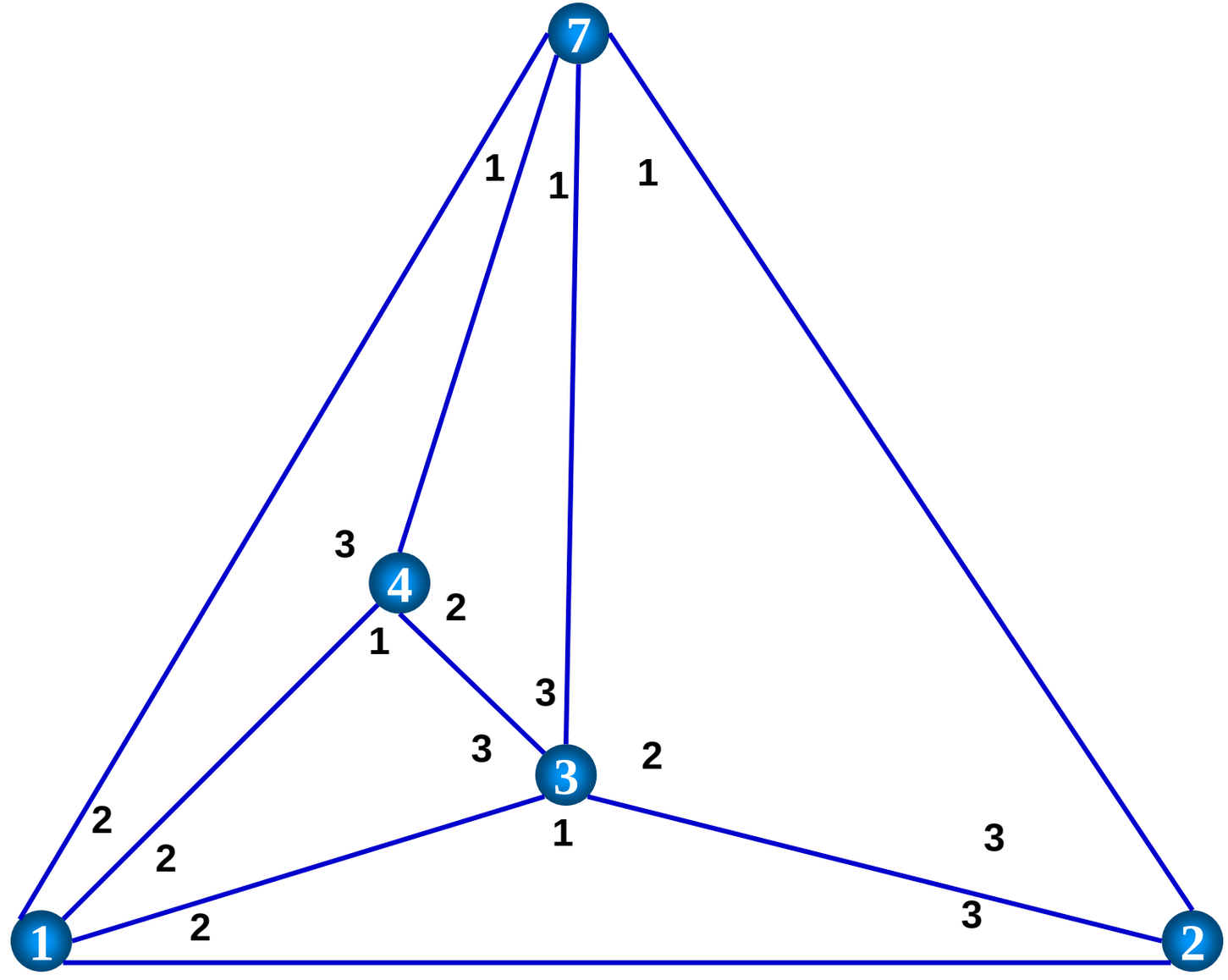
Contraction of edge (v_n,v_{n-2})



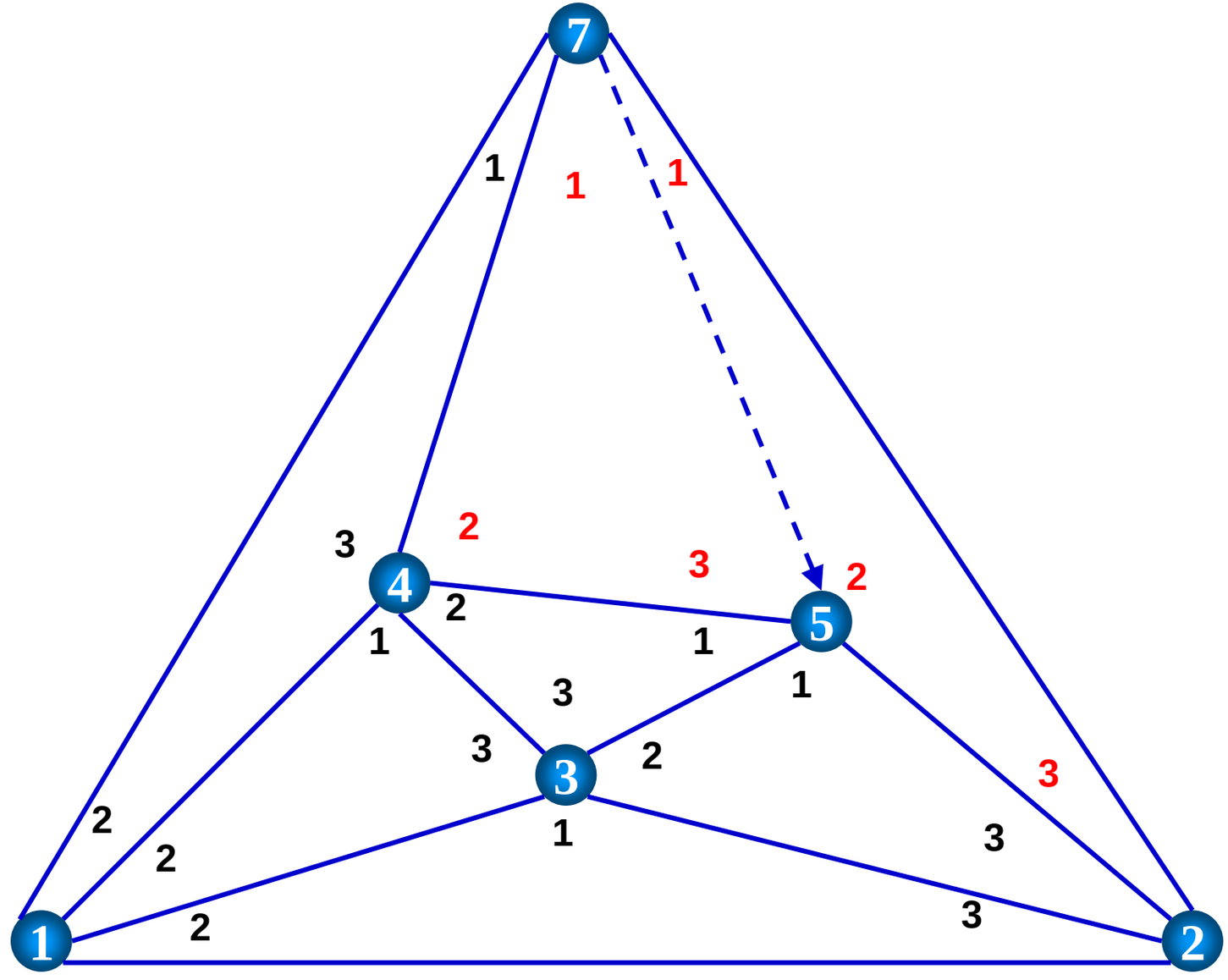
Contraction of edge (v_n,v_{n-2})



If we had this Schnyder labeling…



…we could easily expand edge (v_n,v_{n-2})



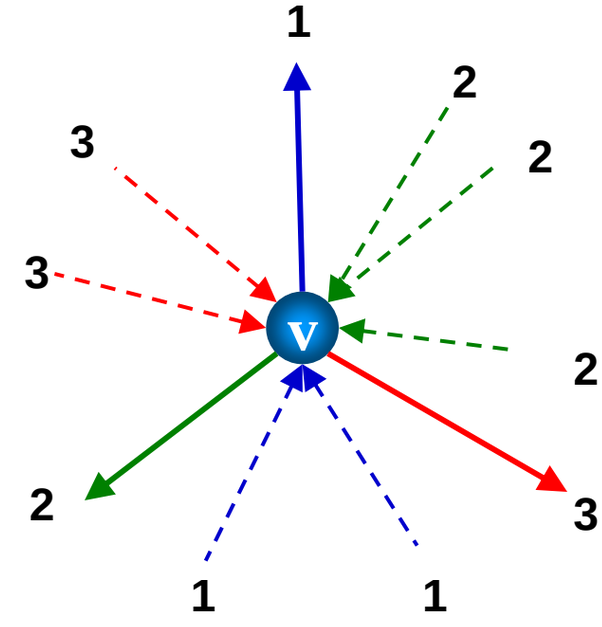
Contractible edges

- Edges (v_n, v_{n-1}), (v_n, v_{n-2}), … are always contractible
  - edges are contractable if their endpoints have exactly two common neighbours
  - this holds in a canonical ordering
    - after the contraction v_n remains the neighbour of all vertices on the outer cycle of G_k (including v_k)
    - therefore, v_n and v_k have the leftmost and rightmost neighbour of v_k on the outer cycle of G_{k-1} in common

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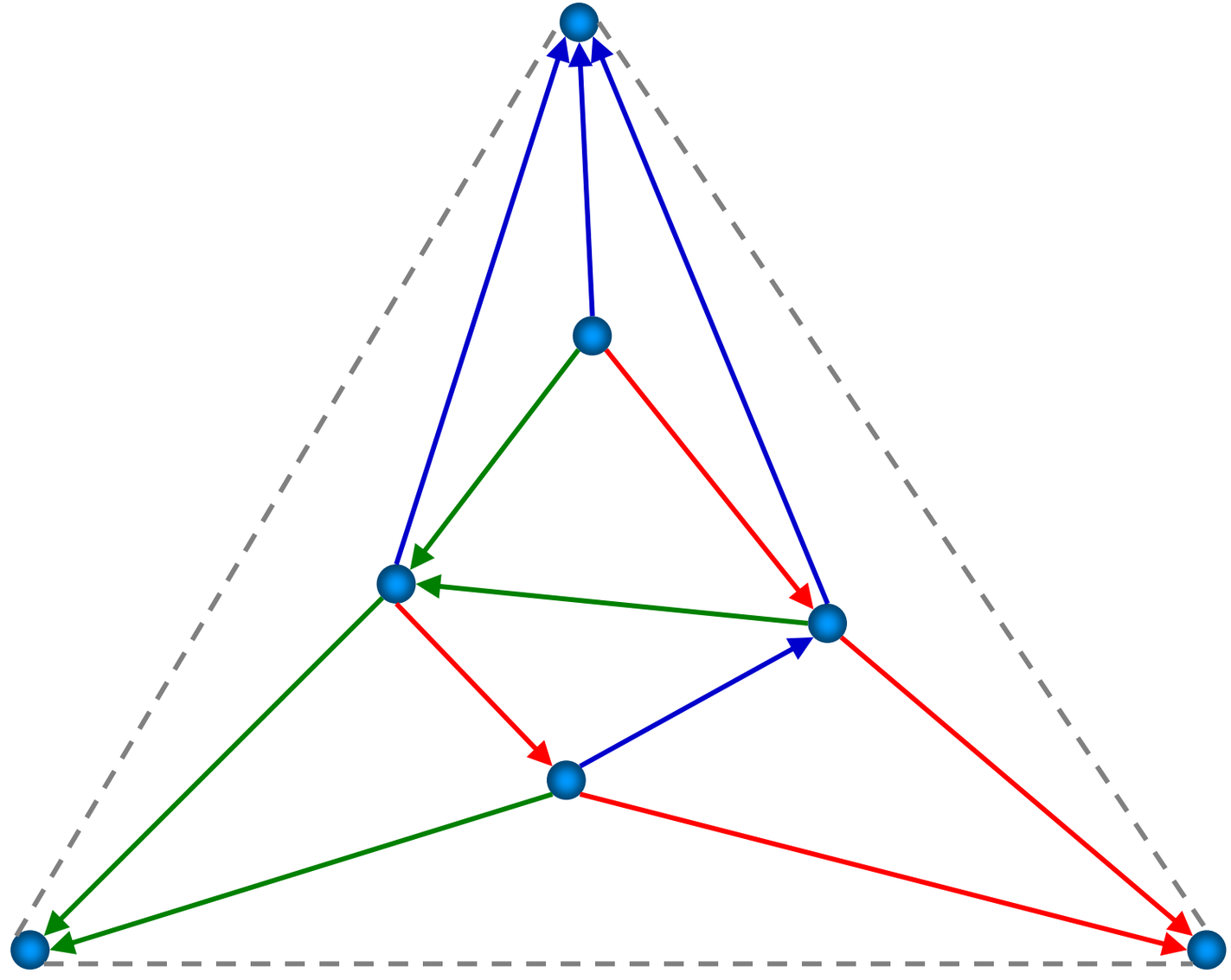
Schnyder realizers

- A *Schnyder realizer* is a partition of the internal edges into sets T_1, T_2 and T_3 such that for each internal node v
  - v has outdegree one in each of the sets
  - the order of the incident edges is the one given in figure
    - dashed edges may be missing



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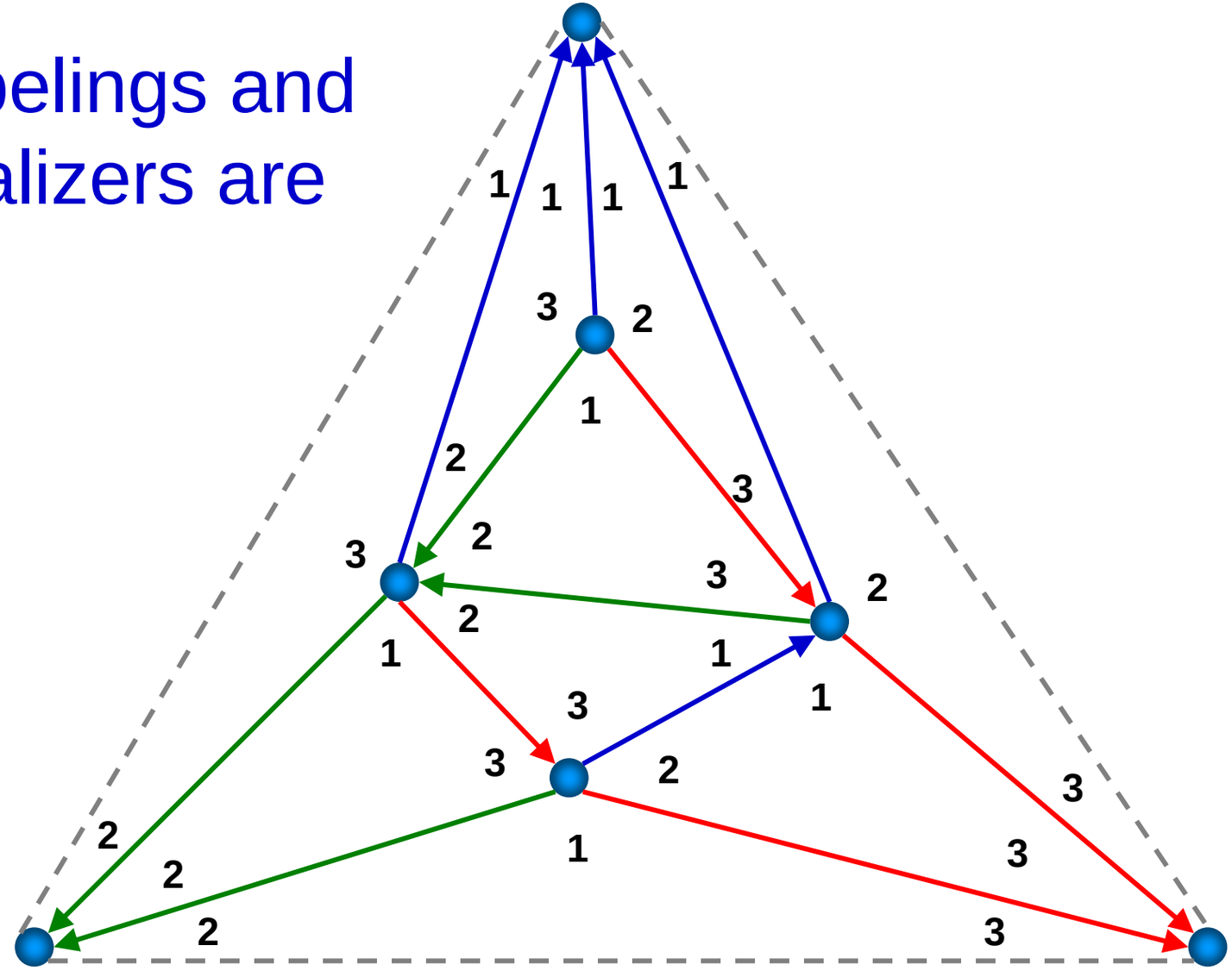
Example of Schnyder realizer



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Schnyder labelings and realizers

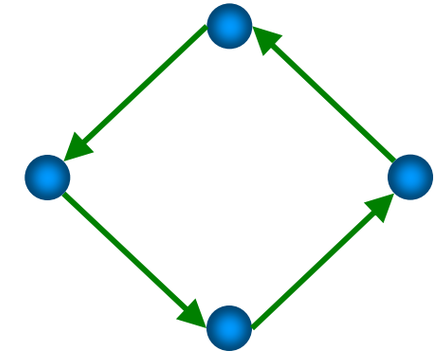
Schnyder labelings and Schnyder realizers are equivalent notions



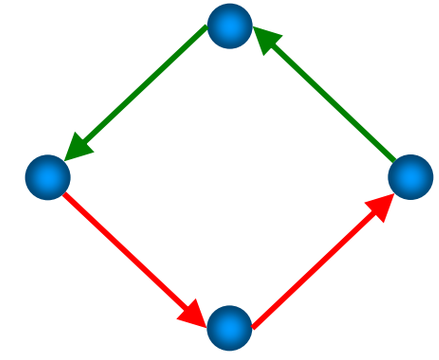
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Properties of Schnyder realizers

- There cannot be a directed cycle of the same color
  - each color forms a tree



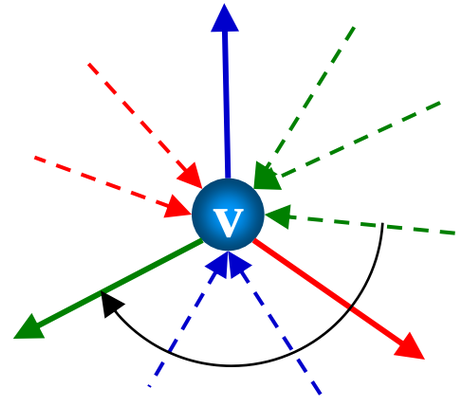
- There cannot be a directed cycle of two colors
  - suppose for a contradiction that there is a cycle formed by green and red edges

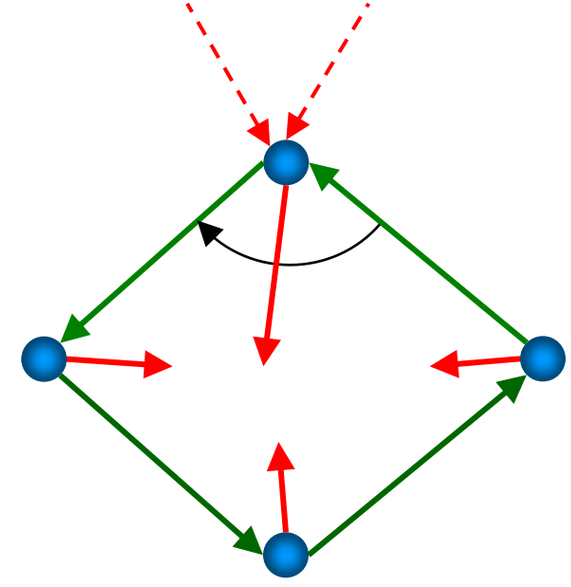


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Cycles of the same color

- between an entering-green edge and the exiting-green edge (clockwise) we have the exiting red edge



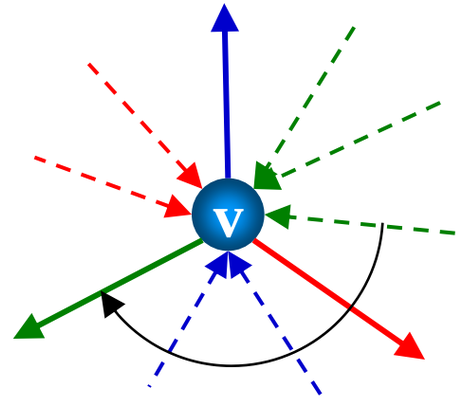


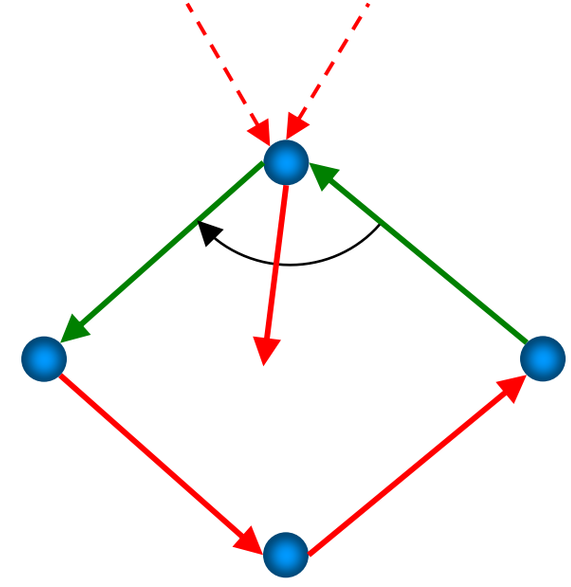
- now the red tree is trapped inside the cycle

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Cycles of two colors

- between an entering-green edge and the exiting-green edge (clockwise) we have the exiting red edge

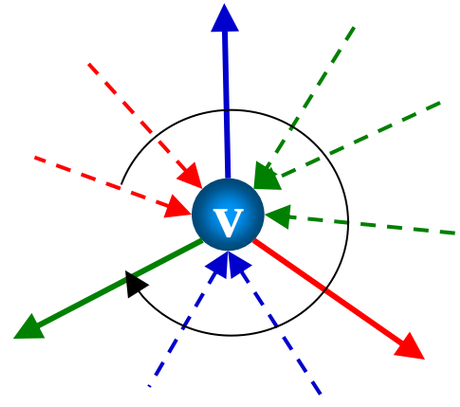


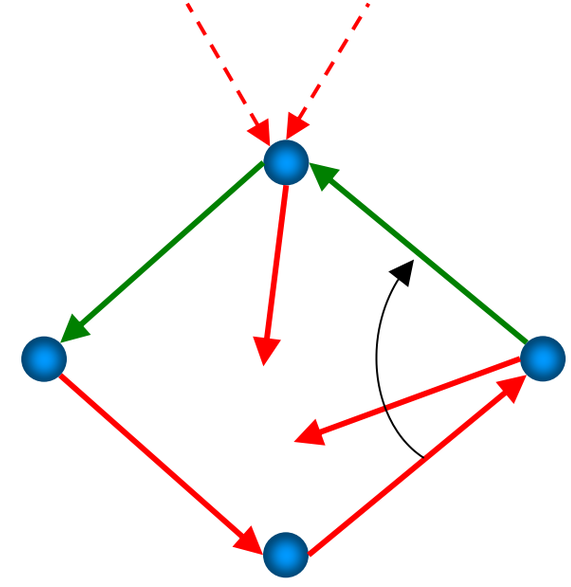


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Cycles of two colors

- between an entering-red edge and the exiting-green edge (clockwise) we have the exiting red edge



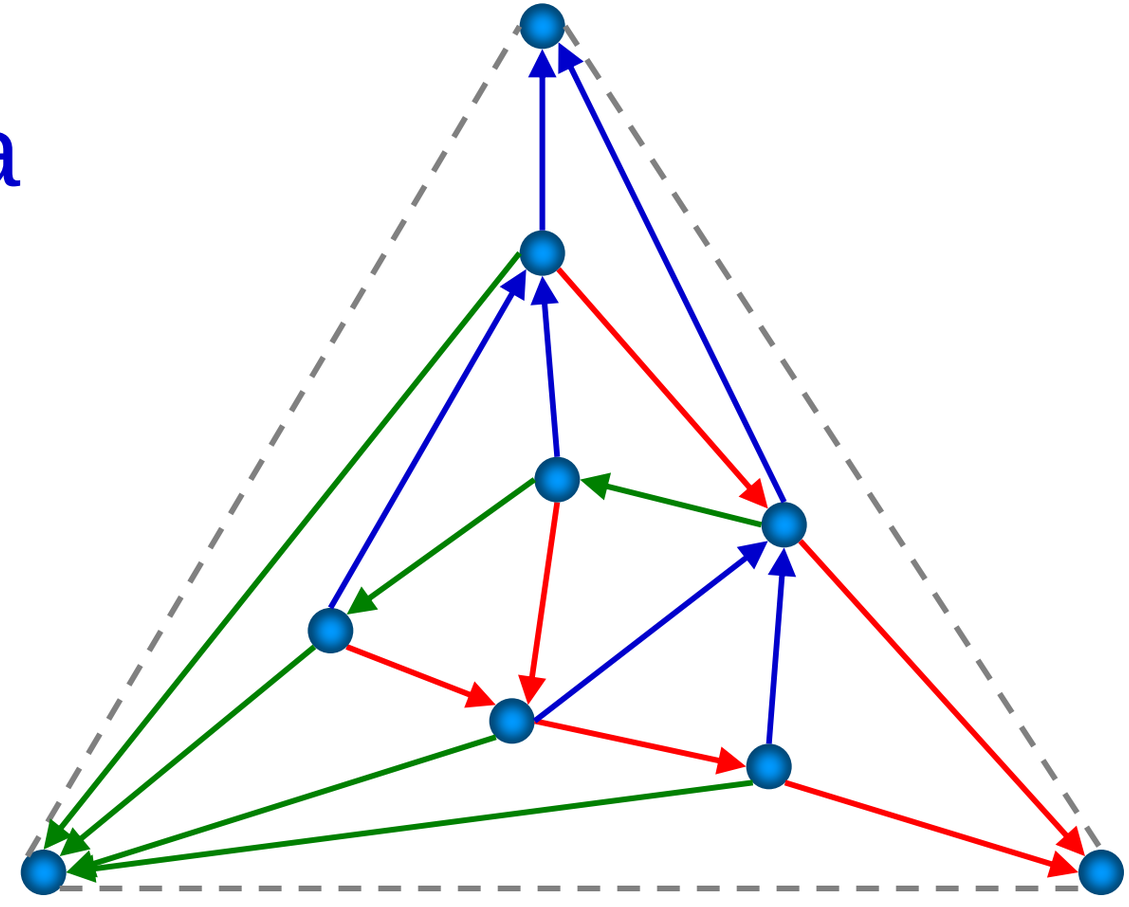


- now the red tree is trapped inside the cycle

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Properties of Schnyder realizers

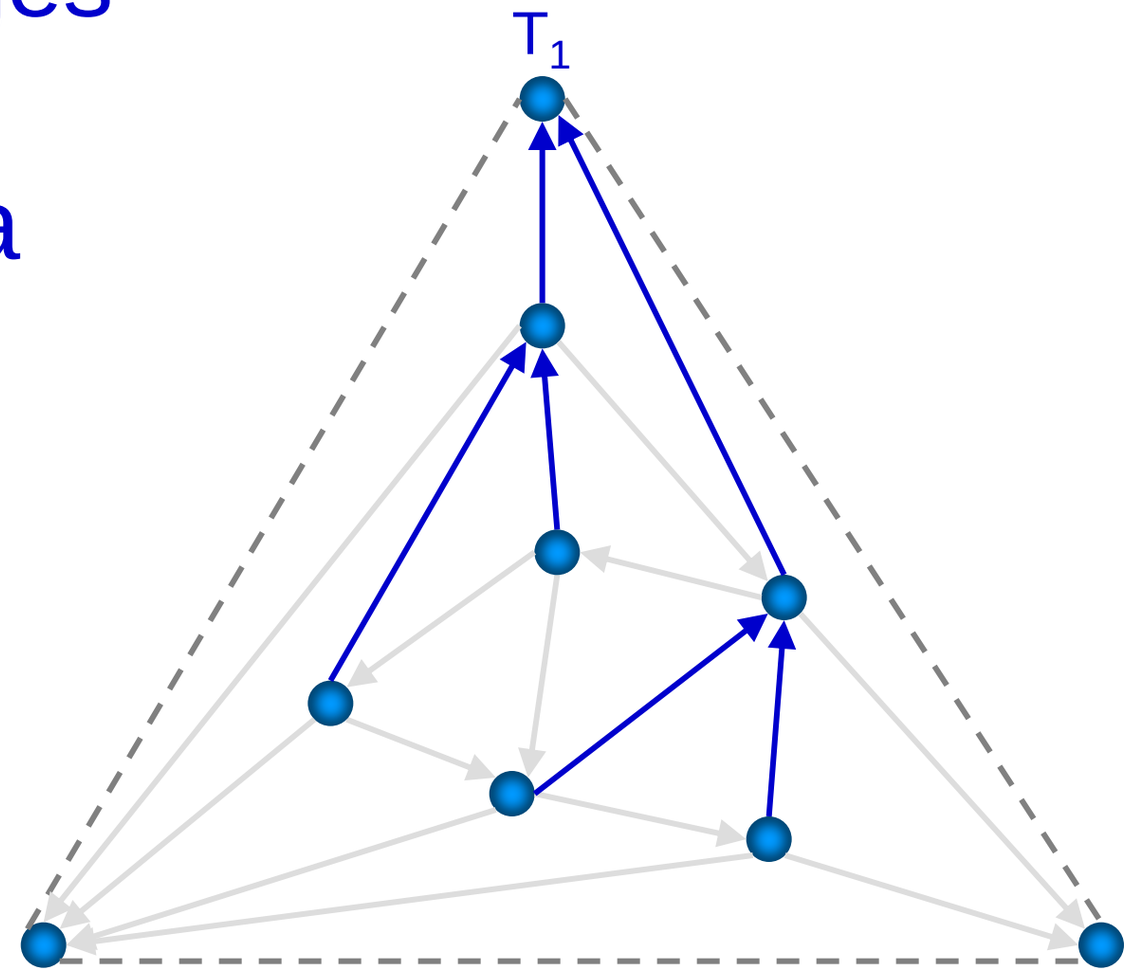
- Any set of edges among T_1, T_2, and T_3 forms a tree touching all internal nodes



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Properties of Schnyder realizers

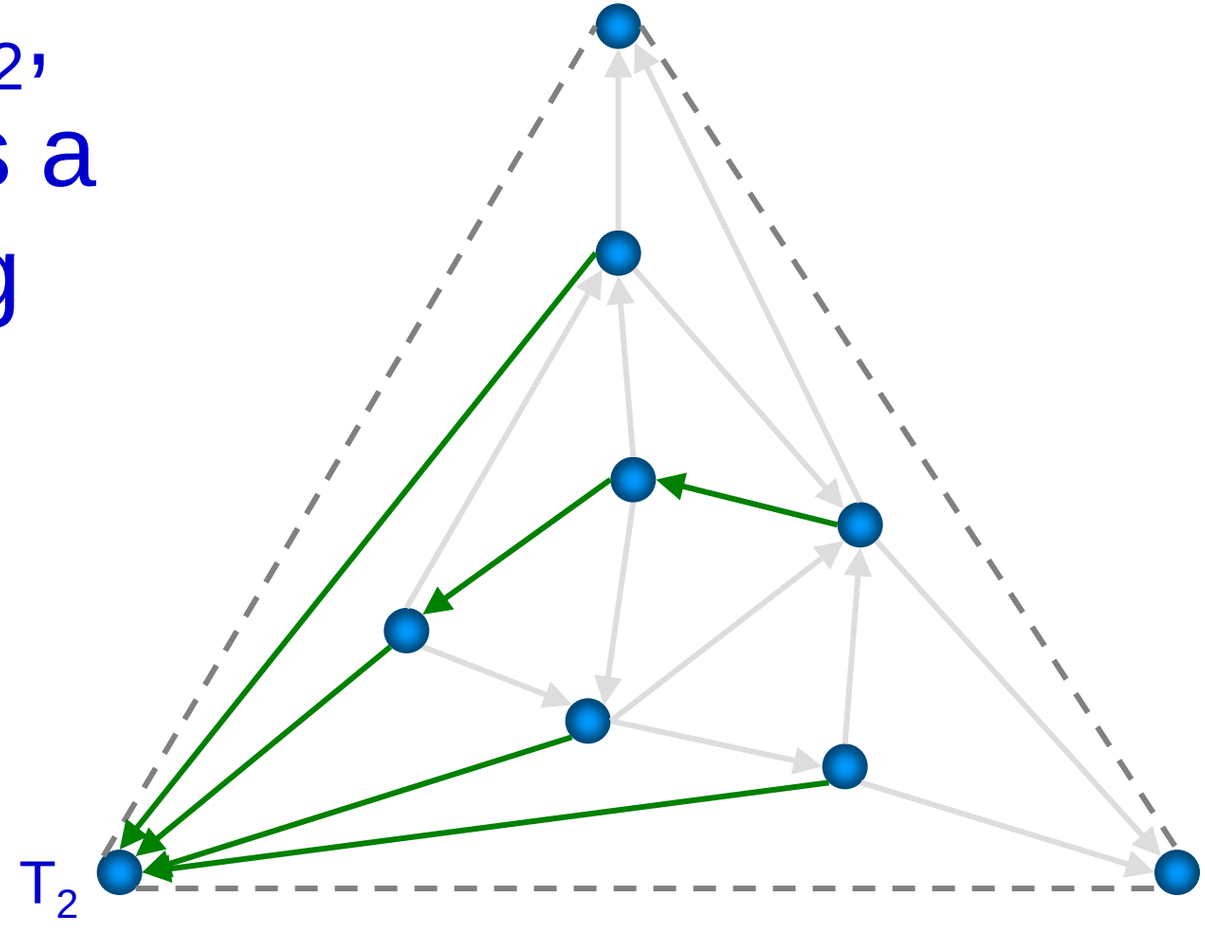
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Properties of Schnyder realizers

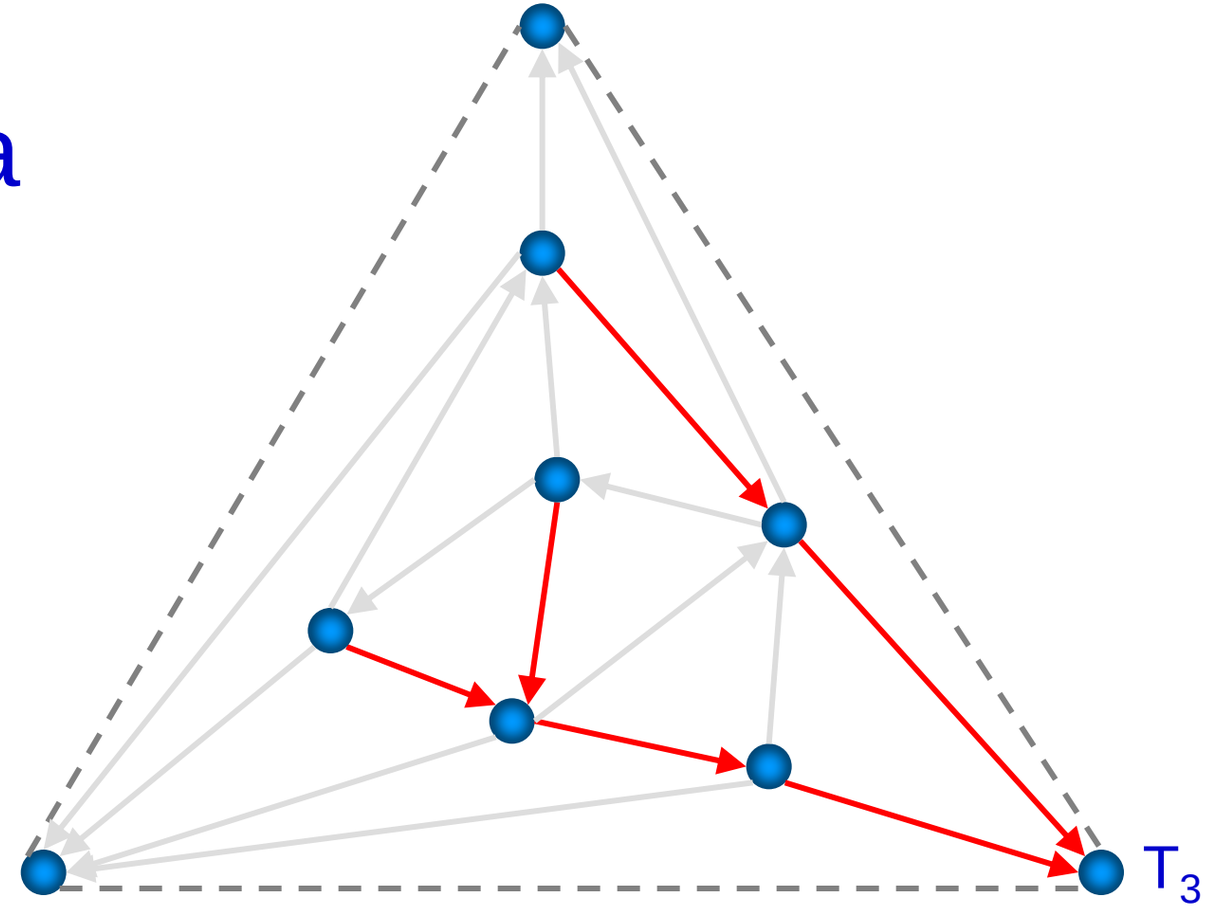
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Properties of Schnyder realizers

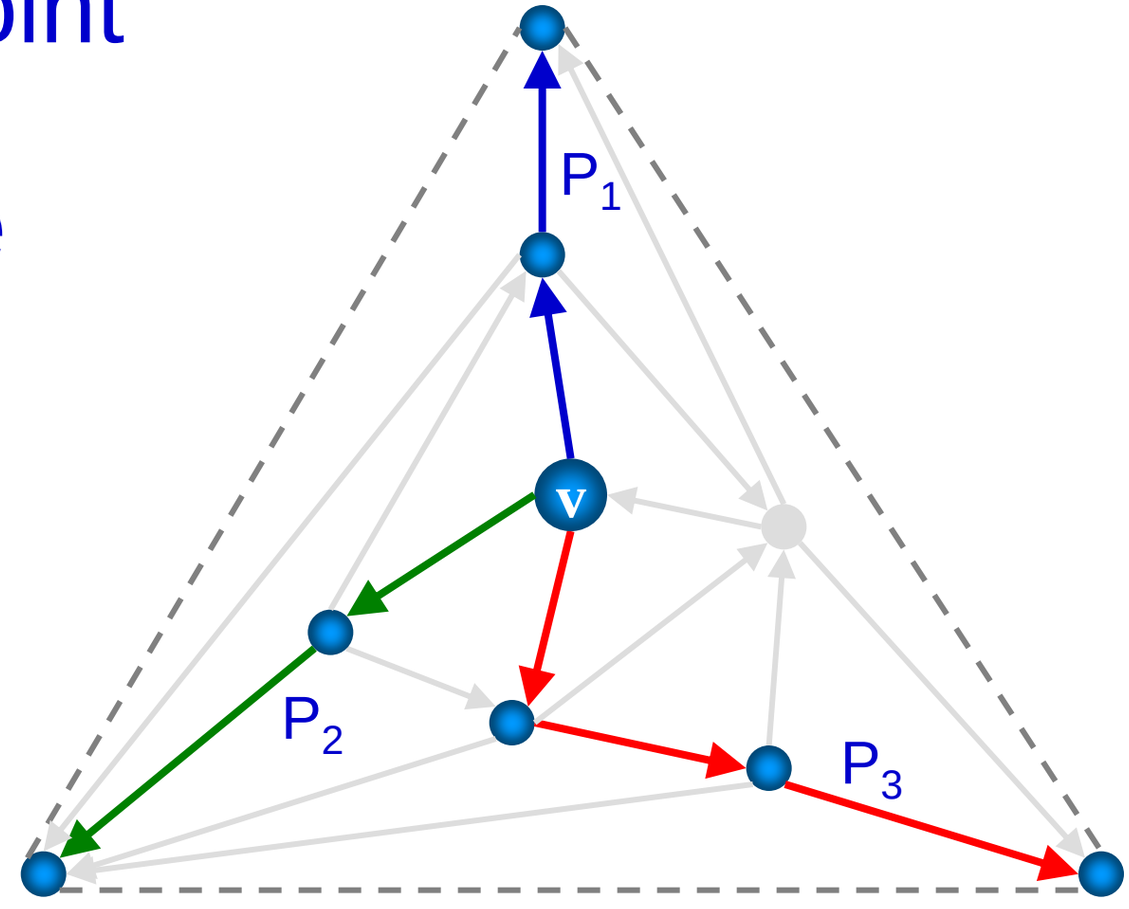
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Properties of Schnyder realizers

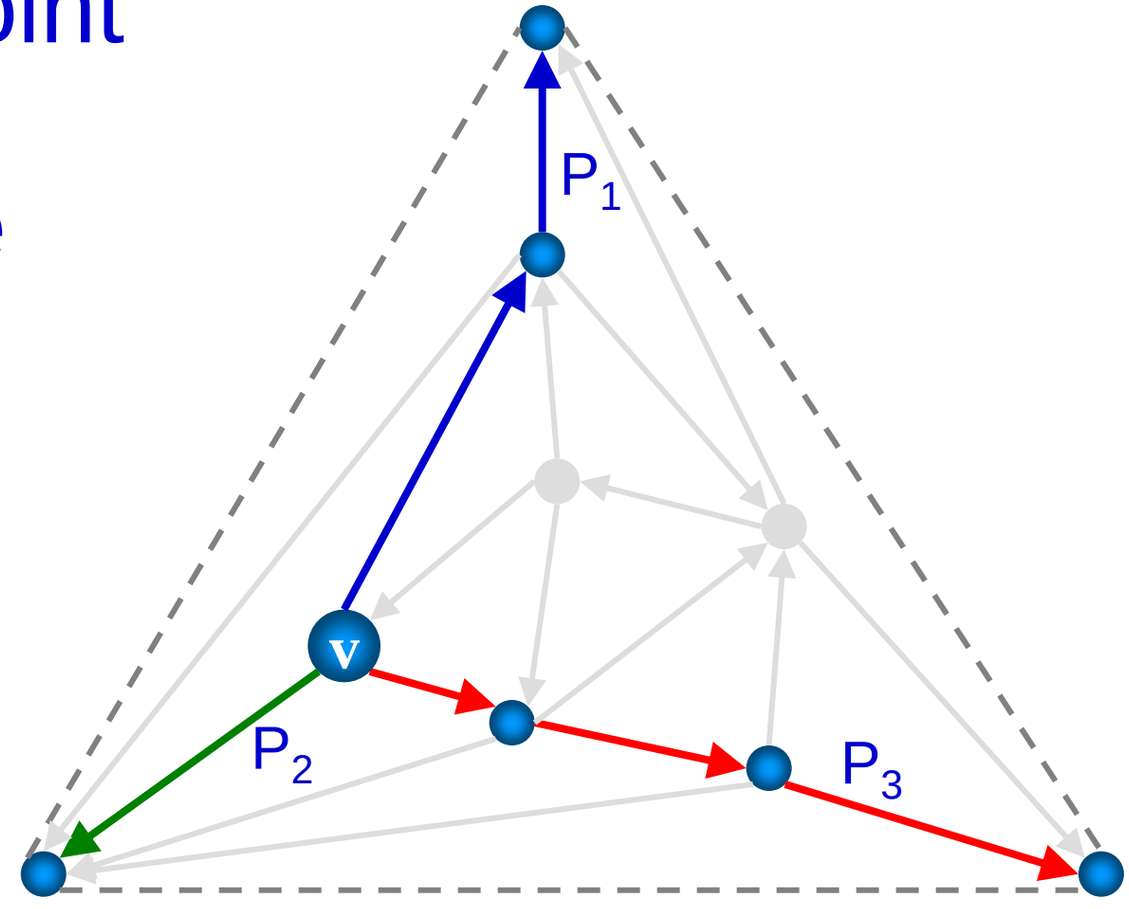
- Any internal node v has three disjoint directed paths linking it to the three external nodes
  - there are no directed cycles of two colors



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Properties of Schnyder realizers

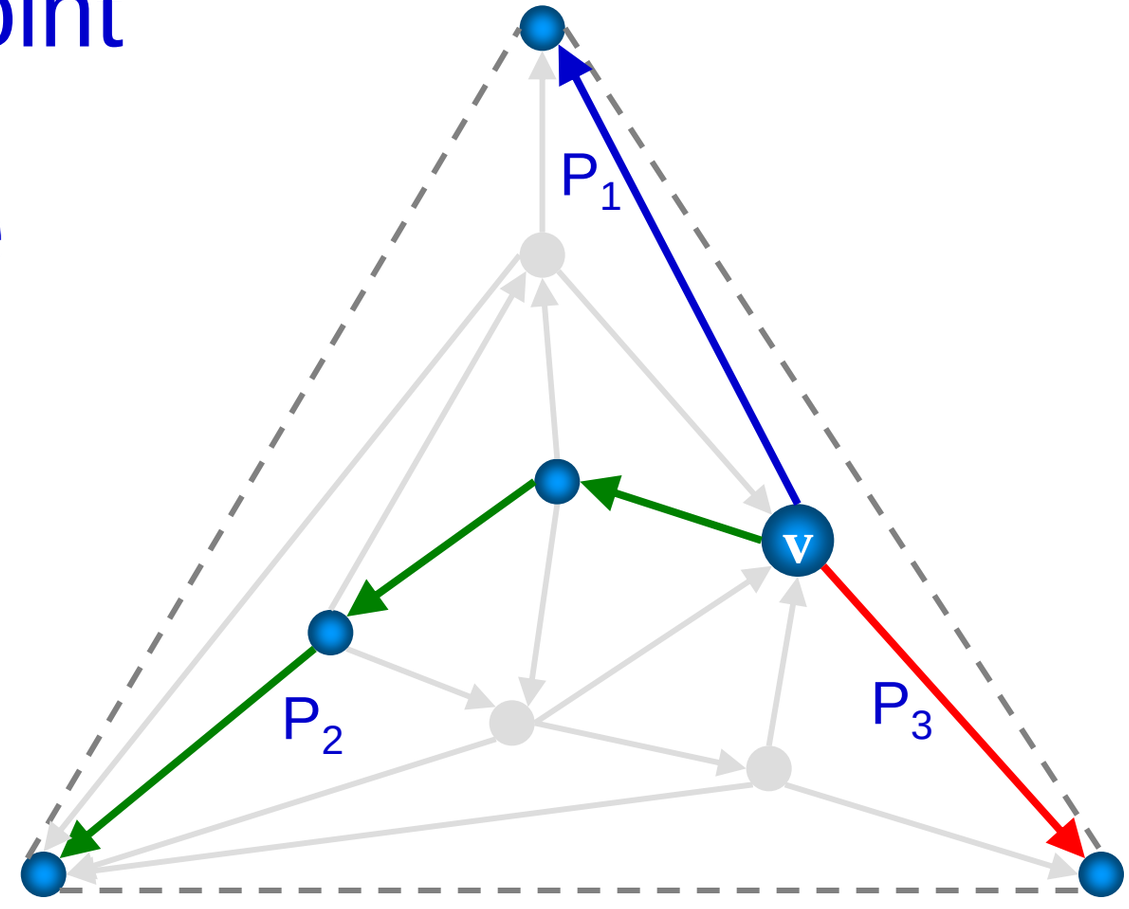
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Properties of Schnyder realizers

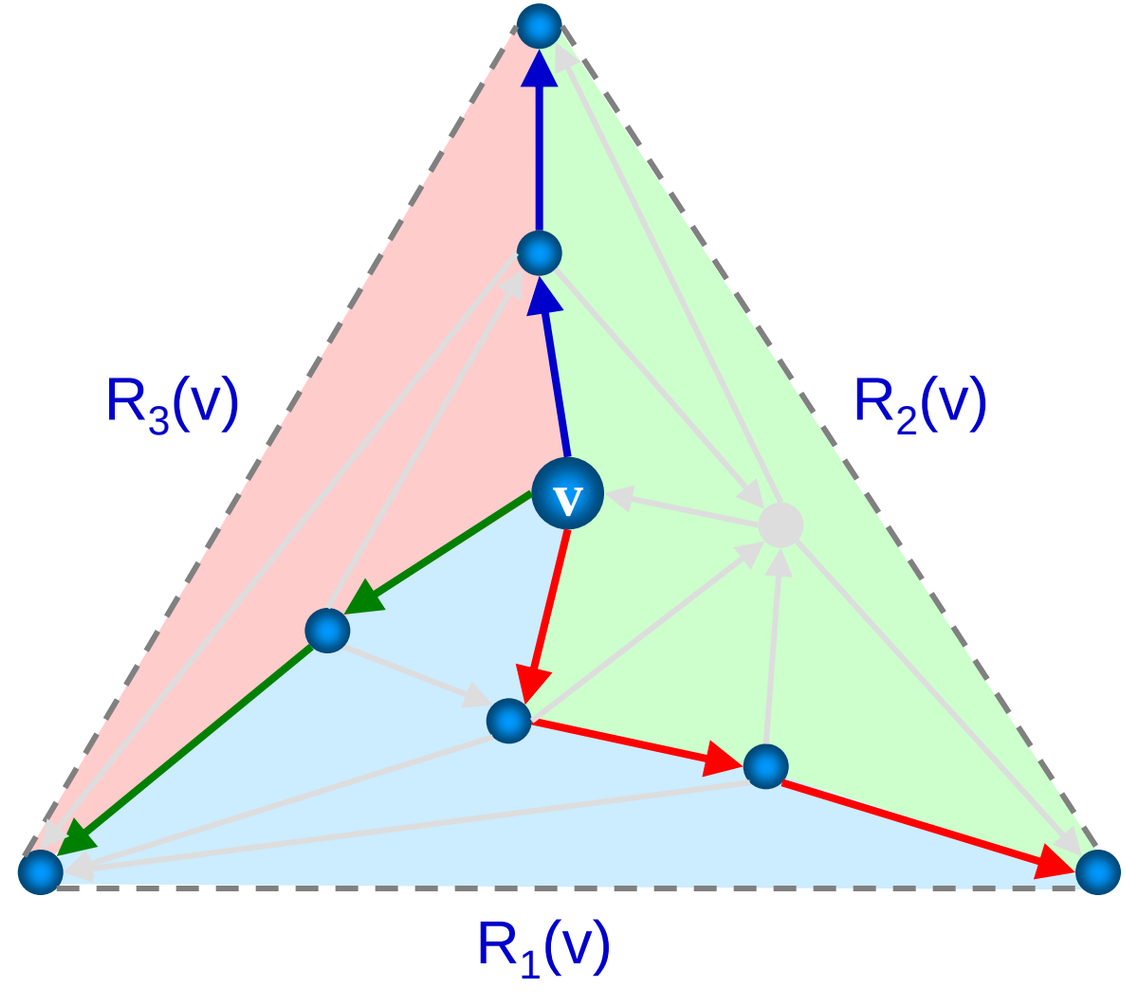
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Properties of Schnyder realizers

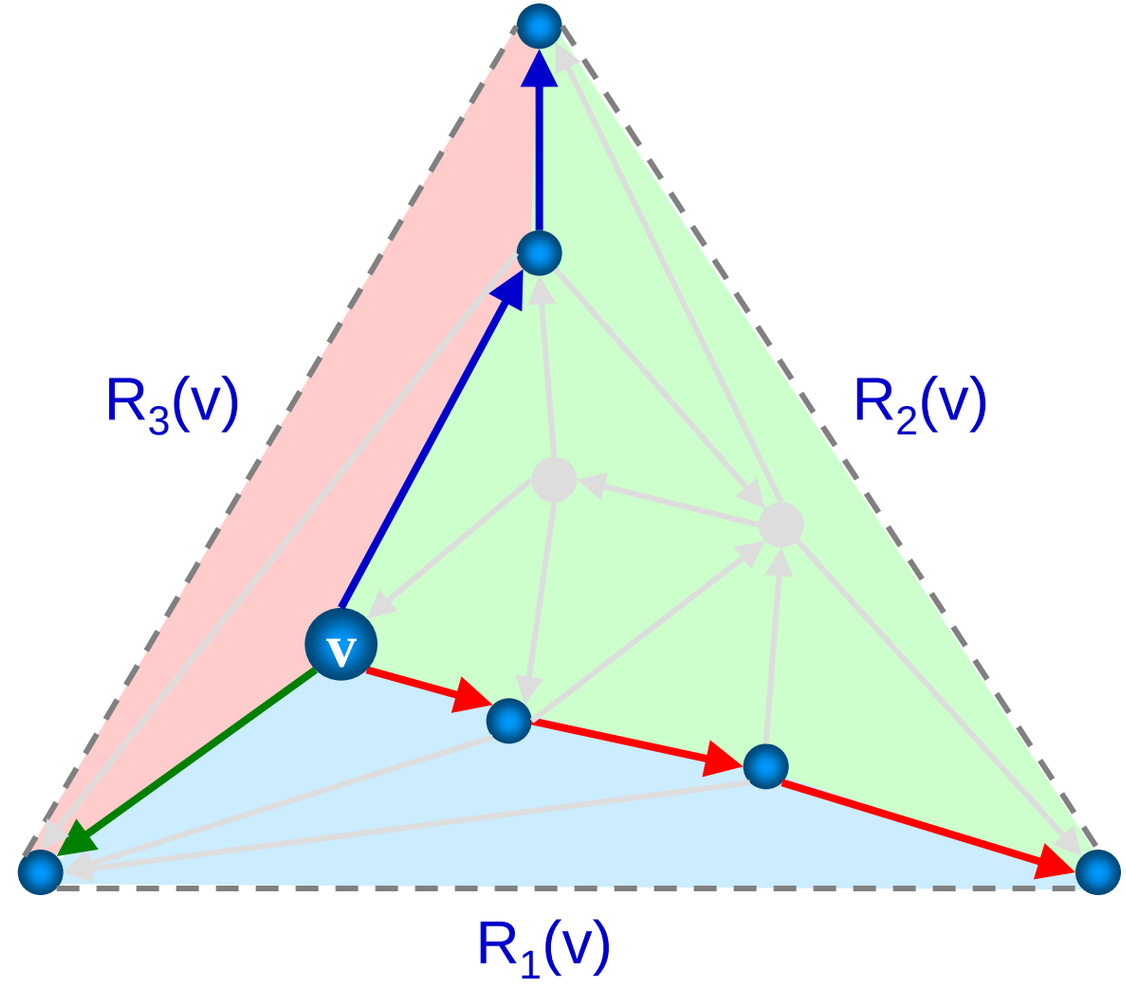
- Any internal node v defines three regions $R_1(v)$, $R_2(v)$ and $R_3(v)$



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Properties of Schnyder realizers

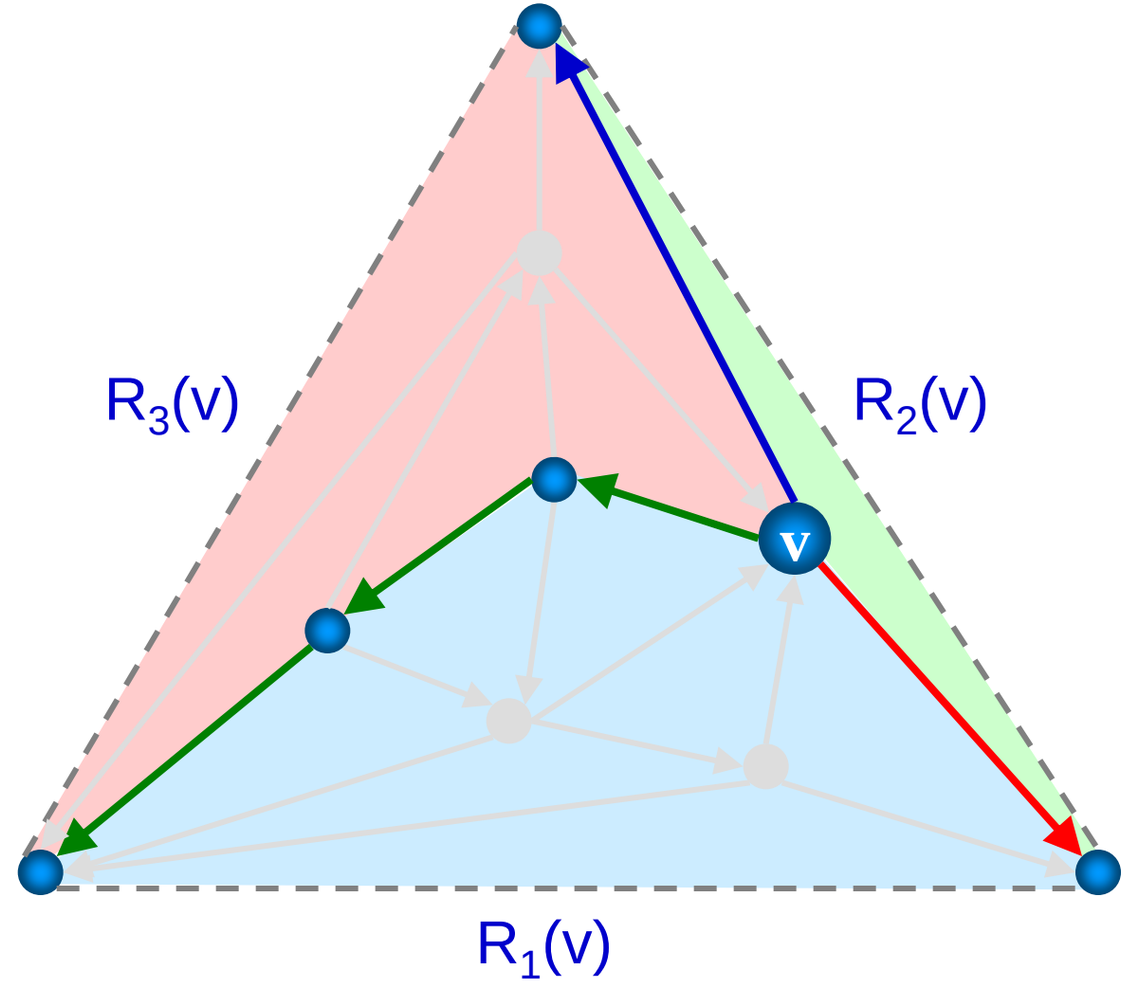
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Properties of Schnyder realizers

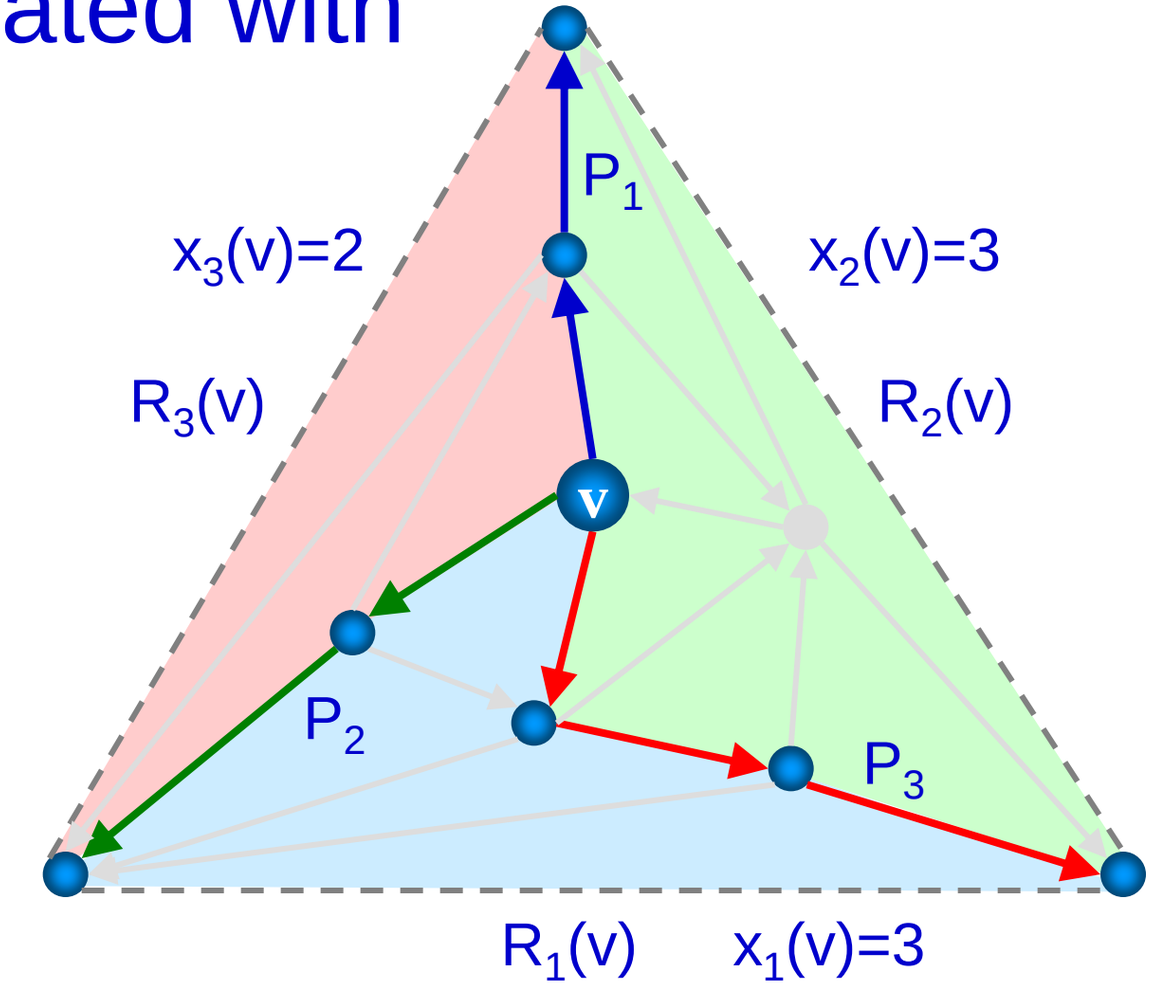
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Properties of Schnyder realizers

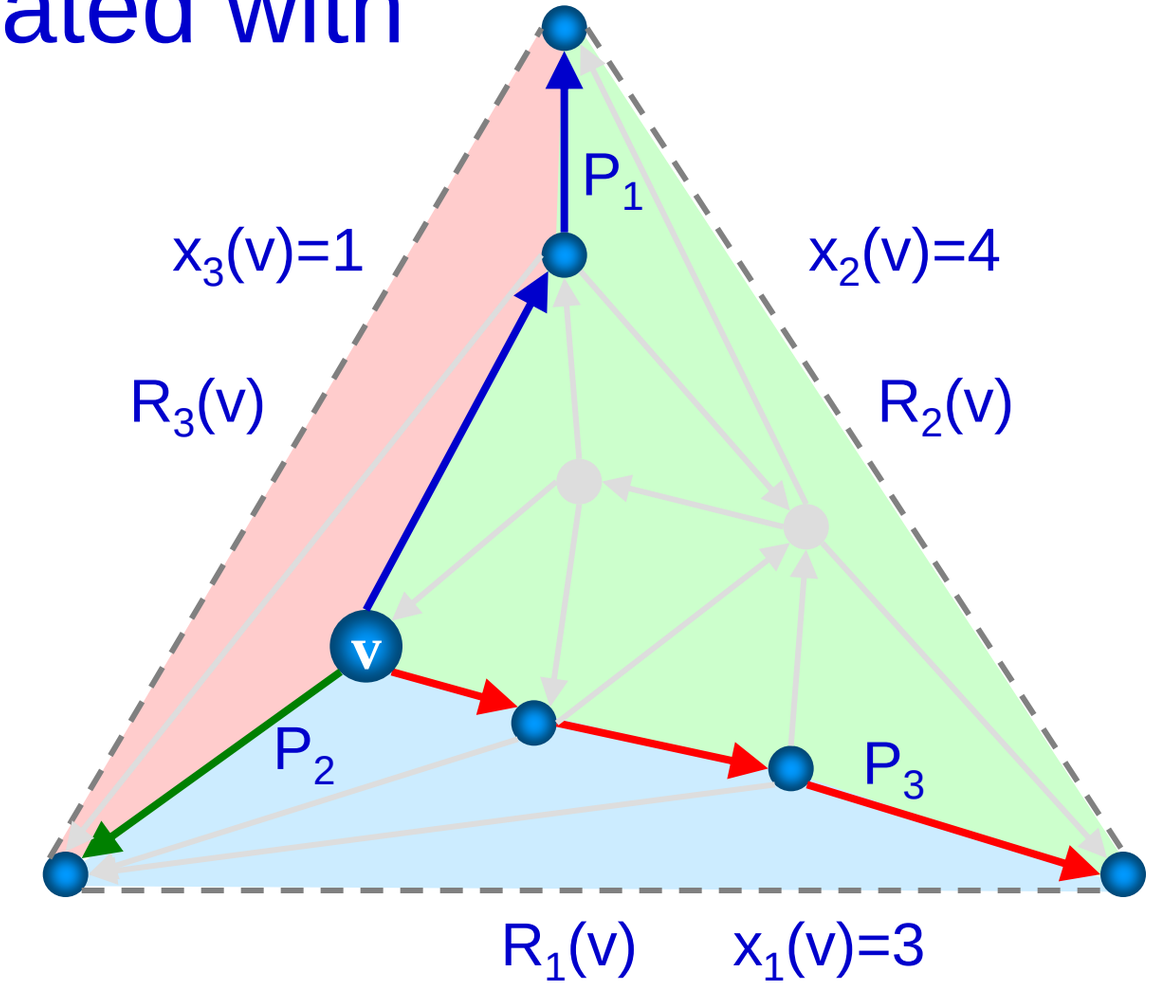
- Each region $R_i(v)$, i=1…3, can be associated with a weight $x_i(v)$
  - count nodes inside $R_i(v)$ and add nodes different from v in path P_{i-1}



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Properties of Schnyder realizers

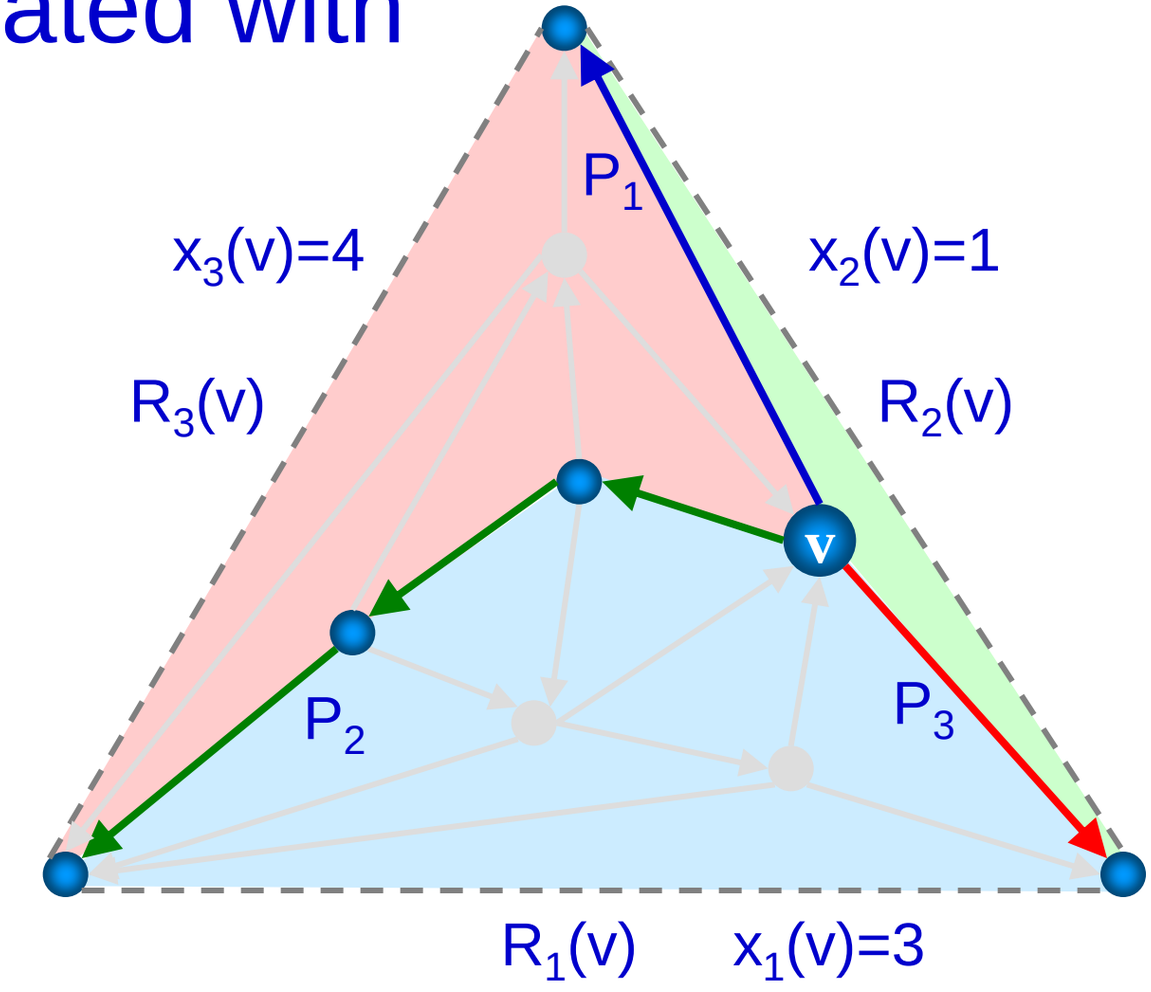
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Properties of Schnyder realizers

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  - count nodes inside $R_i(v)$ and add nodes different from v in path P_{i-1}



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Obtaining a 3D drawing

- For each internal node v we have

$$x_1(v)+x_2(v)+x_3(v) = n - 1$$

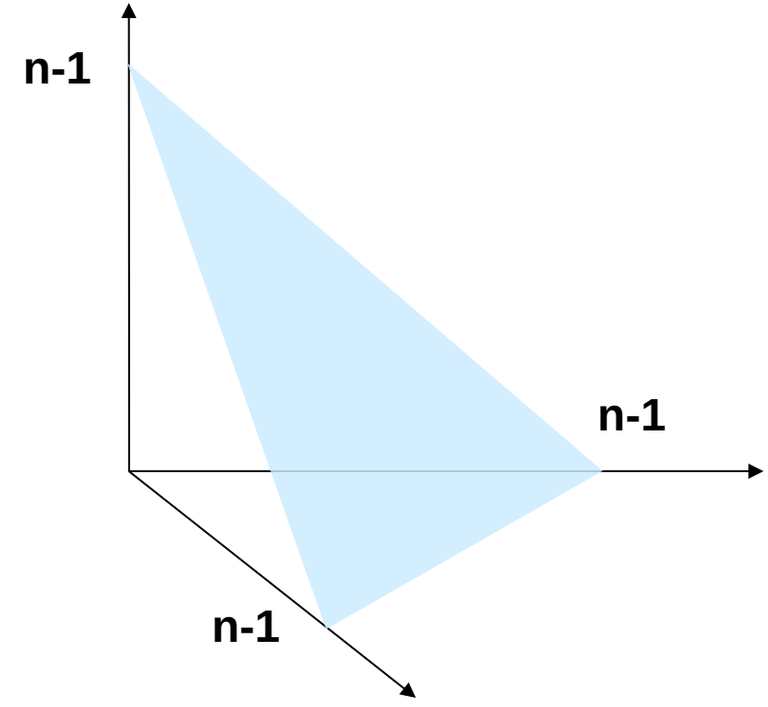
  - this is because we are actually counting all nodes with the exception of v
- Consider for each internal node v the coordinates $(x_1(v), x_2(v), x_3(v))$
- Assign to the external nodes
  - root of T_1 coordinates (n-2,1,0)
  - root of T_2 coordinates (0,n-2,1)
  - root of T_3 coordinates (1,0,n-2)

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Obtaining a 3D drawing

- Since for each node the sum of all its coordinates is equal to n-1, all the nodes lie on plane

$$x_1 + x_2 + x_3 = n - 1$$



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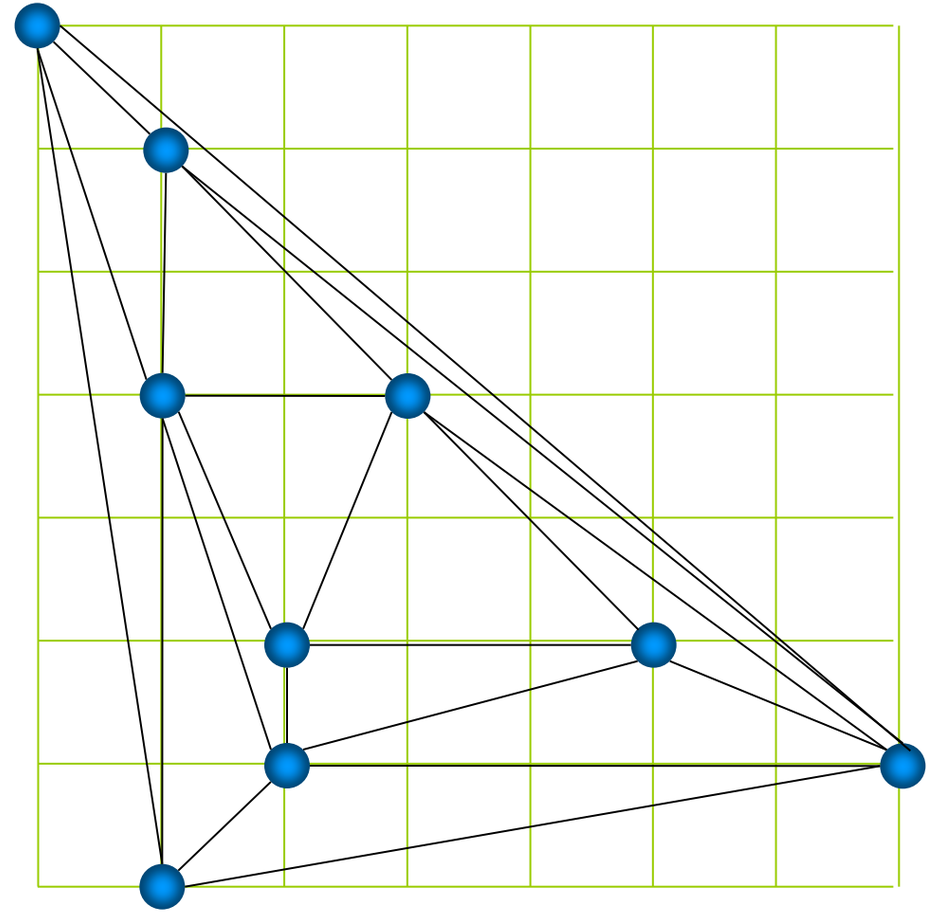
Obtaining a 2D drawing

- It can be proved that the produced 3D drawing does not have intersections
- Also, each node has all integer coordinates
- Projecting such drawing on 2D (i.e., omitting one coordinate) yields a straight-line grid drawing

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An example of drawing

- The same graph of the previous slides



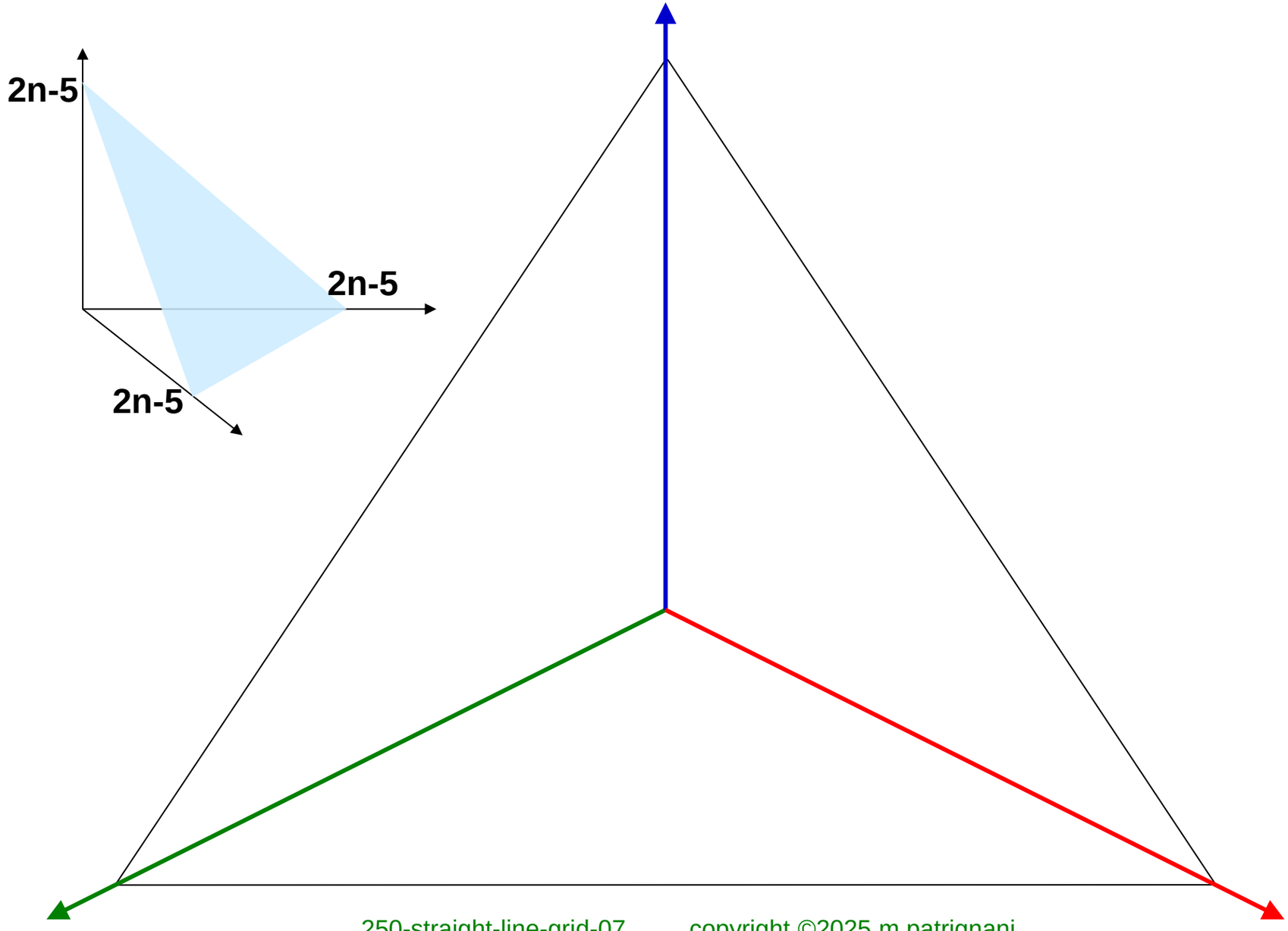
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Proving planarity

- Here we give some hint of the proof that the obtained drawing is planar
- We will consider a simpler model, where the weight $x_i(v)$ is the number of faces of the region $R_i(v)$
  - since there are 2n-4 faces (including the external one) this guarantees a drawing in the (2n-5) $\times$ (2n-5) grid

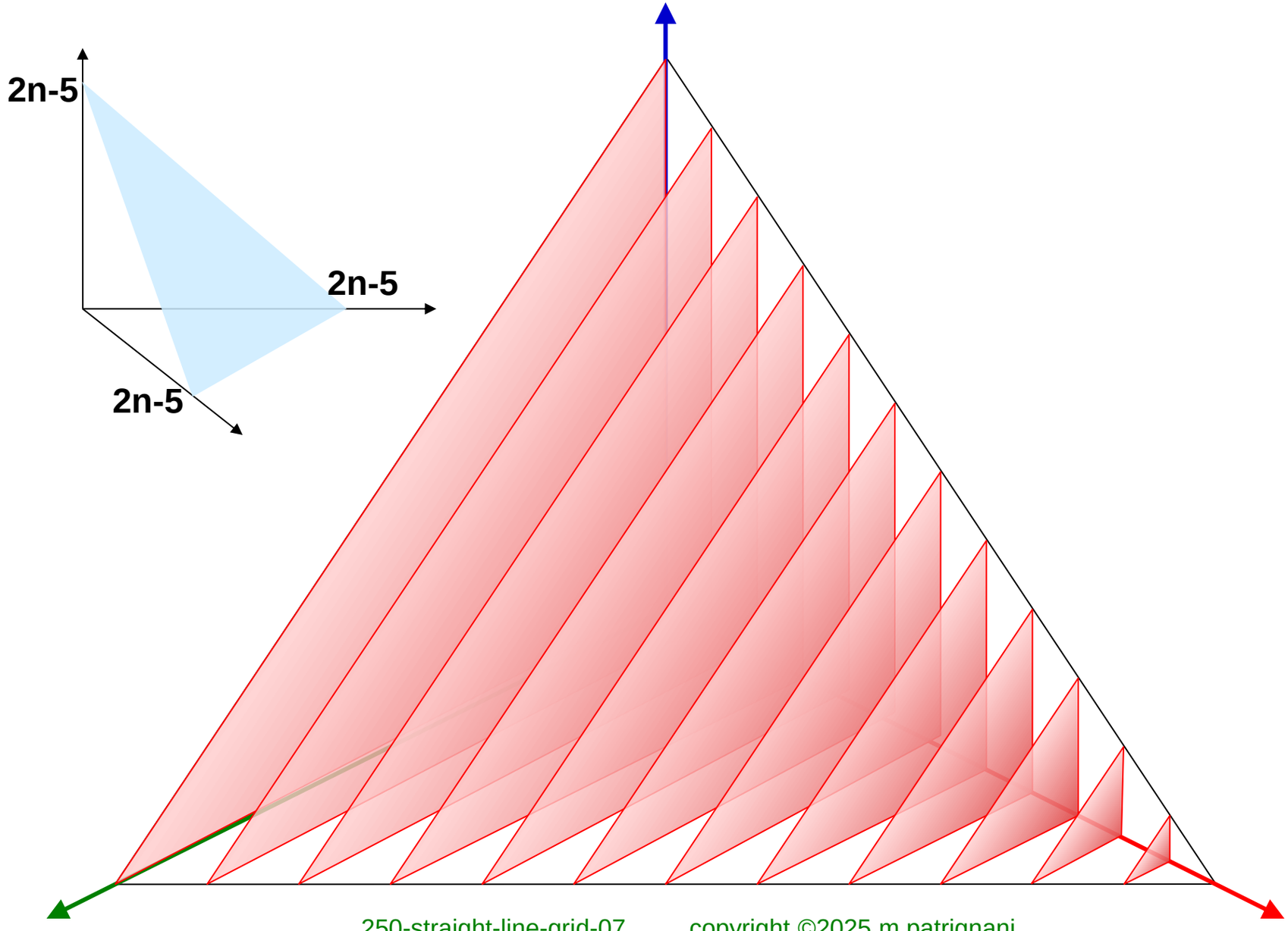
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Drawing on the $x_1+x_2+x_3=2n-5$ plane



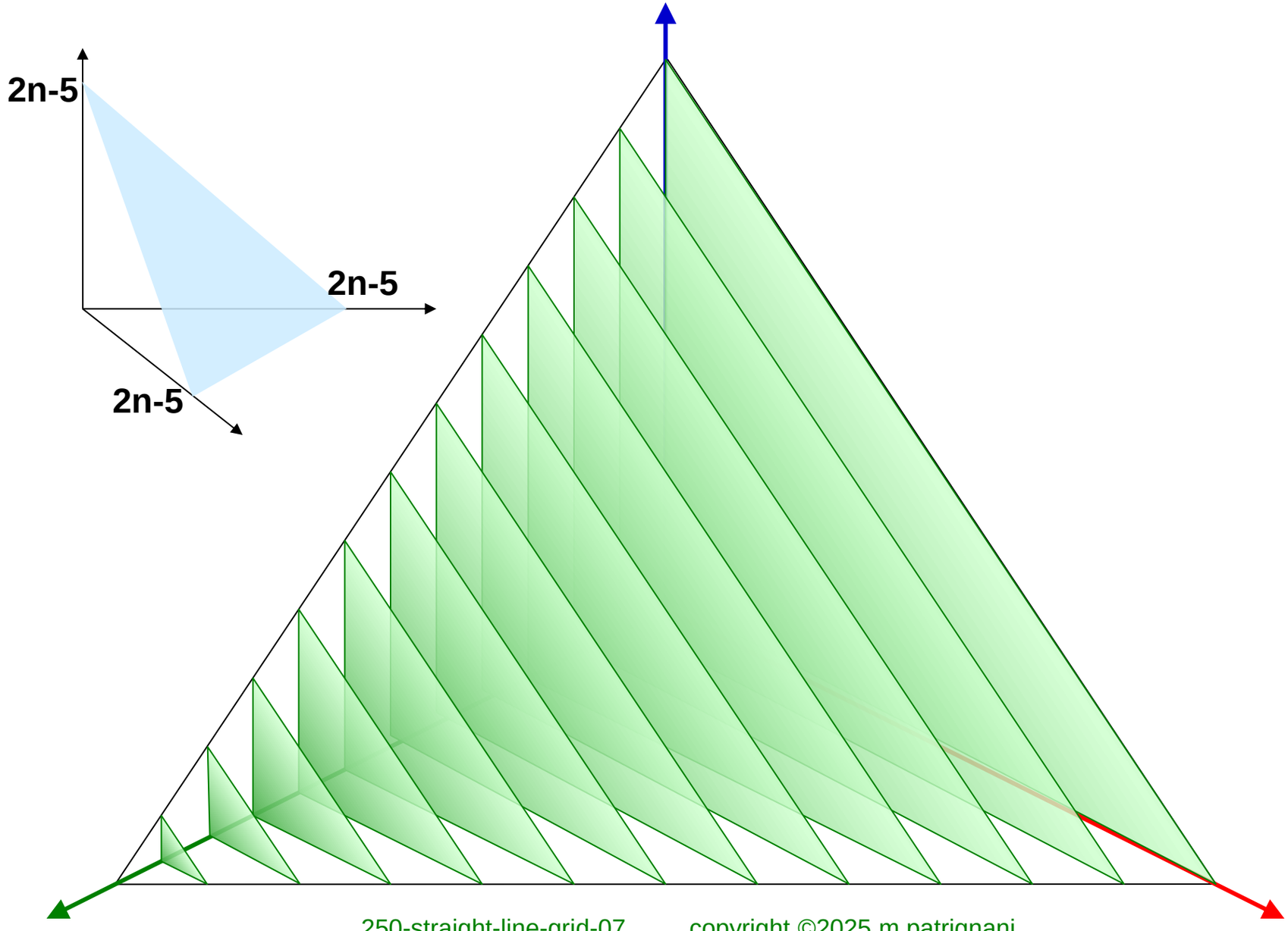
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Drawing on the $x_1+x_2+x_3=2n-5$ plane



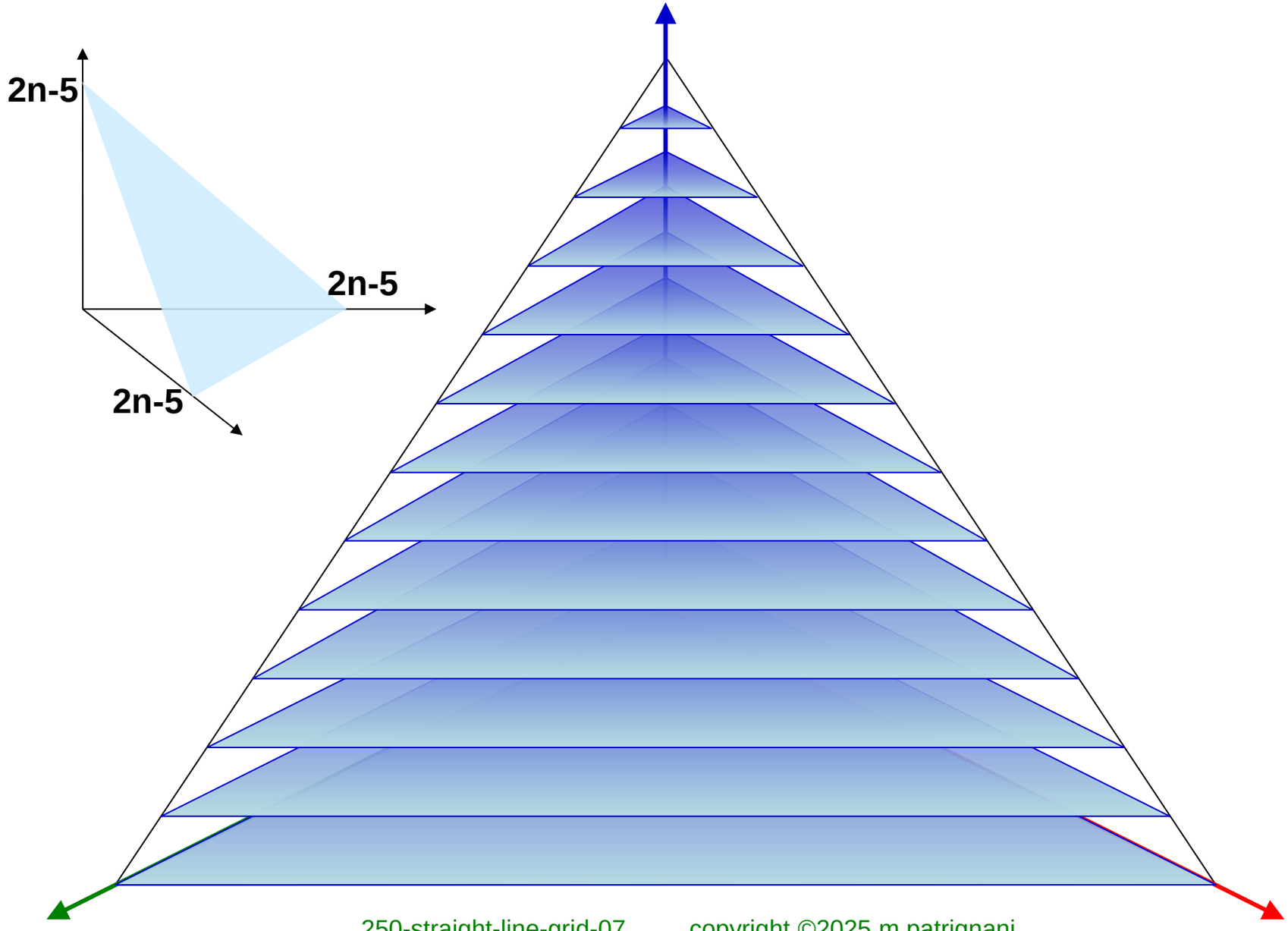
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Drawing on the $x_1+x_2+x_3=2n-5$ plane



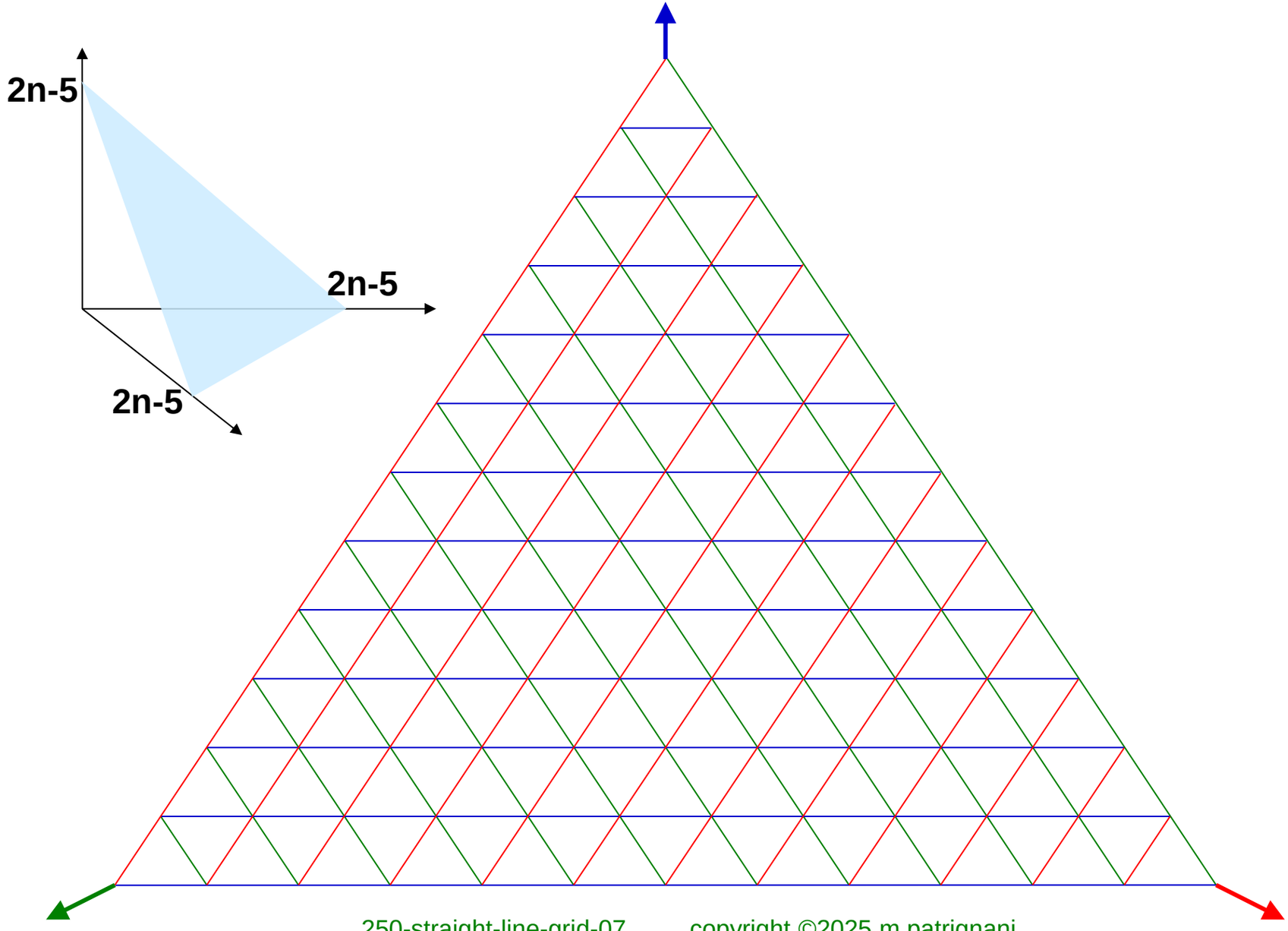
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Drawing on the $x_1+x_2+x_3=2n-5$ plane



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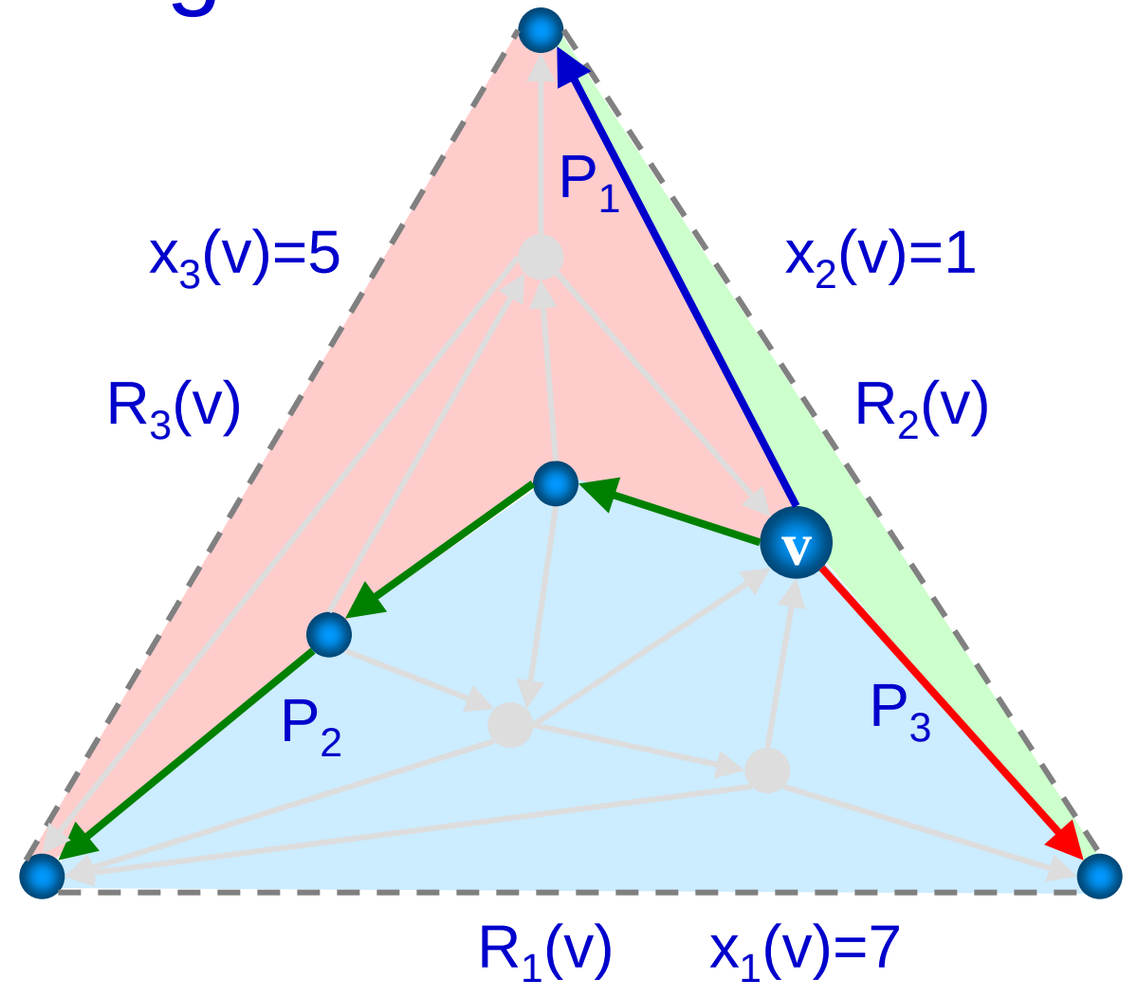
Drawing on the $x_1+x_2+x_3=2n-5$ plane



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Properties of Schnyder realizers

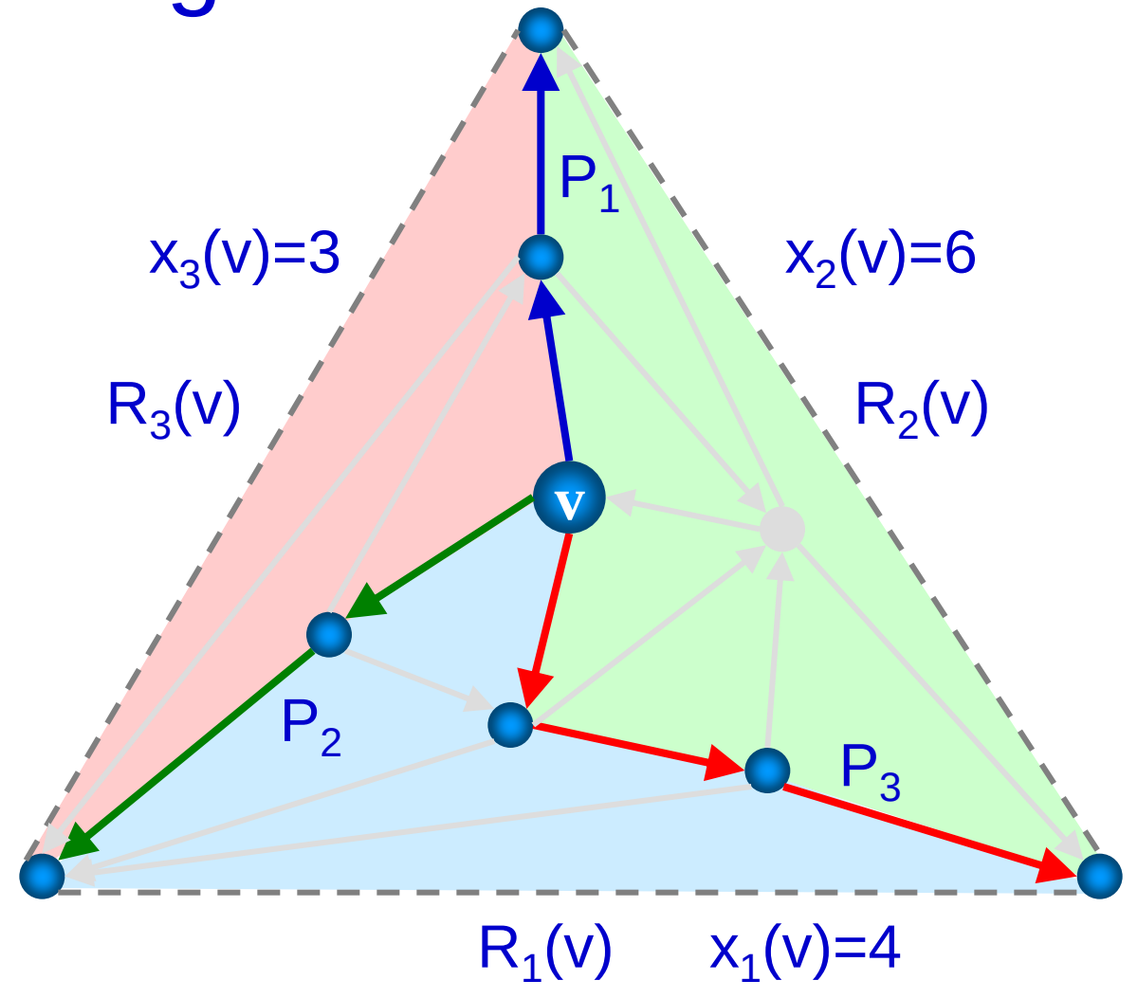
- Moving a node along a green edge
  - the green coordinate $R_2(v)$ increases
  - the other coordinates $R_1(v)$ and $R_3(v)$ decrease



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Properties of Schnyder realizers

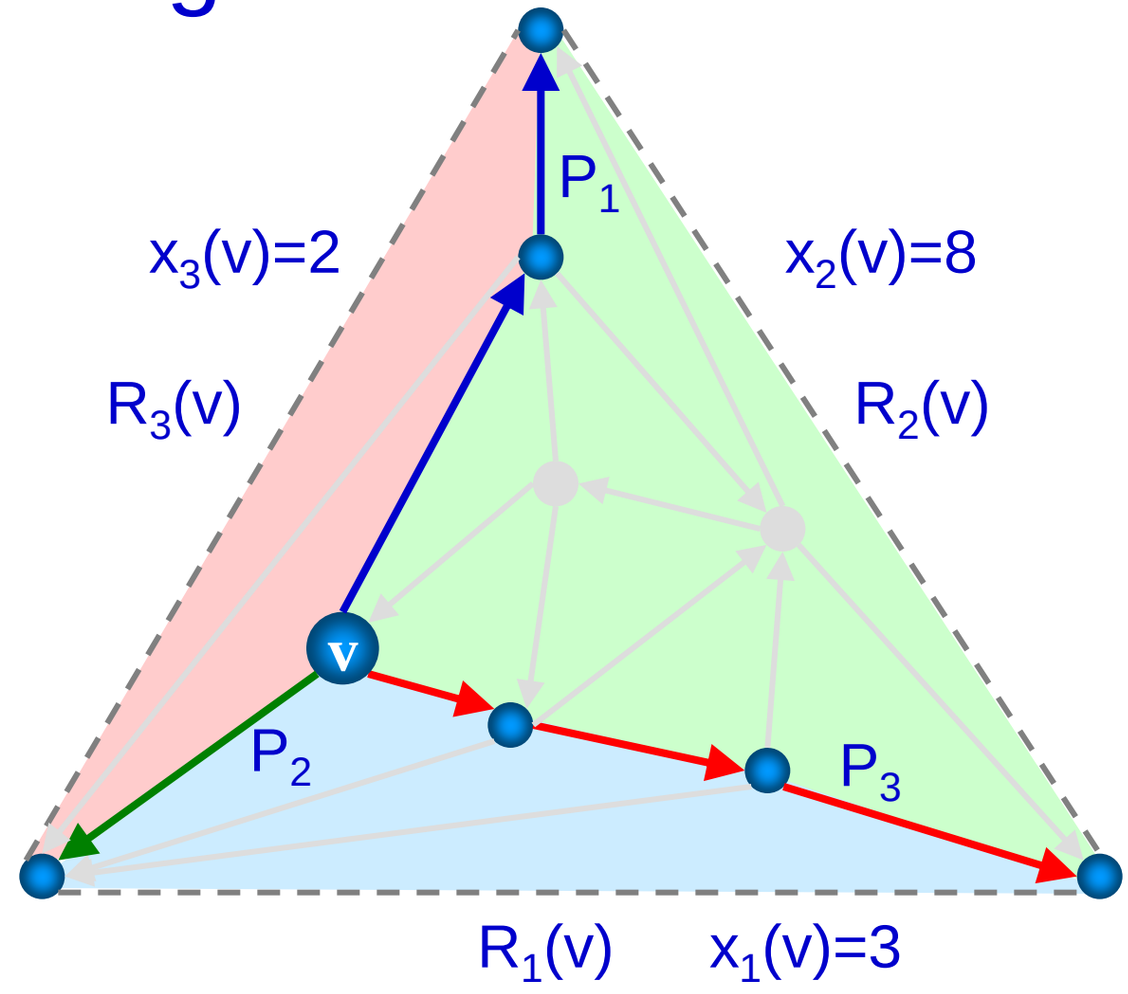
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Properties of Schnyder realizers

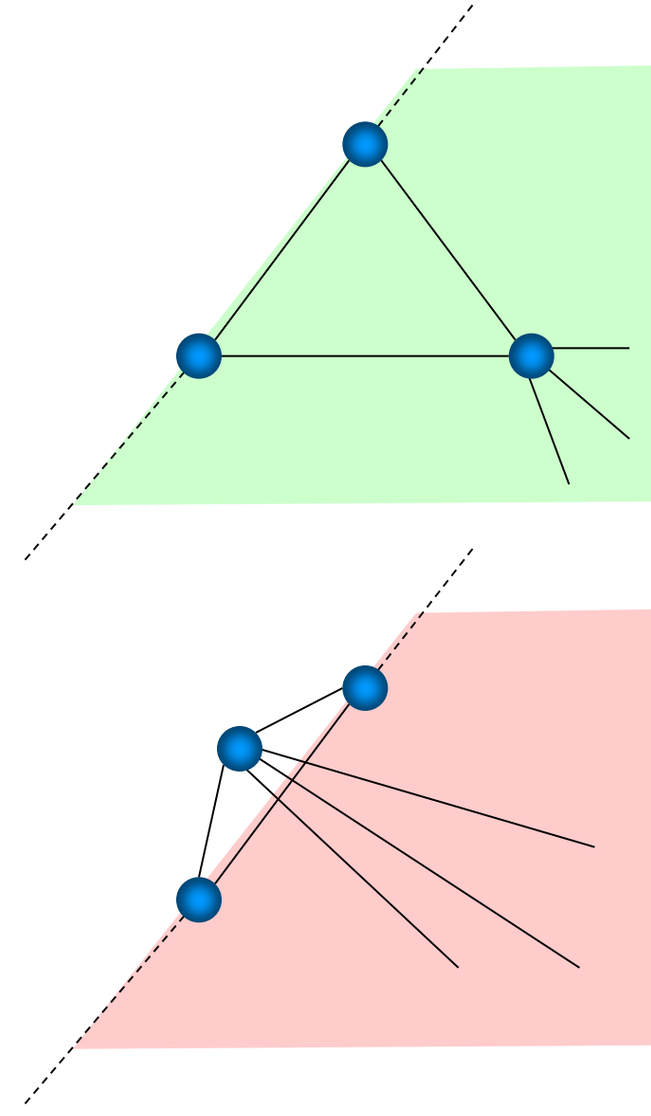
- Moving a node along a green edge
  - the green coordinate $R_2(v)$ increases
  - the other coordinates $R_1(v)$ and $R_3(v)$ decrease



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A planarity condition

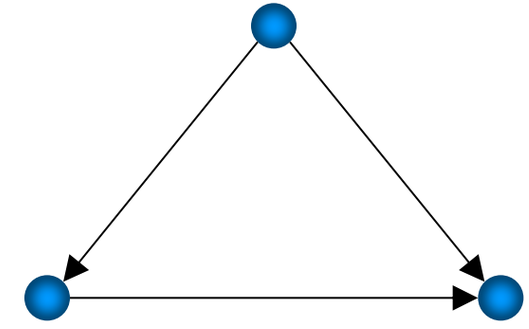
- Each internal face of the drawing is always a triangle
- The drawing is planar if coordinates are such that each face turns counterclockwise as in its topological description
  - intuitively this means that for each node *v* of the face, *v* is on the “right” side of the line passing throught the other two nodes

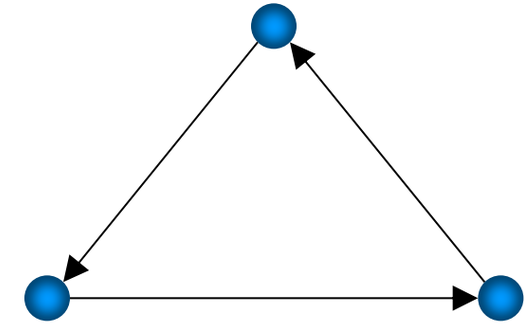


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Faces and directed edges

- Consider a face in the drawing
- Two are the possibilities
  1. the face has a node with two outgoing edges
  2. the face is a directed cycle

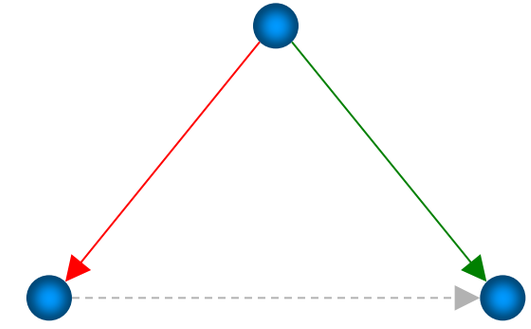




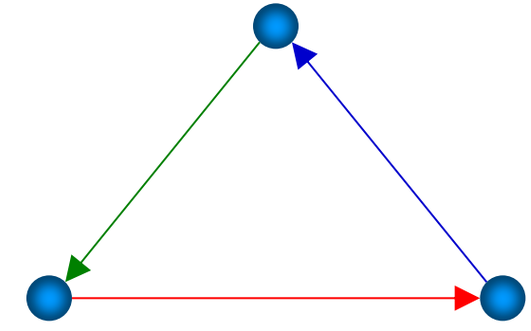
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Faces and colors

1. In the first case the two outgoing edges are necessarily of different colors
   - this is because from each node there is only one outgoing edge per color



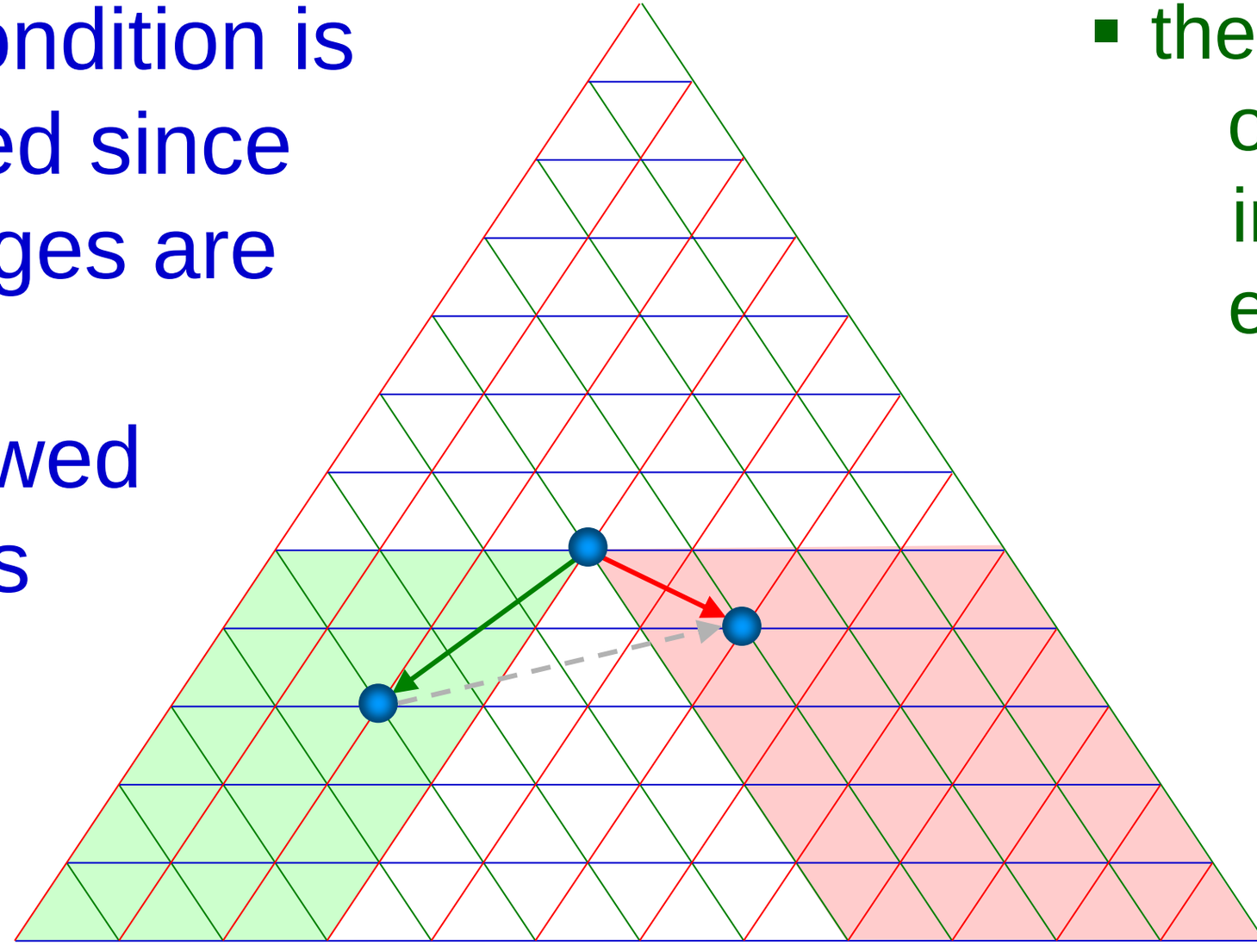
2. In the second case the three edges are of different colors
   - as there cannot be a directed cycle of one or two colors



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First case of the planarity condition

- The condition is satisfied since the edges are in the shadowed regions

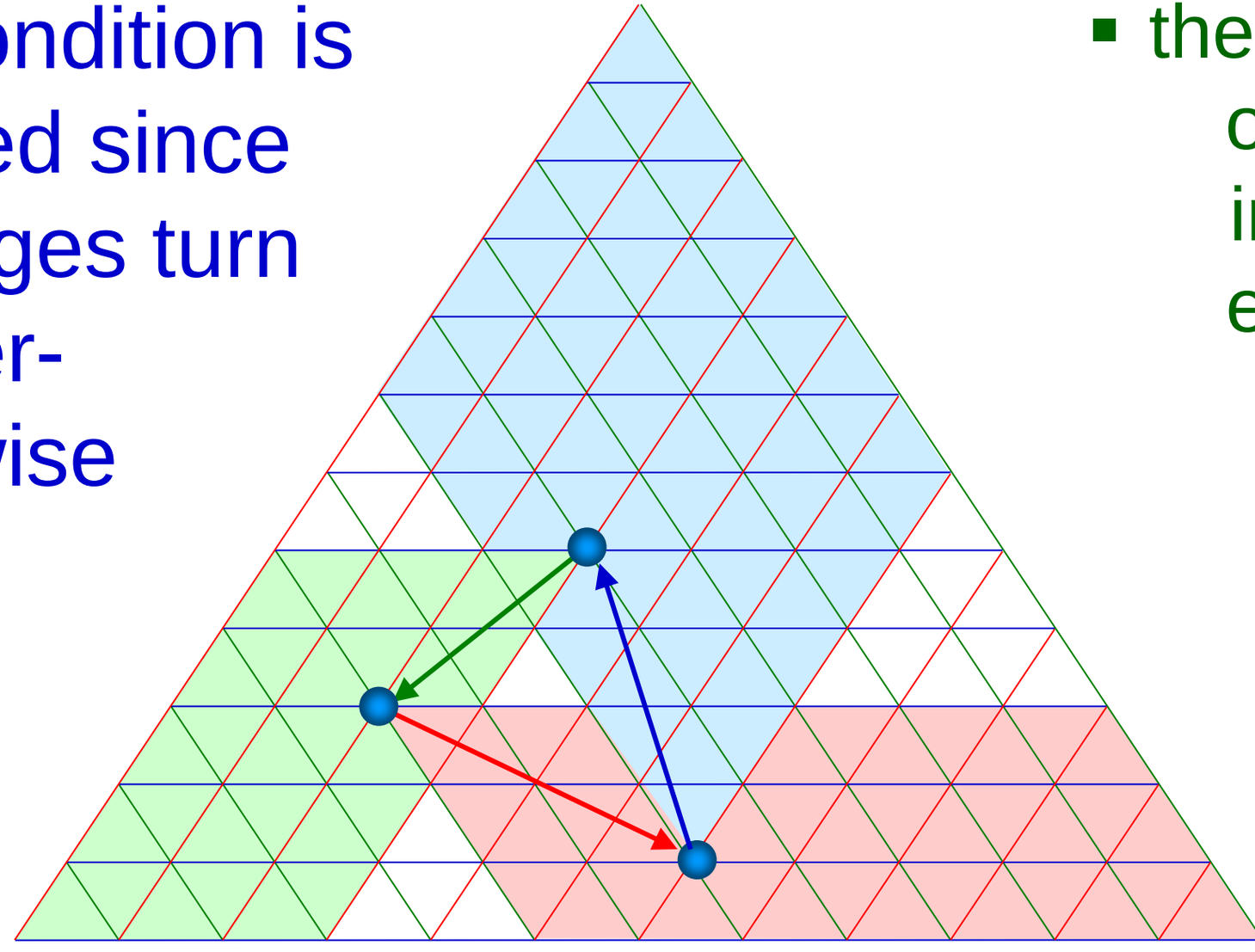


- the triangle is oriented as in the input embedding

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Second case of the planarity condition

- The condition is satisfied since the edges turn counter-clockwise



- the triangle is oriented as in the input embedding

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Bibliography and further readings

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